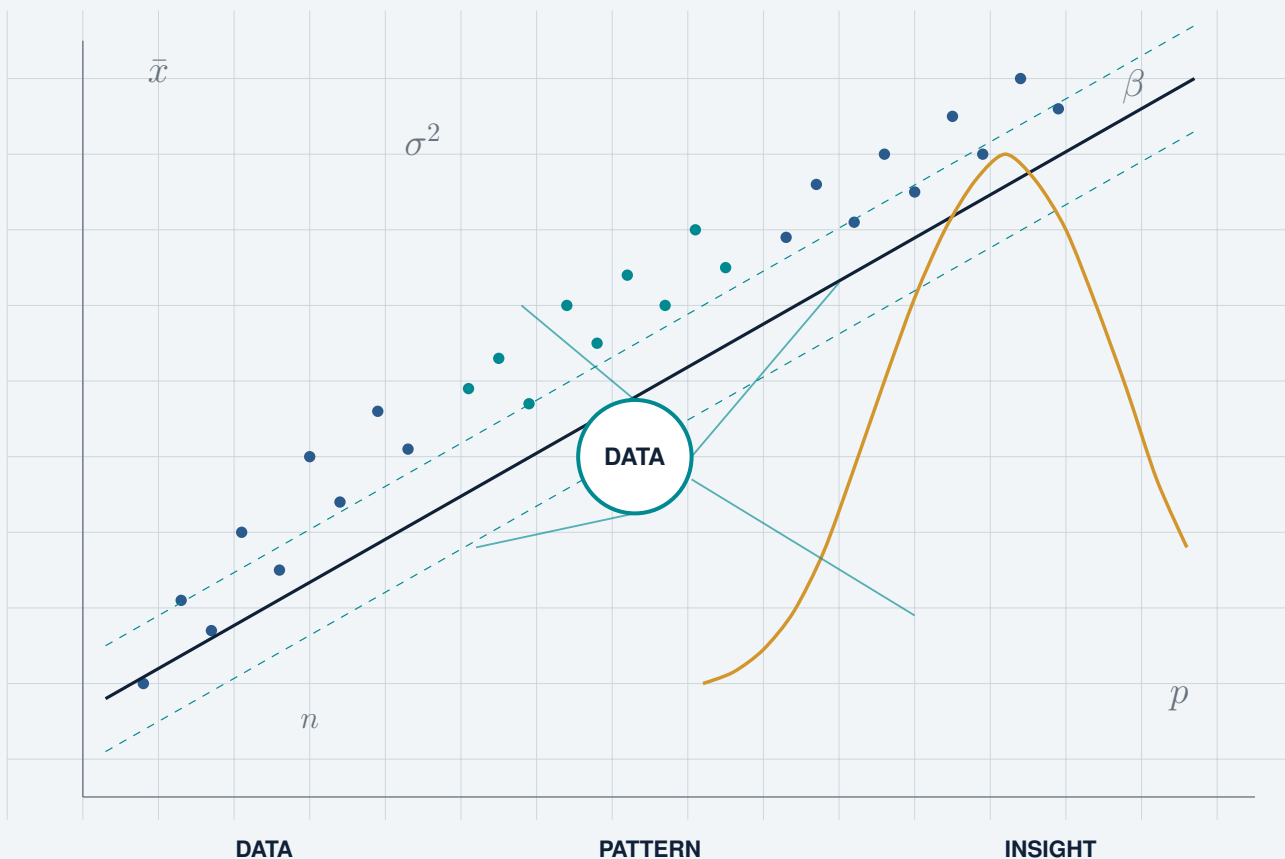


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NEPAL STATISTICAL SOCIETY

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The journal welcomes original research, methodological contributions, empirical studies, review articles, and applications of statistical and data-science methods across diverse fields.

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FROM THE EDITOR-IN-CHIEF

It is with great pleasure that we are presenting the inaugural publication of the *Nepal Journal of Statistics and Data Sciences (NJSD)*, the first journal publication of the **Nepal Statistical Society (NSS)**. This issue marks an important milestone not only for the Society but also for the growing community of statisticians, data scientists, researchers, academics, and practitioners in Nepal.

The establishment of the *Nepal Statistical Society* reflects a collective commitment to promoting the study, teaching, application, and advancement of statistics and data science. The publication of NJSD represents a natural extension of this mission: to create a scholarly platform through which researchers can share methodological innovations, empirical evidence, and applications of statistical and data-science techniques to contemporary problems.

This inaugural issue brings together eight original research contributions covering a diverse range of topics, including structural equation modeling, forecasting, artificial intelligence, GIS and IoT, natural calamities, economic modeling, acceptance sampling, and statistical methodology. The diversity of these contributions reflects the increasingly interdisciplinary nature of statistics and data science and demonstrates how quantitative methods can contribute to understanding problems across education, banking, economics, industry, disaster management, and other fields.

Statistics has traditionally served as a foundation for scientific inquiry by providing systematic approaches to collecting, analyzing, interpreting, and communicating evidence. Today, however, the statistical landscape is changing rapidly. The increasing availability of digital data, advances in computing, artificial intelligence, machine learning, geographic information systems, and Internet of Things technologies have created new opportunities for quantitative research. At the same time, these developments have created new responsibilities concerning data quality, methodological rigor, transparency, reproducibility, privacy, and ethical use of data.

In this changing environment, the role of statistics is not diminished; rather, it becomes even more important. Statistical reasoning provides the principles necessary to distinguish meaningful patterns from random variation, association from causation, prediction from explanation, and information from evidence.

The papers included in this issue illustrate this broader role of statistics.

The contributions employing **structural equation modeling** demonstrate the value of statistical techniques for investigating relationships among complex constructs that cannot always be observed directly. Applications in higher education and the banking sector show how quantitative evidence can inform institutional improvement and organizational decision-making.

The study employing **Box–Jenkins ARIMA modeling for Nepal’s nominal GDP** highlights the continuing importance of statistical forecasting for economic analysis. Reliable forecasts can provide useful information for economic planning, policy formulation, resource allocation, and the assessment of future economic conditions.

The application of **artificial intelligence, GIS, and IoT technologies** to industrial productivity demonstrates another emerging direction for quantitative research. The integration of statistical thinking with computational and digital technologies offers considerable potential for developing more responsive and intelligent decision-support systems.

Research examining **deaths associated with natural calamities using Nepal’s census data** illustrates the social value of statistical analysis. Nepal faces substantial exposure to earthquakes, floods, landslides, and other hazards. Systematic analysis of population and demographic data can help identify patterns of vulnerability and contribute to evidence-based disaster preparedness and risk-reduction policies.

Similarly, research concerning **minimum wage determination, economic growth, and inflation** demonstrates how quantitative modeling can contribute to important socioeconomic policy questions. Statistical evidence can help policymakers move beyond assumptions and anecdotal observations toward more systematic assessments of economic relationships.

The work on **empirical Bayesian acceptance sampling** highlights the practical contribution of statistical methodology to quality assurance and decision-making, while the review of the **history,**

methods, and applications of structural equation modeling provides methodological context that can support researchers in appropriately applying advanced statistical techniques.

Although these studies differ substantially in subject matter and methodology, they share a common foundation: the use of quantitative evidence to understand real-world phenomena and support better decisions.

This is precisely the kind of research culture that the *Nepal Statistical Society* seeks to foster.

The launch of NJSD is therefore more than the publication of another academic journal. It represents the beginning of an institutional platform for bringing together researchers and practitioners who share an interest in statistics and data science. We hope the journal will encourage researchers from Nepal and beyond to contribute rigorous theoretical, methodological, empirical, and interdisciplinary research.

Nepal is experiencing rapid growth in the generation and use of data across government, business, education, development, health, finance, environment, and technology. However, the availability of data alone does not guarantee better decisions. The transformation of data into reliable knowledge requires appropriate research design, sound statistical methods, careful interpretation, and effective communication of evidence.

The Society therefore has an important role to play in strengthening statistical capacity and promoting responsible data practices. NJSD will contribute to this broader objective by providing a forum for scholarly communication and by encouraging research that addresses both national and international challenges.

We extend our sincere appreciation to all authors who have contributed their research to this inaugural issue. We are equally grateful to the reviewers for their time, expertise, and constructive assessments, and to the editorial team and members of the Nepal Statistical Society whose efforts have made this first publication possible.

The publication of this issue is a beginning. Much remains to be done to strengthen statistical research, education, professional practice, and data science in Nepal. We invite researchers, academics, practitioners, and students to participate in this journey by sharing their knowledge, questioning established approaches, developing new methods, and applying quantitative evidence to the challenges facing our society.

As the *Nepal Statistical Society* takes this first step in scholarly publishing through the *Nepal Journal of Statistics and Data Sciences*, we look forward to building a vibrant and inclusive community dedicated to the advancement of statistics and data science.

Bijay Lal Pradhan, PhD

Editor-in-Chief

Nepal Journal of Statistics and Data Sciences

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Perceived Service Quality and Students' Satisfaction in a Higher Education Institution: A Structural Equation Modeling Approach

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ABSTRACT

Background: Increasing demands in terms of efficient operations, cost reduction, and improving productivity have emerged in the sugar industry due to increased competition and sustainable developments. The current pilot study suggests a framework to evaluate the impact that integration of AI, GIS, and IoT would bring on productivity of sugar mills.

Method: This research adopts a mixed frameworks approach to investigate the impact of integrated AI, GIS and IoT technologies on sugar mill productivity. Based on the simulated data that represents more than 20 sugar mills operating in Maharashtra, India, our findings suggest that adoption metrics concerning these technologies positively correlate with performance metrics such as productivity index, cost reduction, downtime reduction, and improvement in sugar recovery.

Result: AI adoption correlated most strongly with productivity index ($r = 0.264$), followed by integration of IoT technologies ($r = 0.119$) and GIS-based monitoring ($r = 0.042$). Random Forest regression provided R^2 value of 0.082. Among the most significant predictors, predictive maintenance score (14.8%) and scores of AI adoption and road quality index (13.5% each) were detected. The average productivity index was at 85.82, while average reductions in cost, downtime, and sugar recovery amounted to 14.12%, 31.92%, and 4.86% correspondingly.

Conclusion: In general, it could be noted that the proposed framework is applicable to various locations where sugar is produced. This study provides a methodology that can be replicated to explore the relationship between technology and productivity in the agro-industrial sector, while emphasizing the importance of predictive maintenance and AI decision-making systems to achieve business outcomes.

Key words: artificial intelligence, geographic information systems, internet of things, sugar industry, productivity analysis, predictive maintenance, digital transformation, operational efficiency

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INTRODUCTION

Education is an important service sector and a cornerstone of national development and advancement (Hill, 1995). In higher education, institutions are responsible not only for delivering academic knowledge but also for providing effective administrative, academic, and student support services. Therefore, the quality of these services is an important consideration in understanding students' overall educational experiences. Service quality is closely associated with customer satisfaction and loyalty in service-oriented organizations, and this relationship is increasingly relevant to higher education institutions (Kasiri et al., 2017). Customer satisfaction is particularly important because of its close connection with perceived service quality (Lupo, 2014).

Perceived service quality in higher education refers to students' overall evaluation of the quality of services provided by an educational institution. Students' perceptions may be influenced by teaching quality, institutional facilities, administrative support, and the responsiveness of the institution to their needs. Quality is also con-

sidered a strategic means of improving organizational performance and meeting established quality standards (DeShields et al., 2005). Service quality is commonly assessed through five dimensions: tangibility, reliability, responsiveness, assurance, and empathy, which collectively represent important aspects of students' experiences within the educational environment (Daniel and Berinyuy, 2010).

Understanding and improving perceived service quality is essential for creating a supportive learning environment and achieving meaningful educational outcomes (Becket and Brookes, 2006). Students' satisfaction may also vary according to their sociodemographic characteristics. Therefore, examining both sociodemographic factors and perceived service quality can provide a broader understanding of students' satisfaction. Therefore, this study aimed to examine the association between students' satisfaction and sociodemographic characteristics and to assess the relationship between perceived service quality and students' satisfaction.

METHODS

An analytical cross-sectional study was conducted among Bachelor's and Master's level students of Birendra Multiple Campus, Bharatpur-10, Chitwan, Nepal, from January to April 2024. The study aimed to examine the association between students' satisfaction and sociodemographic characteristics and to assess the relationship between perceived service quality and students' satisfaction. The study population comprised Bachelor's and Master's level students enrolled at Birendra Multiple Campus.

The sample size was calculated using the single population proportion formula:

$$n = \frac{Z^2 pq}{e^2} \quad (1)$$

where n is the required sample size, Z is the standard normal deviate corresponding to the desired confidence level, p is the estimated proportion of satisfaction, $q = 1 - p$, and e is the margin of error.

For a 95% confidence level, 50% satisfaction level, and 5% margin of error,

Where, $Z = 1.96$, $p = 0.50$, $q = 0.50$, $e = 0.05$

Therefore,

$$n = \frac{(1.96)^2(0.50)(0.50)}{(0.05)^2} \quad (2)$$

$$n = \frac{3.8416 \times 0.25}{0.0025} = 384.16 \approx 384 \quad (3)$$

Thus, calculated sample size was 384 students. Participants were selected using a non-probability purposive sampling technique.

Data were collected using a structured questionnaire. The questionnaire included sections on sociodemographic characteristics, perceived service quality, and students' satisfaction. Sociodemographic characteristics included age, gender, ethnicity, religion, place of residence, family monthly income, faculty, and academic year. Perceived service quality was assessed across five dimensions: tangibility, reliability, responsiveness, assurance, and empathy. Tangibility referred to the physical and observable aspects of institutional services and was measured using five items (T1–T5); reliability assessed the consistency, dependability, and accuracy of service delivery using five items (R1–R5); responsiveness assessed the willingness and promptness of service providers in addressing students' needs using four items (Re1–Re4); assurance assessed competence, courtesy, credibility, trust, and confidence using four items (As1–As4); and empathy assessed the extent to which service providers understand and respond to students' individual needs using five items (Em1–Em5). Students' satisfaction was assessed using nine items (Sa1–Sa9). All service-quality and satisfaction items were measured using a five-point Likert scale ranging from strongly agree (5) to strongly disagree (1).

The collected data were checked for completeness, coded, and analyzed using SPSS, R, and Jamovi software. Descriptive statistics were used to summarize the characteristics of the participants and study variables. Frequencies and percentages were calculated for categorical variables, while means and standard deviations were

calculated for scale-level variables. The Chi-square test was used to examine the association between students' satisfaction and selected sociodemographic characteristics. Structural Equation Modeling (SEM) was employed to examine the relationships between the dimensions of perceived service quality and students' satisfaction.

Structural Equation Modeling

Structural Equation Modeling (SEM) was used to examine the hypothesized relationship between perceived service quality and students' satisfaction while accounting for measurement error in the latent constructs. The analysis consisted of two related components: the measurement model, which assessed whether the observed questionnaire items adequately represented their respective latent constructs, and the structural model, which evaluated the hypothesized relationships among the constructs (Pradhan, 2017). The measurement model was expressed as

$$\mathbf{x} = \mathbf{\Lambda}_x \boldsymbol{\xi} + \boldsymbol{\delta} \quad (4)$$

where $\mathbf{x}$ and $\mathbf{y}$ denote observed indicators, $\boldsymbol{\xi}$ and $\boldsymbol{\eta}$ represent latent constructs, $\mathbf{\Lambda}_x$ and $\mathbf{\Lambda}_y$ are factor-loading matrices, and $\boldsymbol{\delta}$ and $\boldsymbol{\epsilon}$ represent measurement errors. The structural relationship was specified as

$$\eta = \beta \xi + \zeta \quad (5)$$

where η represents students' satisfaction, ξ represents perceived service quality, β is the structural path coefficient representing the effect of perceived service quality on students' satisfaction, and ζ is the structural disturbance term.

The adequacy of the measurement model was assessed using standardized factor loadings, Composite Reliability (CR), and Average Variance Extracted (AVE), which are commonly used to evaluate the reliability and validity of latent constructs in confirmatory measurement models (Pradhan, 2017; Pradhan et al., 2023), with

$$CR = \frac{\left(\sum_{i=1}^k \lambda_i\right)^2}{\left(\sum_{i=1}^k \lambda_i\right)^2 + \sum_{i=1}^k \theta_i} \quad (6)$$

and

$$AVE = \frac{\sum_{i=1}^k \lambda_i^2}{\sum_{i=1}^k \lambda_i^2 + \sum_{i=1}^k \theta_i} \quad (7)$$

where λ_i denotes the standardized factor loading and θ_i denotes the corresponding measurement-error variance.

Overall Model Fit and Structural Path Assessment

Overall model fit was evaluated by comparing the observed covariance matrix $\mathbf{S}$ with the model-implied covariance matrix $\boldsymbol{\Sigma}(\boldsymbol{\theta})$. The chi-square goodness-of-fit statistic was used to test the null hypothesis that the model-implied covariance matrix does not differ from the observed covariance matrix:

$$\chi^2 = (N - 1)F_{\min} \quad (8)$$

with

$$F_{\min} = F[\mathbf{S}, \boldsymbol{\Sigma}(\boldsymbol{\theta})] \quad (9)$$

where N is the sample size and $F_{\min}$ is the minimum value of the fitting function. The chi-square statistic was interpreted together with its degrees of freedom because it is sensitive to sample size. The relative chi-square was calculated as

$$\frac{\chi^2}{df} \quad (10)$$

with smaller values indicating less discrepancy between the hypothesized and observed covariance structures.

Incremental model fit was assessed using the Comparative Fit Index (CFI) and Tucker–Lewis Index (TLI). CFI compares the proposed model with an independence or baseline model and can be expressed as

$$CFI = 1 - \frac{\max(\chi_m^2 - df_m, 0)}{\max(\chi_b^2 - df_b, 0)} \quad (11)$$

where the subscripts m and b denote the proposed and baseline models, respectively. TLI evaluates improvement in fit while accounting for model complexity and can be expressed as

$$TLI = \frac{\left(\frac{\chi_b^2}{df_b}\right) - \left(\frac{\chi_m^2}{df_m}\right)}{\left(\frac{\chi_b^2}{df_b}\right) - 1} \quad (12)$$

Approximate fit was assessed using the Root Mean Square Error of Approximation (RMSEA),

$$RMSEA = \sqrt{\frac{\max(\chi^2 - df, 0)}{df(N-1)}} \quad (13)$$

which reflects the degree of model misfit per degree of freedom. The Standardized Root Mean Square Residual (SRMR) was also reported as a measure of the average standardized difference between the observed and model-implied correlations/covariances. Values of CFI and TLI closer to 1 and lower values of RMSEA and SRMR indicate better fit. As commonly used guidelines, CFI and TLI values around 0.95 or higher and RMSEA and SRMR values around 0.06 and 0.08 or lower, respectively, indicate relatively good fit, although these values were not treated as rigid universal cutoffs.

The hypothesized relationship between perceived service quality and students' satisfaction was assessed using the estimated structural path coefficient. Its standard error was used to obtain the test statistic,

$$z = \frac{\hat{\beta}}{SE(\hat{\beta})} \quad (14)$$

and the corresponding two-sided p -value. A $(1 - \alpha) \times 100\%$ confidence interval was calculated as

$$\hat{\beta} \pm z_{\alpha/2} SE(\hat{\beta}) \quad (15)$$

The relationship was considered statistically significant at the 5% significance level if $p < 0.05$, or equivalently,

if the 95% confidence interval for the path coefficient did not include zero. The magnitude and direction of β were used to assess the strength and direction of the hypothesized relationship.

RESULTS

Among the 392 students, the mean age was 22.12 ± 4.34 years, with the largest proportion aged 20–25 years (46.9%), followed by those aged < 20 years (42.6%). Females constituted 55.4% of the participants, while 74.5% belonged to the Brahmin/Chhetri ethnic group.

Most participants were Hindu (93.4%), while 5.4% were Buddhist and the remaining participants followed other religions. In terms of marital status, the majority were unmarried (87.8%), whereas 11.7% were married and 0.5% reported other marital statuses. This distribution reflects the predominantly young student population included in the study.

Regarding occupation, half of the participants (50.8%) were associated with agriculture, followed by those abroad (20.7%) and those engaged in service (19.9%). The remaining 8.7% were classified under other occupations. Participants were almost equally distributed by place of residence, with 50.3% living in rural areas and 49.7% in urban areas. Regarding monthly family income, the largest proportion (37.5%) reported an income of 50,000–1 lakh, while 33.4% reported less than 50,000 and 29.1% reported 1 lakh or more.

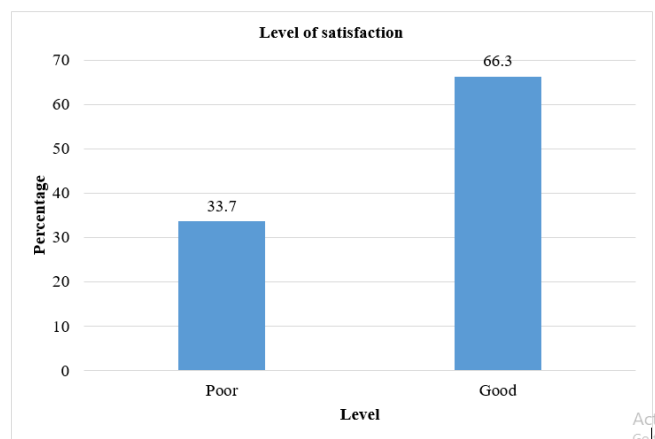
By educational stream, Science students constituted the largest proportion (35.97%), followed by Arts (28.83%), Management (28.06%), and Education (7.14%). Most participants were pursuing Bachelor's degrees (79.34%), whereas 20.66% were enrolled in Master's degree programs. Among the individual programs, MA had the highest representation (13.01%), followed by BBA (11.48%), BBS (8.93%), and BIT (8.16%). The remaining participants were distributed across BSc Physical Group, BSc Biology Group, BSc Microbiology, BSc Geology, BSc CSIT, B.Ed., BCA, and MBS programs, as presented in Table ??.

Overall, 66.3% of the students reported satisfaction with the various aspects of their academic environment. The estimated proportion of satisfied students was 66.3%, with a 95% confidence interval ranging from 61.41% to 71.00%. This suggests that approximately two-thirds of the students were satisfied with their academic environment, while a considerable proportion remained dissatisfied (Figure 1).

Table 2 presents the association between students' level of satisfaction and selected sociodemographic and educational characteristics. A statistically significant association was observed between age and satisfaction ($\chi^2 = 17.65$, $p = 0.001$). The proportion of students reporting good satisfaction increased with age, from 61.7% among those aged below 20 years to 64.1% among those aged 20–25 years and 90.5% among those aged 25–30 years. All participants aged 30–35 years and above 35 years re-

Table 1. Sociodemographic characteristics of students
($n = 392$)

Variables	Frequency	Percentage (%)
Age (years)		
< 20	167	42.6
20–25	184	46.9
25–30	21	5.4
30–35	10	2.6
≥ 35	10	2.6
Mean ± SD	22.12 ± 4.34	
Gender		
Male	175	44.6
Female	217	55.4
Ethnicity		
Brahmin/Chhetri	292	74.5
Dalit/Janajati	50	12.8
Terai/Madhesi	20	5.1
Others	30	7.7
Religion		
Hindu	366	93.4
Buddhist	21	5.4
Christian	2	0.5
Others	3	0.8
Marital status		
Unmarried	344	87.8
Married	46	11.7
Others	2	0.5
Occupation		
Agriculture	199	50.8
Service	78	19.9
Abroad	81	20.7
Others	34	8.7
Place of residence		
Urban	195	49.7
Rural	197	50.3
Monthly income		
< 50,000	131	33.4
50,000–1 lakh	147	37.5
≥ 1 lakh	114	29.1
Stream of education		
Science	141	35.97
Management	110	28.06
Education	28	7.14
Arts	113	28.83
Level of education		
Bachelor degree	311	79.34
Master degree	81	20.66
Faculty/Program		
BSc Physical group	31	7.91
BSc Biology group	25	6.38
BSc Microbiology	28	7.14
BSc Geology	30	7.65
BSc CSIT	27	6.89
B.Ed.	28	7.14
BIT	32	8.16
BCA	30	7.65
BBA	45	11.48
BBS	35	8.93
MBS	30	7.65
MA	51	13.01

**Figure 1.** Overall level of satisfaction.

ported good satisfaction. However, these older age groups had relatively small numbers of participants, and therefore the observed differences should be interpreted with caution.

No statistically significant association was found between gender and satisfaction ($\chi^2 = 0.172$, $p = 0.678$). Good satisfaction was reported by 67.4% of male students and 65.4% of female students, indicating only a small difference between the two groups. Similarly, satisfaction did not differ significantly by ethnicity ($\chi^2 = 2.141$, $p = 0.710$). The proportion reporting good satisfaction ranged from 66.1% among Brahmin/Chhetri students to 70.0% among both Dalit/Janajati and Terai/Madhesi students.

Religion was also not significantly associated with satisfaction ($\chi^2 = 1.200$, $p = 0.753$). Good satisfaction was reported by 66.4% of Hindu students and 61.9% of Buddhist students. However, the very small numbers in the Christian and other religious groups should be considered when interpreting these results. Likewise, marital status was not significantly associated with satisfaction ($\chi^2 = 3.324$, $p = 0.190$). Although married students reported a higher proportion of good satisfaction than unmarried students (76.1% vs. 64.8%), the difference was not statistically significant.

Occupation was not significantly associated with satisfaction ($\chi^2 = 3.312$, $p = 0.346$). Good satisfaction was highest among participants associated with other occupations (69.1%), followed by agriculture (68.8%), service (61.5%), and those abroad (55.9%). Similarly, place of residence did not show a statistically significant association with satisfaction ($\chi^2 = 3.175$, $p = 0.075$). Rural students reported a higher proportion of good satisfaction than urban students (70.6% vs. 62.1%); however, this difference did not reach the conventional level of statistical significance.

Monthly family income was also not significantly associated with satisfaction ($\chi^2 = 0.880$, $p = 0.640$). The proportion of students reporting good satisfaction was 66.4% among those with a monthly income below 50,000, 68.7% among those with an income of 50,000–1 lakh, and 63.2% among those with an income of 1 lakh or more. Thus, satisfaction levels were relatively similar across the

Table 2. Association of satisfaction with sociodemographic variables

Variables	Level of satisfaction		Chi-square	p-value
	Poor	Good		
Age (years)				
< 20	64 (38.3)	103 (61.7)	17.65	0.001*
20–25	66 (35.9)	118 (64.1)		
25–30	2 (9.5)	19 (90.5)		
30–35	0 (0.0)	10 (100)		
> 35	0 (0.0)	10 (100)		
Gender				
Male	57 (32.6)	118 (67.4)	0.172	0.678
Female	75 (34.6)	142 (65.4)		
Ethnicity				
Brahmin/Chhetri	99 (33.9)	193 (66.1)	2.141	0.710
Dalit/Janajati	15 (30.0)	35 (70.0)		
Terai/Madhesi	6 (30.0)	14 (70.0)		
Others	5 (31.3)	11 (68.8)		
Religion				
Hindu	123 (33.6)	243 (66.4)	1.200	0.753*
Buddhist	8 (38.1)	13 (61.9)		
Christian	0 (0.0)	2 (100)		
Others	1 (33.3)	2 (66.7)		
Marital status				
Unmarried	121 (35.2)	223 (64.8)	3.324	0.190
Married	11 (23.9)	35 (76.1)		
Others	0 (0.0)	2 (100)		
Occupation				
Agriculture	62 (31.2)	137 (68.8)	3.312	0.346
Service	30 (38.5)	48 (61.5)		
Abroad	15 (44.1)	19 (55.9)		
Others	25 (30.9)	56 (69.1)		
Place of residence				
Urban	74 (37.9)	121 (62.1)	3.175	0.075
Rural	58 (29.4)	139 (70.6)		
Monthly income (Rs.)				
< 50,000	44 (33.6)	87 (66.4)	0.880	0.640
50,000–1 lakh	46 (31.3)	101 (68.7)		
≥ 1 lakh	42 (36.8)	72 (63.2)		
Stream of education				
Science	56 (35.0)	104 (65.0)	19.63	< 0.0001
Management	62 (42.5)	84 (57.5)		
Education	10 (25.5)	29 (74.4)		
Arts	4 (8.5)	43 (91.5)		
Level of education				
Bachelor degree	128 (36.8)	220 (63.2)	13.41	< 0.001
Master degree	4 (9.1)	40 (90.9)		

*Likelihood ratio test value.

income categories.

In contrast, educational characteristics showed significant associations with satisfaction. A statistically significant association was observed between educational stream and satisfaction ($\chi^2 = 19.63$, $p < 0.0001$). Good satisfaction was highest among Arts students (91.5%), followed by Education (74.4%), Science (65.0%), and Management students (57.5%). This indicates considerable variation in satisfaction across different educational streams.

Similarly, level of education was significantly associated with satisfaction ($\chi^2 = 13.41$, $p < 0.001$). A substantially

higher proportion of Master's-level students reported good satisfaction (90.9%) compared with Bachelor's-level students (63.2%). This difference suggests that postgraduate students in the study population had more favorable perceptions of their academic environment than undergraduate students.

Overall, the findings demonstrate that age, educational stream, and level of education were significantly associated with students' satisfaction. In contrast, gender, ethnicity, religion, marital status, occupation, place of residence, and monthly family income did not show statistically significant associations with satisfaction. Among

the significant factors, educational stream and level of education showed particularly notable differences in the proportion of students reporting good satisfaction.

Structural Equation Modeling

The Structural Equation Modeling (SEM) results indicated that the proposed model demonstrated a good overall fit to the observed data. The SRMR value of 0.046 and RMSEA value of 0.047 indicated a close fit between the hypothesized model and the observed data. The χ^2/df ratio of 3.23 was within the generally acceptable range, suggesting an adequate balance between model fit and complexity. Furthermore, the CFI (0.963), TLI (0.950), NFI (0.962), and GFI (0.943) exceeded the commonly accepted threshold of 0.90, indicating satisfactory to good model fit. The AGFI value of 0.931 further supported the adequacy of the model after accounting for model complexity (Table 3).

The structural model demonstrated that all five dimensions of perceived service quality were positively and significantly associated with students' satisfaction. Tangibility showed a significant positive association with satisfaction ($\beta = 0.324$, 95% CI: 0.1017–0.546, $p = 0.006$), indicating that better physical facilities and observable aspects of institutional services were associated with higher student satisfaction. Similarly, reliability was positively associated with satisfaction ($\beta = 0.295$, 95% CI: 0.0966–0.4941, $p = 0.006$), suggesting that consistent and dependable service delivery contributed to greater satisfaction. Responsiveness also demonstrated a significant positive relationship with students' satisfaction ($\beta = 0.446$, 95% CI: 0.335–0.506, $p = 0.040$). Assurance was positively associated with satisfaction ($\beta = 0.249$, 95% CI: –0.0994–0.597, $p = 0.004$), while empathy showed the strongest positive association among the five dimensions ($\beta = 0.531$, 95% CI: 0.1904–0.8708, $p = 0.003$) (Table 4).

The value of adjusted R^2 in SEM is 0.769, which indicates that 76.9% of the variation in satisfaction is explained by the service quality domains included in the model, while the remaining variation in the dependent variable is due to other factors that are not included in the model. This indicates that the model has relatively strong explanatory power (Table 5).

The reliability and validity analysis demonstrated that all constructs exhibited satisfactory measurement properties, consistent with the evaluation of latent constructs through confirmatory factor analysis and related reliability and validity measures (Pradhan, 2017; Pradhan et al., 2023). Cronbach's alpha values ranged from 0.70 to 0.89 across Tangibility, Reliability, Responsiveness, Assurance, Empathy, and Satisfaction, with all values meeting the recommended threshold of 0.70, indicating acceptable internal consistency. Similarly, all Composite Reliability (CR) values exceeded 0.70, further supporting the reliability of the constructs. The Average Variance Extracted (AVE) values were above 0.50 for all constructs, providing evidence of satisfactory convergent validity and indicating that the constructs adequately explained the variance in their respective indicators (Table 6).

The discriminant validity analysis showed that the diagonal values, representing the square roots of AVE,

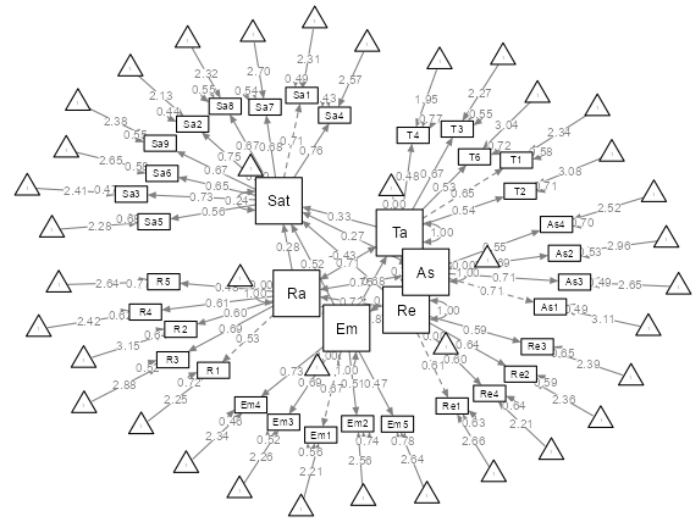


Figure 2. Structural equation model

ranged from 0.714 to 0.762 across the six constructs. These values were greater than the corresponding inter-construct correlations, providing evidence that the constructs were sufficiently distinct from one another. The correlations among the constructs were positive and statistically significant at the 0.01 level, ranging from 0.532 to 0.673, with the strongest association observed between Empathy and Satisfaction ($r = 0.673$). The results support adequate discriminant validity, indicating that Tangibility, Reliability, Responsiveness, Assurance, Empathy, and Satisfaction represent related but distinct constructs in the measurement model (Table 7).

The SEM results showed that all observed items had positive factor loadings on their respective latent constructs, indicating that the items adequately represented the underlying domains. For Tangibility, the loadings ranged from 0.461 to 0.674; for Reliability, from 0.483 to 0.692; and for Responsiveness, from 0.589 to 0.637. The Assurance items showed loadings ranging from 0.546 to 0.713, while Empathy loadings ranged from 0.468 to 0.734. The Satisfaction items demonstrated loadings between 0.562 and 0.758, with Sa4 showing the highest loading (0.758) (Figure 2).

DISCUSSION

The participants were predominantly young students, with a mean age of 22.12 ± 4.34 years, and represented different educational streams and levels. The present study found that 66.3% of students were satisfied with various aspects of the academic environment (95% CI: 61.41%–71.00%), with the median score being 26 (out of 45) for satisfaction. In comparison, Singh and Dhakal (2025) reported an overall median score of 147.0 out of 200, with 89.9% of students reporting high satisfaction. This finding was higher than the level reported by Shrestha et al. (2019), who reported an overall score of 131.25. Similarly, Mirghani and Elnour (2017) found that 74.5% of students viewed the academic environment positively. However, the present finding was substantially higher than that reported by Kasemy et al. (2022), who

Table 3. Model fitting criteria in structural equation modeling

Indicator	Calculated value	Recommended value
SRMR	0.046	Below 0.08 or 0.10
RMSEA	0.047	< 0.05
Chi-Square/df	3.23	Between 2 and 5
CFI	0.963	> 0.90
TLI	0.950	> 0.90
NFI	0.962	> 0.90
GFI	0.943	> 0.90
AGFI	0.931	> 0.90

Table 4. Structural model equation

Independent variables	β	95% Confidence Interval		Estimate	SE	z value	p-value
		Lower	Upper				
Tangibility	0.324	0.1017	0.5460	0.387	0.141	2.75	0.006
Reliability	0.295	0.0966	0.4941	0.420	0.152	2.76	0.006
Responsiveness	0.446	0.3350	0.5060	0.589	0.336	1.75	0.040
Assurance	0.249	-0.0994	0.5970	0.304	0.219	2.39	0.004
Empathy	0.531	0.1904	0.8708	0.618	0.211	2.93	0.003

Table 5. Adjusted R-squared (R^2)

Dependent variable	Independent variables	R^2
Satisfaction	Tangibility, Reliability, Responsiveness, Assurance, Empathy	0.769

found an overall score indicating a low level of satisfaction. In contrast, [Abusaad et al. \(2015\)](#) reported a neutral level of satisfaction. These differences may reflect variations in institutional settings, student characteristics, and educational environments.

The findings showed that age, stream of education, and level of education were significantly associated with students' satisfaction, suggesting that students' experiences and expectations may vary according to their demographic and academic characteristics. In the present study, quality of teaching was the highest-rated aspect of the academic environment ([Singh and Dhakal, 2025](#)), whereas ? identified learning facilities and resources as the highest-rated domain. Furthermore, students' satisfaction was significantly associated with academic year ($p = 0.038$), which is consistent with ?, who also reported a significant association ($p = 0.02$).

A key finding of the study was the significant positive relationship between perceived service quality and students' satisfaction. All five dimensions of service quality, namely tangibility, reliability, responsiveness, assurance, and empathy, were positively associated with students' satisfaction. Among these dimensions, empathy showed the strongest association, followed by responsiveness, tangibility, reliability, and assurance. This finding suggests that students' satisfaction is not determined solely by physical facilities or academic resources but is also closely related to how effectively and respectfully institutional services respond to their needs. In particular, attention to students' individual concerns and timely re-

sponses may play an important role in creating a more satisfying academic experience.

The structural model further indicated that perceived service quality explained 76.9% of the variance in students' satisfaction, demonstrating a substantial relationship between the quality of institutional services and students' overall satisfaction. This finding emphasizes the importance of considering multiple dimensions of service quality when evaluating the student experience. Improvements in institutional facilities, consistency of services, responsiveness of staff, professional support, and individualized attention may therefore be important areas for institutions seeking to enhance student satisfaction.

The significant associations observed for age, educational stream, and level of education also suggest that a one-size-fits-all approach to student services may not adequately address the needs of all students. Different groups of students may have different academic expectations, support requirements, and experiences within the institution. Institutional authorities should therefore consider differences across student groups when planning academic and support services.

CONCLUSIONS

The study concludes that more than half of the students were satisfied with the academic environment, although satisfaction varied across different student characteristics. Age, stream of education, and level of education were associated with students' satisfaction. Perceived service quality was also positively associated with satisfaction

Table 6. Internal and convergent reliabilities

Latent	Observed	Cronbach's α	Loading	z	p -value	Error	AVE	CR
Tangibility	T1	0.70	0.647	0	–	0.0871	0.53	0.71
	T2	–	0.536	8.76	< 0.001	0.0871	–	–
	T3	–	0.674	10.47	< 0.001	0.0901	–	–
	T4	–	0.481	7.99	< 0.001	0.0962	–	–
	T5	–	0.530	8.67	< 0.001	0.0794	–	–
Reliability	R1	0.73	0.526	0	–	0.1227	0.51	0.72
	R2	–	0.604	8.39	< 0.001	0.1227	–	–
	R3	–	0.692	9.04	< 0.001	0.1331	–	–
	R4	–	0.613	8.46	< 0.001	0.1204	–	–
	R5	–	0.483	7.26	< 0.001	0.1216	–	–
Responsiveness	Re1	0.71	0.609	0	–	0.0989	0.57	0.74
	Re2	–	0.637	10.05	< 0.001	0.0989	–	–
	Re3	–	0.589	9.47	< 0.001	0.1013	–	–
	Re4	–	0.600	9.61	< 0.001	0.1074	–	–
Assurance	As1	0.75	0.712	0	–	0.0804	0.54	0.76
	As2	–	0.686	12.37	< 0.001	0.0804	–	–
	As3	–	0.713	12.83	< 0.001	0.0845	–	–
	As4	–	0.546	9.94	< 0.001	0.0849	–	–
Empathy	Em1	0.74	0.665	0	–	0.0881	0.58	0.75
	Em2	–	0.509	8.98	< 0.001	0.0881	–	–
	Em3	–	0.690	11.75	< 0.001	0.0867	–	–
	Em4	–	0.734	12.35	< 0.001	0.0876	–	–
	Em5	–	0.468	8.32	< 0.001	0.0902	–	–
Satisfaction	Sa1	0.89	0.711	0	–	0.0746	0.53	0.89
	Sa2	–	0.747	14.04	< 0.001	0.0746	–	–
	Sa3	–	0.728	13.70	< 0.001	0.0725	–	–
	Sa4	–	0.758	14.25	< 0.001	0.0681	–	–
	Sa5	–	0.562	10.60	< 0.001	0.0719	–	–
	Sa6	–	0.652	12.28	< 0.001	0.0698	–	–
	Sa7	–	0.678	12.76	< 0.001	0.0672	–	–
	Sa8	–	0.670	12.61	< 0.001	0.0680	–	–
	Sa9	–	0.670	12.61	< 0.001	0.0661	–	–

Table 7. Discriminant validity using the Fornell–Larcker criterion

Domains	Tan.	Rel.	Resp.	Assu.	Emp.	Satis.
Tangibility	0.728	0.580**	0.538**	0.532**	0.532**	0.594**
Reliability	0.580**	0.714	0.584**	0.569**	0.647**	0.647**
Responsiveness	0.538**	0.584**	0.755	0.560**	0.636**	0.659**
Assurance	0.532**	0.569**	0.560**	0.735	0.644**	0.670**
Empathy	0.532**	0.647**	0.636**	0.644**	0.762	0.673**
Satisfaction	0.594**	0.647**	0.659**	0.670**	0.673**	0.728

Note: Diagonal elements in bold represent the square roots of the Average Variance Extracted (AVE). Off-diagonal elements represent Pearson correlations between the constructs. ** Significant at the 0.01 level (2-tailed).

across all five dimensions, namely tangibility, reliability, responsiveness, assurance, and empathy. The findings indicate that students' satisfaction depends not only on physical facilities and academic resources but also on the reliability and responsiveness of institutional services, the competence and professionalism of service providers, and the extent to which students' individual needs are understood and addressed. Therefore, higher education institutions should regularly assess students' perceptions of service quality and prioritize improvements in these areas to create a supportive, responsive, and student-centered academic environment.

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Modeling and Forecasting Nepal's Nominal GDP with An Updated Box Jenkins ARIMA Analysis

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ABSTRACT

Background: Gross domestic product (GDP) is one of the principal indicators used to summarize the scale and evolution of economic activity and provides an important information base for macroeconomic planning, fiscal management, investment decisions, and the allocation of public resources. Reliable GDP forecasts are particularly valuable for economies experiencing structural transformation and recurrent external shocks. This study focuses the ARIMA analysis of the annual nominal GDP from 1990/91 to 2024/25. The study aims to model and forecast Nepal's annual nominal GDP using the univariate Box–Jenkins time-series methodology.

Method: The study uses 35 annual observations of Nepal's nominal GDP at current prices, expressed in million Nepalese rupees. The univariate Box Jenkins approach was applied in R to identify and estimate appropriate autoregressive integrated moving average (ARIMA) models. The GDP series was logarithmically transformed to reduce scale effects and stabilize variation. Stationarity was examined using graphical analysis, autocorrelation and partial autocorrelation correlograms, the Augmented Dickey–Fuller (ADF) test, and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test. Six pre-specified ARIMA models were estimated using maximum likelihood and compared using Akaike information criterion (AIC), corrected Akaike information criterion (AICc), Bayesian information criterion (BIC), log-likelihood, root mean square error (RMSE), mean absolute deviation (MAD), and residual Ljung–Box diagnostics. An automated model-selection procedure was also used as a robustness check.

Results: The GDP series exhibited a strong non-stationary trend in its original form. After logarithmic transformation and first differencing, the KPSS test supported stationarity, although the ADF test continued to provide mixed evidence regarding the presence of a unit root. Among the six pre specified candidate models, ARIMA(1, 1, 2) produced the lowest AIC (-98.7618) and AICc (-97.3825). The residual Ljung Box test was statistically insignificant at the 5% level, indicating no strong evidence of remaining autocorrelation in the residuals. However, the autoregressive coefficient was estimated at the boundary of the parameter space, resulting in an unavailable standard error and suggesting a potential near unit root estimation issue. The automated model selection procedure identified the more parsimonious ARIMA(0, 1, 0) model with drift and reported lower information criteria than the restricted candidate model. Forecasts based on ARIMA(1, 1, 2) indicate a continued increase in Nepal's nominal GDP from NPR 6,353,083 million in 2025/26 to NPR 7,907,673 million in 2029/30, although the prediction intervals widen substantially as the forecast horizon increases.

Conclusion: The findings demonstrate both the usefulness and limitations of univariate ARIMA modeling for forecasting Nepal's aggregate GDP. Within the restricted set of pre-specified models, ARIMA(1, 1, 2) was identified as the preferred model according to AIC and AICc, while the automated analysis suggested that the simpler ARIMA(0, 1, 0) model with drift provides an important parsimonious benchmark. The results indicate that Nepal's nominal GDP is expected to continue increasing over the forecast period, although forecast uncertainty increases over time. The study highlights the importance of comparing alternative specifications and suggests that future research should evaluate multivariate, hybrid, and other forecasting approaches alongside univariate ARIMA models to improve the robustness and accuracy of GDP forecasts for Nepal.

Key words: ARIMA; Box–Jenkins methodology; GDP forecasting; Nepal; Time series analysis; Nominal GDP.

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INTRODUCTION

Gross domestic product (GDP) is a central summary measure of economic activity. In nominal terms, GDP represents the monetary value of final goods and services produced within an economy at current prices. Changes in nominal GDP can reflect changes in the volume of production, changes in prices, or both. Consequently, nominal GDP and real GDP answer different analytical questions. Real GDP is generally more appropriate when the objective is to study changes in the volume of output, whereas nominal GDP remains highly relevant for fis-

cal aggregates, current-price national accounts, debt and revenue ratios, public expenditure planning, and other decisions formulated in monetary terms.

GDP can be compiled through the production, expenditure, and income approaches. Under the production approach, value added is aggregated across institutional sectors and industries, with the appropriate treatment of taxes and subsidies on products. Under the expenditure approach, GDP is associated with final consumption, gross capital formation, and net exports. Under the income approach, it corresponds to the income gener-

ated through production, including compensation of employees, taxes less subsidies on production and imports, and operating surplus or mixed income. Although these approaches are conceptually related, official national accounts are compiled within an accounting framework designed to provide internally consistent macroeconomic aggregates.

For Nepal, reliable information about the expected trajectory of GDP is important because public sector planning, infrastructure expenditure, fiscal policy, development programmes, and private investment decisions all depend partly on expectations concerning the future size of the economy. Forecasts cannot eliminate uncertainty, particularly when an economy is exposed to policy changes, natural disasters, international price movements, migration and remittance dynamics, or other structural shocks. Nevertheless, transparent statistical forecasts provide a useful baseline against which alternative scenarios and policy assumptions can be assessed.

The Box–Jenkins framework remains a useful starting point for such an exercise because it provides a systematic sequence for identifying non-stationarity, transforming the series where appropriate, examining autocorrelation structures, estimating competing ARIMA specifications, diagnosing residuals, and generating forecasts (Box and Jenkins, 1976; Box et al., 2015; Hyndman et al., 2025). ARIMA models are particularly attractive when the primary objective is to construct a transparent univariate benchmark using the historical dynamics of the series itself.

Recent forecasting research also shows why a classical ARIMA benchmark remains relevant even when more complex methods are available. Nepal-specific work by Chaudhary and Uprety (2023) compares ARIMA, artificial neural networks, and hybrid ARIMA–ANN specifications for GDP forecasting and reports improved predictive performance for ANN and hybrid approaches. Similarly, recent international work continues to compare conventional ARIMA specifications with machine-learning and hybrid methods rather than assuming that a single class of models is universally superior (Atif, 2025; Sinu et al., 2024). These developments again suggest that a carefully diagnosed ARIMA model should be interpreted as an important baseline for macroeconomic forecasting.

The objectives of this study are therefore fourfold:

1. to describe the historical evolution of Nepal's annual nominal GDP from 1990/91 to 2024/25;
2. to investigate the time-series properties of the GDP series using graphical, ADF, and KPSS evidence;
3. to estimate and compare a set of candidate ARIMA models using likelihood-based information criteria, in-sample error measures, and residual diagnostics; and
4. to generate five-year forecasts for 2025/26–2029/30 and interpret the associated forecast uncertainty.

A further contribution of the updated study is methodological transparency. The candidate-model comparison

and the automated `auto.arima()` result do not point to exactly the same preferred specification. Instead of suppressing this difference, the study reports both results and discusses the implications for parsimony and model uncertainty.

REVIEW OF RELATED LITERATURE

ARIMA and the Box–Jenkins tradition

The autoregressive integrated moving-average model is one of the most widely used approaches to univariate time-series modeling. The classical Box–Jenkins framework emphasizes an iterative cycle of identification, estimation, diagnostic checking, and forecasting (Box and Jenkins, 1976; Box et al., 2015). The autoregressive component represents dependence on previous values of the series, the integrated component represents differencing used to address non-stationarity, and the moving-average component represents dependence on previous innovations.

A major advantage of the framework is that model choice is not based solely on one numerical criterion. A useful model should balance goodness of fit with parsimony and should leave residuals that are approximately consistent with a white-noise process. Information criteria such as AIC and BIC provide systematic penalties for unnecessary complexity, while residual diagnostics help determine whether meaningful serial dependence remains after estimation.

More recent Nepal-focused research has expanded the methodological comparison. Upadhyay and Pradhan (2023) applied the autoregressive integrated moving average (ARIMA) methodology to forecast Nepal's GDP, providing an earlier Nepal-specific benchmark for univariate GDP forecasting. Chand (2024) applied ARIMA modeling to Nepal's per-capita GDP using annual data from 1960 to 2022 and identified an ARIMA(2,2,1) specification for long-horizon prediction. The study demonstrates that even closely related macroeconomic indicators may require different orders of integration and different dynamic structures.

A particularly relevant recent contribution is the Nepal Rastra Bank study by Chaudhary and Uprety (2023). The authors examine GDP forecasting using standalone ARIMA and ANN models as well as additive and multiplicative hybrid ARIMA–ANN models. Their findings emphasize that Nepal's GDP dynamics may contain both linear and nonlinear components and that hybrid or neural-network approaches can improve predictive accuracy in some settings. This result does not make ARIMA obsolete; instead, it reinforces the role of ARIMA as a transparent benchmark against which more flexible approaches can be compared.

Recent empirical work has also combined ARIMA forecasting with econometric approaches in examining Nepal's macroeconomic conditions. Pradhan et al. (2025) investigated the relationship between foreign direct investment and economic growth in Nepal using ARDL and ARIMA forecasting models. The study illustrates the value of combining explanatory econometric models with time-series forecasting approaches when

analyzing Nepal's economic dynamics.

ARIMA-based forecasting has also been applied to other important macroeconomic and economic indicators in Nepal. Pradhan and Koirala [Pradhan and Koirala \(2024\)](#) analyzed and forecast international tourist arrivals in Nepal, demonstrating the broader applicability of time-series forecasting techniques to Nepalese economic planning. Such applications highlight the continued relevance of statistical forecasting methods for economic indicators beyond GDP. The application of statistical forecasting in Nepal extends beyond macroeconomic aggregates to tourism and agricultural-related indicators. Pradhan and Koirala [Pradhan and Koirala \(2024\)](#) examined trends and forecast international tourist arrivals in Nepal, while Poudel et al. [Poudel et al. \(2025\)](#) applied statistical forecasting methods to cereal grain requirements for livestock and fisheries feed. These studies demonstrate the wider use of time-series forecasting for planning and resource allocation in Nepalese contexts.

GDP forecasting studies across countries: China, India, Greece, Sudan, Bangladesh, Kenya, and Pakistan also report considerable variation in preferred ARIMA specifications. ([Dritsaki, 2015](#); [Hassan, 2020](#); [Maity and Chatterjee, 2012](#); [Miah et al., 2019](#); [Ning et al., 2010](#); [Wabomba et al., 2016](#); [Zakai, 2014](#)). The diversity of reported models supports the principle that ARIMA order should be determined empirically rather than selected by analogy alone.

A recent multi country study further illustrates the continuing use of ARIMA as a baseline in comparisons involving regime switching and deep learning approaches ([Atif, 2025](#)). The broader methodological lesson is that additional complexity should be justified by improved predictive performance and robust diagnostics rather than assumed to be superior a priori.

Taken together, the literature provides three motivations for the present analysis. First, ARIMA remains a useful and interpretable benchmark for macroeconomic forecasting. Second, updated data can materially change the preferred model. Third, comparison with parsimonious alternatives and automated selection procedures is important, especially when a more complex model produces boundary estimates or other numerical warnings.

DATA AND METHODOLOGY

Data source and study period

The study uses annual nominal GDP data for Nepal covering fiscal years 1990/91 to 2024/25, yielding 35 observations. GDP is measured at current prices and expressed in million Nepalese rupees. The updated dataset used in the empirical analysis is stored in the study's online data repository and was read directly into R for cleaning and time-series construction. The analysis should be interpreted with the understanding that official national-account estimates may be revised as statistical methodologies and source data are updated. The study therefore documents the exact sample used for estimation rather than combining multiple data vintages.

For time-series modeling, the fiscal-year label was converted to the first calendar year of the fiscal period. Thus,

1990/91 was indexed as 1990, 1991/92 as 1991, and so forth. The annual frequency is one observation per year:

$$GDP_t, \quad t = 1990, 1991, \dots, 2024. \quad (1)$$

Because nominal GDP rises strongly over the sample and its scale changes substantially over time, the principal ARIMA analysis uses the natural logarithm:

$$y_t = \ln(GDP_t). \quad (2)$$

The logarithmic transformation does not remove every possible source of non-stationarity, but it places the series on a scale in which changes can be interpreted approximately as proportional changes and can reduce the influence of level dependent variance.

The ARIMA model

Let y_t denote a time series. An ARIMA(p, d, q) model can be written as

$$\phi(B)(1 - B)^d y_t = c + \theta(B)\varepsilon_t, \quad (3)$$

where B is the backshift operator, $(1 - B)^d$ denotes differencing of order d , $\phi(B)$ is the autoregressive polynomial, $\theta(B)$ is the moving-average polynomial, and ε_t is an innovation term.

For example, after one difference, the ARIMA(1, 1, 2) model considered in this study can be represented conceptually as

$$(1 - \phi_1 B)(1 - B)y_t = (1 + \theta_1 B + \theta_2 B^2)\varepsilon_t. \quad (4)$$

The exact sign convention for MA parameters depends on the software parameterization, so the fitted coefficients are interpreted according to the implementation used in R.

The Box-Jenkins procedure applied in this study contains four broad stages:

1. **Identification:** inspect the series and assess stationarity using plots, ACF/PACF patterns, and formal unit-root or stationarity tests;
2. **Estimation:** estimate plausible ARIMA specifications by maximum likelihood;
3. **Diagnostic checking:** examine residual behavior using plots, ACF/PACF, normality testing, and Ljung-Box statistics;
4. **Forecasting:** generate out-of-sample forecasts and prediction intervals from the selected specification.

Stationarity tests

Two complementary tests were used. The ADF test has the null hypothesis that the series contains a unit root. Failure to reject the null is therefore interpreted as evidence consistent with non-stationarity. The KPSS test reverses the hypotheses: its null is stationarity around the specified deterministic component, and rejection suggests non-stationarity ([Dickey and Fuller, 1979](#); [Kwiatkowski et al., 1992](#)).

Using both tests is useful because the tests are based on opposite null hypotheses. Nevertheless, their results

do not always agree perfectly in small samples or in the presence of structural changes. Accordingly, the present study interprets the ADF and KPSS evidence jointly rather than treating either test as mechanically decisive.

Candidate model estimation and comparison

The following six ARIMA models were estimated on the log-transformed series:

(1, 1, 1), (1, 1, 2), (2, 1, 1), (2, 1, 2), (6, 1, 1), (6, 1, 2).

Models were estimated using maximum likelihood. Their performance was compared using:

- Akaike information criterion (AIC);
- corrected Akaike information criterion (AICc);
- Bayesian information criterion (BIC);
- log-likelihood;
- root mean squared error (RMSE);
- mean absolute deviation (MAD); and
- Ljung–Box residual test statistics and p -values.

The information criteria balance fit against model complexity, while RMSE and MAD summarize in-sample residual magnitude. The Ljung–Box test evaluates whether residual autocorrelation remains after accounting for the fitted model. A model is not selected solely because it minimizes one error measure; model adequacy also requires reasonable residual behavior and parameter estimation.

Automated model-selection robustness check

As a robustness exercise, the study also applies `auto.arima()` with exhaustive search settings. This procedure searches over a broader set of admissible specifications according to an information criterion. The automated result is not identical to the best model in the restricted candidate set. Reporting this difference is important because it reveals whether the manually specified candidates omitted a simpler or statistically preferred alternative.

Forecasting and uncertainty

Five annual forecasts are generated for 2025/26–2029/30. The study reports point forecasts together with 80% and 90% prediction intervals. Prediction intervals are particularly important because the uncertainty surrounding a time-series forecast generally increases as the horizon extends. The forecasts should therefore be interpreted as conditional statistical projections based on the historical dynamics and model assumptions, not as guaranteed future outcomes.

RESULTS AND DISCUSSION

Historical pattern of Nepal's nominal GDP

The annual GDP series displays a pronounced long-run upward movement over the study period. This pattern is expected for nominal GDP because both real economic

expansion and changes in the general price level contribute to increases in current-price output. The visual trend nevertheless indicates that direct modeling of the untransformed level series with a stationary ARMA process would be inappropriate without further treatment.

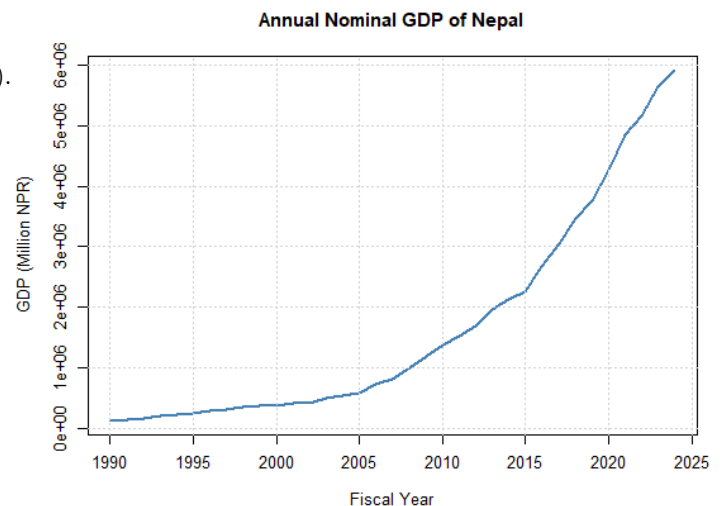


Figure 1. Annual nominal GDP of Nepal, 1990/91–2024/25.

The figure should be discussed in relation to both scale and trend. A steadily rising nominal series does not by itself imply a stable growth process because the increments can vary substantially over time. For this reason, the analysis proceeds with logarithmic transformation and differencing rather than assuming that the visible upward path represents a deterministic trend that can simply be extrapolated.

Stationarity analysis

Table 1 presents the formal stationarity results obtained from the R analysis.

The evidence for the GDP level is consistent across the two tests. The ADF statistic has a very high reported p -value, so the unit-root null is not rejected. Conversely, the KPSS test rejects the null of level stationarity. These findings support the conclusion that the level series is non-stationary.

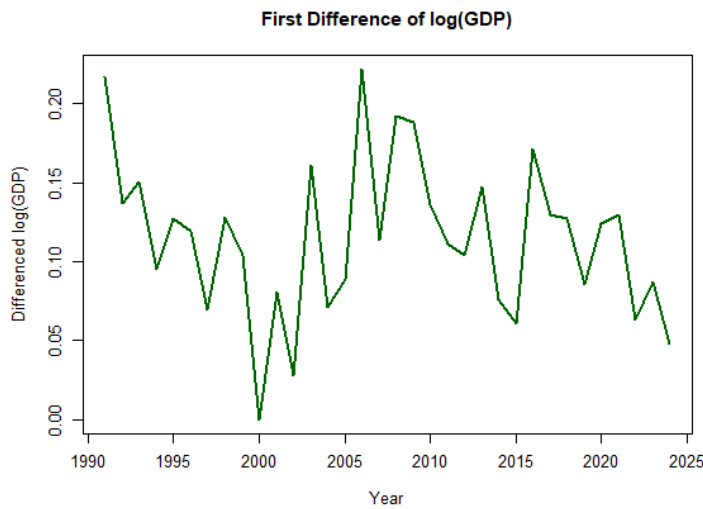
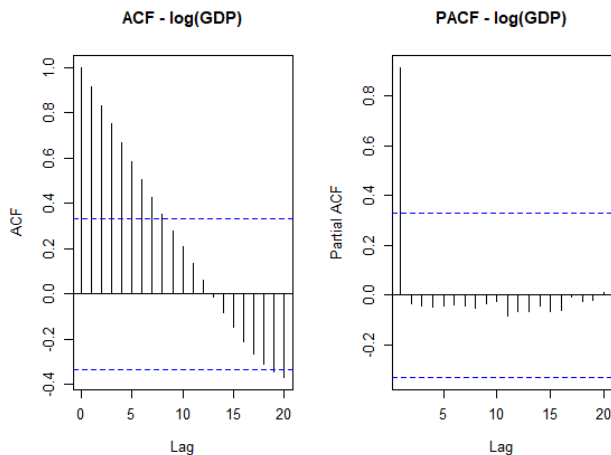
The log-transformed series also remains non-stationary according to the reported tests. The ADF test does not reject a unit root, and the KPSS test rejects stationarity. Consequently, the logarithm alone is insufficient to produce a stationary process.

After first differencing the log GDP series, the evidence becomes mixed. The KPSS test no longer rejects stationarity at the 5% level, whereas the ADF test still fails to reject its unit-root null. This divergence should not be concealed. It means that the formal tests do not provide unanimous confirmation of stationarity after one difference. In the present study, first differencing is nevertheless retained for the candidate ARIMA models because the transformed series is substantially more stable visually and the model-selection and residual diagnostics are conducted on the resulting ARIMA($p, 1, q$) specifications.

Table 1. Stationarity tests for Nepal's nominal GDP series

Test	Statistic	<i>p</i> -value	Interpretation at 5%
ADF (GDP level)	-0.1236	0.9900	Fails to reject unit root
KPSS (GDP level)	0.8527	0.0100	Rejects stationarity
ADF (log GDP)	-2.8638	0.2377	Fails to reject unit root
KPSS (log GDP)	0.9792	0.0100	Rejects stationarity
ADF (first difference of log GDP)	-1.7889	0.6548	Fails to reject unit root
KPSS (first difference of log GDP)	0.1105	0.1000	Does not reject stationarity

This choice should be interpreted as an empirical modeling decision rather than as a claim that every stationarity test gives the same conclusion.

**Figure 2.** First-differenced log GDP**Figure 3.** log GDP ACF-PACF diagnostics.

The ACF and PACF plots are used as identification aids rather than as the sole determinants of model order. With only 35 annual observations, over parameterization can quickly consume degrees of freedom. Candidate models are therefore evaluated jointly using information criteria, residual diagnostics, and parsimony.

Comparison of candidate ARIMA models

Table 2 summarizes the results for the six pre-specified ARIMA candidates.

The model comparison shows that no single candidate dominates every statistic. ARIMA(1,1,2) produces the lowest AIC and AICc among the six pre-specified models, while ARIMA(6,1,2) produces the smallest RMSE and MAD and the largest residual Ljung-Box *p*-value among the candidates shown. The latter model, however, is substantially more parameterized and has less favorable AICc and BIC values. This trade-off illustrates why model selection should not be based on in-sample error alone.

ARIMA(1,1,1), ARIMA(2,1,1), and ARIMA(2,1,2) all have Ljung-Box *p*-values below 0.05 in the comparison table, suggesting remaining residual autocorrelation under the particular test settings used. By contrast, ARIMA(1,1,2), ARIMA(6,1,1), and ARIMA(6,1,2) have *p*-values above 0.05. Among these diagnostically acceptable candidates, ARIMA(1,1,2) offers the strongest AIC and AICc performance and uses fewer parameters than the two high-order alternatives.

For these reasons, ARIMA(1,1,2) is retained as the preferred model *within the restricted candidate set*. It does not mean that ARIMA(1,1,2) is necessarily the globally optimal model among all possible ARIMA specifications.

Automated model-selection robustness check

The automated search produces the following result:

ARIMA(0, 1, 0) with drift,

with estimated drift 0.1146 and standard error 0.0085. The model reports

$AIC = -103.98$, $AICc = -103.59$, $BIC = -100.92$.

These information criteria are lower than those reported for ARIMA(1,1,2). The automated result therefore provides strong evidence in favor of a more parsimonious specification. In economic terms, ARIMA(0,1,0) with drift corresponds to a random-walk-type process in which the differenced log GDP series has a positive average increment.

This finding has two implications. First, the restricted candidate list did not include the automated procedure's preferred simple specification. Second, the empirical evidence does not justify presenting ARIMA(1,1,2) as the only defensible forecasting model. A rigorous interpretation is that ARIMA(1,1,2) performs best among the manually pre-selected candidates, whereas ARIMA(0,1,0)

Table 2. Comparison of pre-specified ARIMA models fitted to log GDP

Model	AIC	AICc	BIC	LogLik	RMSE	MAD	Q-stat	p-value
ARIMA(1,1,1)	-97.0311	-96.2311	-92.4520	51.5155	0.0507	0.0390	22.7112	0.0038
ARIMA(1,1,2)	-98.7618	-97.3825	-92.6563	53.3809	0.0484	0.0373	11.8279	0.1064
ARIMA(2,1,1)	-96.0920	-94.7127	-89.9866	52.0460	0.0498	0.0379	18.5984	0.0095
ARIMA(2,1,2)	-93.8271	-91.6843	-86.1953	51.9136	0.0501	0.0380	19.3147	0.0037
ARIMA(6,1,1)	-95.0360	-89.2760	-82.8251	55.5180	0.0442	0.0318	7.2785	0.0635
ARIMA(6,1,2)	-97.1465	-89.6465	-83.4093	57.5733	0.0411	0.0305	3.6588	0.1605

with drift is the preferred automated information criterion benchmark.

The coexistence of these results is itself informative. Macroeconomic time series can sometimes be forecast competitively by simple persistence plus drift models, especially when long-run trends dominate short-run cyclical structure. Future research should therefore compare forecasts from the two models using an explicit holdout or rolling origin evaluation rather than relying exclusively on in-sample criteria.

Parameter estimates of ARIMA(1,1,2)

The estimated parameters for the selected restricted-candidate model are shown in Table 3.

Table 3. Parameter estimates for ARIMA(1,1,2)

Parameter	Estimate	Std. Error	z-value	p-value
AR(1)	0.999999			
MA(1)	-0.798887	0.179635	-4.4473	8.696×10^{-6}
MA(2)	0.404909	0.247498	1.6360	0.1018

The MA(1) coefficient is strongly different from zero according to the reported asymptotic z-test, whereas the MA(2) coefficient is not statistically significant at the conventional 5% level. More importantly, the AR(1) estimate is essentially equal to one and its standard error is unavailable because the fitted parameter is at or extremely close to the boundary of the parameter space. The R output consequently reports NaN for the standard error and associated test statistics.

This numerical feature is a material limitation of the ARIMA(1,1,2) specification. A boundary estimate can indicate weak identification or near-cancellation between the integrated and autoregressive components. The result does not automatically invalidate the model's fitted values or residual diagnostics, but it weakens the case for treating the estimated coefficient structure as stable and precisely measured. This issue provides an additional reason to report the ARIMA(0,1,0) with drift benchmark prominently.

The estimated innovation variance for ARIMA(1,1,2) is 0.002641, and the log-likelihood is 53.38. The in-sample training RMSE is 0.04837 on the log scale, while the MAE is 0.03735. These values summarize fit within the estimation sample,

Residual diagnostics

The residual analysis provides a second assessment of model adequacy. The Shapiro–Wilk test reports

$$W = 0.95294, \quad p = 0.1395.$$

Thus, at the 5% significance level, the normality null is not rejected. This result does not prove that the residual distribution is exactly normal; rather, it indicates that the available sample does not provide strong evidence against normality.

The `checkresiduals()` Ljung–Box diagnostic reports

$$Q^* = 6.7952, \quad df = 4, \quad p = 0.1471,$$

using a total of seven lags and model degrees of freedom equal to three. Since the p -value exceeds 0.05, there is no statistically significant evidence of residual autocorrelation under this diagnostic configuration.

The residual plot should be interpreted jointly with the formal tests. Visually, the residuals fluctuate around zero without a persistent trend, while the ACF does not show an obvious sequence of dominant autocorrelations. The histogram and Q–Q diagnostic can be used to assess whether the Shapiro–Wilk result is consistent with the observed distribution. Taken together, the diagnostics support the conclusion that the fitted ARIMA(1,1,2) model captures a substantial part of the serial dependence in the differenced log GDP series. Nevertheless, the boundary AR estimate remains a separate parameter-estimation concern.

Five-year GDP forecasts

Table 4 reports forecasts generated from the ARIMA(1,1,2) model and transformed back to the original GDP scale.

The point forecasts imply continued growth in nominal GDP throughout the five-year projection period. The forecast rises from approximately NPR 6.353 trillion in 2025/26 to approximately NPR 7.908 trillion in 2029/30. Because the model is fitted to nominal GDP, these increases can not be interpreted directly as forecasts of real economic growth. Future inflation and price-level movements are embedded in the projected current price aggregate.

The uncertainty surrounding the projections increases rapidly with the forecast horizon. For 2025/26, the 90% prediction interval ranges from NPR 5.838 trillion to NPR 6.913 trillion. By 2029/30, the 90% interval expands from NPR 5.386 trillion to NPR 11.611 trillion. The wide interval at longer horizons demonstrates why

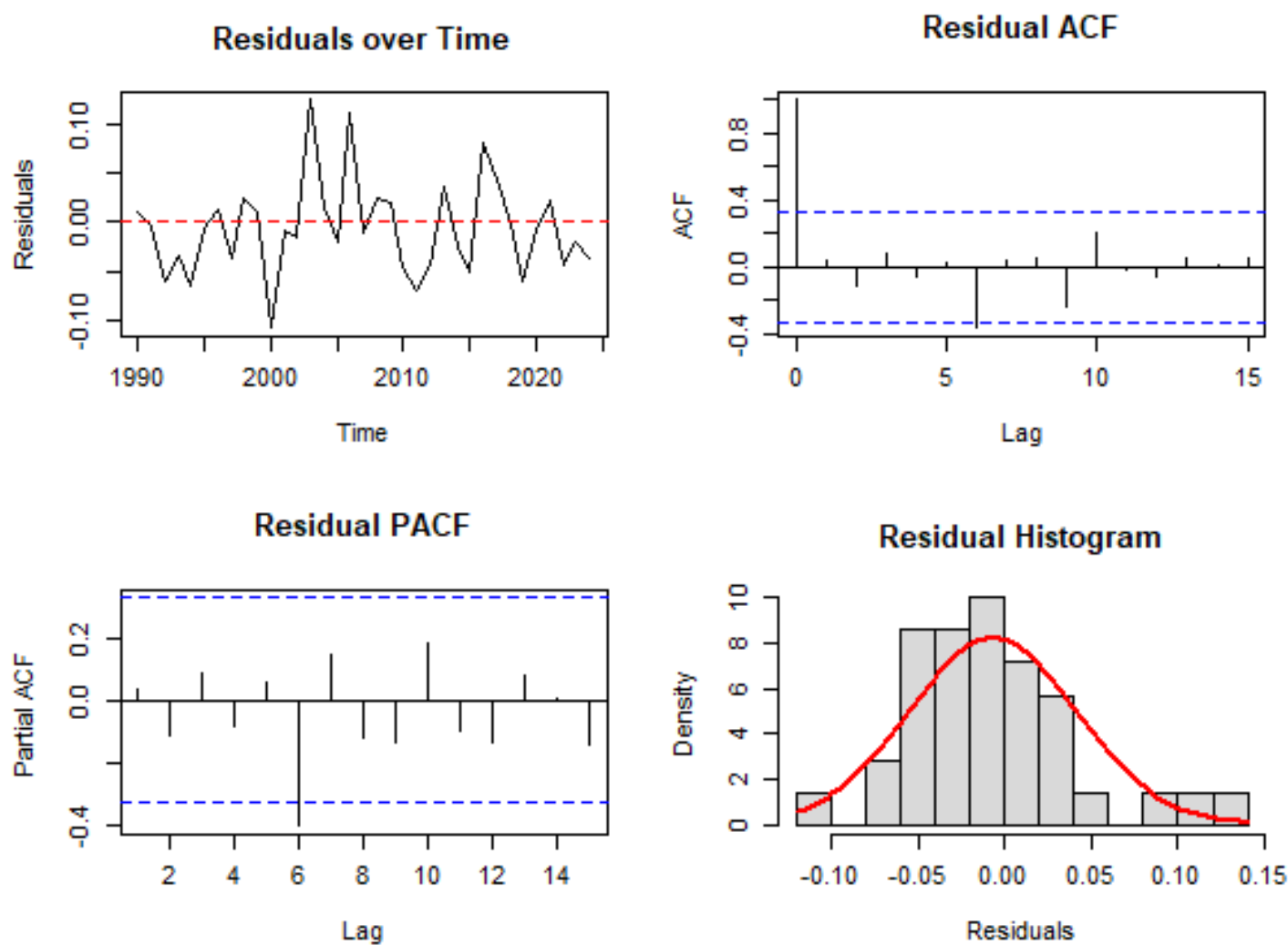


Figure 4. Residual diagnostics for the ARIMA(1,1,2) model.

Table 4. Forecasts of Nepal's nominal GDP, 2025/26–2029/30 (million NPR)

Fiscal year	Forecast	Lower 80%	Upper 80%	Lower 90%	Upper 90%
2025/26	6,353,083	5,948,129	6,785,608	5,838,099	6,913,495
2026/27	6,710,434	6,054,036	7,438,001	5,879,923	7,658,252
2027/28	7,087,885	6,055,846	8,295,805	5,791,631	8,674,261
2028/29	7,486,567	5,986,150	9,363,059	5,618,381	9,975,949
2029/30	7,907,673	5,862,380	10,666,536	5,385,524	11,610,994

point forecasts alone can be misleading. The central projection suggests continued expansion, but the range of statistically plausible outcomes under the model assumptions becomes increasingly broad.

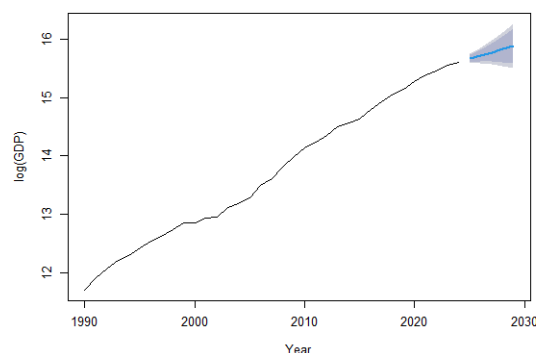


Figure 5. Five-year forecast of Nepal's nominal GDP from ARIMA(1,1,2).

The automated search favors ARIMA(0,1,0) with drift, with substantially lower AIC, AICc, and BIC than the manually selected ARIMA(1,1,2). Moreover, the ARIMA(1,1,2) model has an AR coefficient effectively equal to one with an unavailable standard error. Therefore, this study does not claim that the more complicated model is unambiguously superior.

The findings are broadly consistent with recent forecasting literature in one important respect: simple linear models remain useful benchmarks, but alternative approaches can sometimes improve accuracy. The Nepal Rastra Bank hybrid ARIMA ANN study is particularly relevant because it suggests that Nepal's GDP may contain nonlinear components not fully captured by a purely linear ARIMA structure (Chaudhary and Uprety, 2023). Thus, the present model should be viewed as a transparent baseline and a contribution to reproducible national macroeconomic forecasting rather than as a universal forecasting solution.

A further methodological issue concerns validation. The current results emphasize in-sample information criteria and residual diagnostics. These are necessary for model development but do not replace genuine out-of-sample forecast evaluation. A future extension should reserve one or more recent observations for testing, or use rolling-origin cross-validation, and compare ARIMA(1,1,2), ARIMA(0,1,0) with drift, and alternative methods using common forecast accuracy measures such as RMSE, MAE, MASE, and scale-appropriate percentage errors.

The policy implication is not that the forecast path should be treated as a precise prediction of Nepal's future economy. Instead, the projections provide a quantitative baseline for planning. Ministries, central-bank analysts, development agencies, and researchers can use such a baseline when constructing alternative fiscal, revenue, expenditure, or macroeconomic scenarios.

The results also support the use of parsimonious benchmark models in forecasting exercises. Since the automated ARIMA(0,1,0) with drift model outperforms the restricted candidate models on the reported information criteria, future work should compare this simple benchmark explicitly with higher-order ARIMA models. More complex models should be retained only when they demonstrate clear improvements in out-of-sample accuracy or other relevant forecasting objectives.

Finally, GDP is influenced by factors that are not contained in a univariate historical series. Remittances, tourism, agricultural conditions, public investment, credit conditions, external demand, inflation, exchange rates, and extraordinary shocks may all affect future macroeconomic outcomes. Extensions using ARIMAX, VAR/VECM frameworks, dynamic factor models, machine learning, or hybrid methods may therefore complement the univariate approach.

CONCLUSION

The GDP level and log GDP series show strong evidence of non-stationarity. After first differencing the log series, the KPSS result is consistent with stationarity, whereas

the ADF test continues to fail to reject a unit root. The evidence is therefore mixed and is reported transparently.

Among the six pre-specified ARIMA candidates, ARIMA(1,1,2) has the lowest AIC and AICc and passes the reported residual autocorrelation diagnostic at the 5% level. The residual normality test also does not reject normality. However, ARIMA(6,1,2) has smaller in-sample RMSE and MAD, illustrating that different selection criteria can favor different models. More importantly, the ARIMA(1,1,2) autoregressive coefficient is estimated at a boundary value and has an unavailable standard error.

The automated model-selection procedure identifies ARIMA(0,1,0) with drift and produces substantially lower AIC, AICc, and BIC values than the restricted-candidate winner. Consequently, the most defensible interpretation is not that ARIMA(1,1,2) is universally optimal. Rather, ARIMA(1,1,2) is the preferred specification within the manually selected candidate set, while ARIMA(0,1,0) with drift provides a more parsimonious and information-criterion-favored benchmark.

Forecasts from ARIMA(1,1,2) indicate a continued increase in nominal GDP from NPR 6.353 trillion in 2025/26 to NPR 7.908 trillion in 2029/30. The prediction intervals, however, widen markedly over time, emphasizing that the projections should be used as conditional planning baselines rather than precise future outcomes.

Future research should extend the analysis in three directions: first, through rolling or holdout out-of-sample forecast evaluation; second, through explicit comparison with the random-walk-with-drift benchmark; and third, through multivariate, machine-learning, and hybrid approaches that can incorporate nonlinearities and additional macroeconomic information. Such comparisons would provide stronger evidence about whether additional model complexity produces meaningful forecasting gains for Nepal's economy.

DATA AVAILABILITY

The dataset used for this study covers annual nominal GDP of Nepal from fiscal year 1990/91 to 2024/25. The cleaned study dataset is available from the author's online repository:

https://raw.githubusercontent.com/bijayprad/MPPhil/refs/heads/main/data/National%20Data/GDP1990_2025.csv

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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AI-Driven Productivity Impact Analysis Using GIS and IoT Data in Sugar Industries: A Pilot Study

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ABSTRACT

Background: Increasing demands in terms of efficient operations, cost reduction, and improving productivity have emerged in the sugar industry due to increased competition and sustainable developments. The current pilot study suggests a framework to evaluate the impact that integration of AI, GIS, and IoT would bring on productivity of sugar mills.

Method: This research adopts a mixed frameworks approach to investigate the impact of integrated AI, GIS and IoT technologies on sugar mill productivity. Based on the simulated data that represents more than 20 sugar mills operating in Maharashtra, India, our findings suggest that adoption metrics concerning these technologies positively correlate with performance metrics such as productivity index, cost reduction, downtime reduction, and improvement in sugar recovery.

Result: AI adoption correlated most strongly with productivity index ($r = 0.264$), followed by integration of IoT technologies ($r = 0.119$) and GIS-based monitoring ($r = 0.042$). Random Forest regression provided R^2 value of 0.082. Among the most significant predictors, predictive maintenance score (14.8%) and scores of AI adoption and road quality index (13.5% each) were detected. The average productivity index was at 85.82, while average reductions in cost, downtime, and sugar recovery amounted to 14.12%, 31.92%, and 4.86% correspondingly.

Conclusion: In general, it could be noted that the proposed framework is applicable to various locations where sugar is produced. This study provides a methodology that can be replicated to explore the relationship between technology and productivity in the agro- industrial sector, while emphasizing the importance of predictive maintenance and AI decision-making systems to achieve business outcomes.

Key words: artificial intelligence, geographic information systems, internet of things, sugar industry, productivity analysis, predictive maintenance, digital transformation, operational efficiency

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INTRODUCTION

The sugar industry is a fundamental agro-industrial industry facing continuous pressure for productivity improvement, while at the same time sustainability is becoming a critical issue throughout the entire value chain, from cultivation to production processes [Higgins et al. \(2004\)](#). The sector which already has a value of more than \$50 billion, per year, has suffered with several inefficient processes at production level such as logistics during harvesting and transportation, and resource over-consumption [Solomon \(2016\)](#). Traditional methods are insufficient to cope with such issues; therefore, relevant stakeholders have sought to embrace digital transformation and the usage of Artificial Intelligence (AI), Geographic Information System (GIS) and Internet of Things (IoT) seems to have high potential to enable productivity gains [Misra et al. \(2026\)](#). Nevertheless, studies conducted till to date, are limited to assessing such technologies individually and they have not evaluated the combination of all three technologies together.

Some recent studies report on the application of AI to predict equipment failure and estimate yield [Martin et al. \(2026\)](#), whereas others investigate the usage of GIS in relation to the study of sugarcane patterns [Mar-](#)

[tin et al. \(2026\)](#). IoT provides a system for monitoring equipment and surrounding environmental conditions [Misra et al. \(2020\)](#), and yet other studies address sustainability and technology implementation within agro-industries, but none focus on investigating a framework that combines AI, GIS, and IoT applications together within the manufacturing processes, for instance some literature address logistic and transportation improvement without integrating AI driven decisions [Higgins and Laredo \(2006\)](#) and other authors focus on the sustainability issue of sugar manufacturing without a clear indication of the implementation of the IoT to improve efficiency of production [Eggleston and Lima \(2015\)](#). In this paper, we hypothesize that an integration between AI, GIS and IoT may positively improve productivity factors of sugar manufacturing. To investigate this, we design a pilot study focusing on the analysis of simulated data of 20 sugar mills located in different area of Maharashtra. Our research methodology combines statistical and machine learning techniques, using Random Forest classification, to demonstrate how such factors influence the productivity of sugar factories. The present work aims to answer the following three questions: 1) Which are the individual and synergistic effects of using AI, GIS and

IoT on productivity? 2) Which factors are more influential for operational productivity? 3) Provide a framework that could be replicated on other agro-industrial scenarios. Our main contribution is threefold: Firstly, this paper presents empirical results regarding the synergistic effect of the adoption of AI, GIS and IoT on sugar manufacturing, bridging the gap between the proposed benefits and the actual implementation outcomes. Secondly, it defines a data driven approach, adaptable to other agro-industrial scenarios, paving the path toward digital transformation on a larger scale. 22 Thirdly, it provides actionable information and guidance for industry stakeholders, identifying predictive maintenance and AI as key factors to achieve optimal operation performance and indicating how these can be used in practice, noting for example how the simple inclusion of AI is correlated with 26.4 productivity improvements. The rest of the paper is organized as follows. Section 2 surveys the literature available on technology adoption in the sugar industry; Section 3 describes our methodology and dataset; Section 4 presents the findings of the experiment; Section 5 discusses the implications derived from these findings, and Section 6 concludes and recommends future work. Our aim is to provide both a theoretical contribution by systematically examining these components and a pragmatic, step-by-step path for implementation into agribusinesses that enables the discourse on digitalisation in agro-industries.

The application of digital technologies such as Artificial Intelligence (AI), Geographic Information System (GIS) and Internet of Things (IoT) on agriculture and agro-industry has been increasing over the past decade. In particular to sugarcane cultivation and sugar production, several studies have applied AI, GIS and IoT separately for different applications. For example, [Misra et al. \(2026\)](#) had conducted a study of AI application on water use and productivity prediction for sugar cane growth, while [Vinayaka and Prasad \(2024\)](#) presented the application of remote sensing and AI in yield prediction. Both these works have limited the study towards the agricultural domain only, and the industrial production aspect is neglected. GIS has widely been applied for spatial analysis of sugarcane farming for land use, land suitable and allocation of resources. [Kulwant and Patel \(2024\)](#) applied the integration of GIS with satellite imagery to better monitor sugarcane cultivation condition and soil property. However, for industrial application, the potential for utilizing GIS on logistics of transportation and site allocation for the production facility has not yet been exploited and research was focused on other aspect [Tamrakar and Yadav \(2025\)](#). As for IoT, many researchers has focused on the aspect of real time monitoring of production equipment and environment for sugar industry, where [Misra et al. \(2020\)](#) described a study on the application of IoT sensor on the precision farming, AS Olayinka and KE

[Olayinka et al. \(2026\)](#) explored IoT as digital twin in sugar manufacturing for the production to optimize productivity. This shows that a joint approach on integrating IoT with AI and GIS to improve the overall productivity from agriculture till production level had yet to be

conducted. Few of the study had also focused on integrating these technologies, although in limited manner, such as [Martin et al. \(2026\)](#) has designed an AI based system for the prediction of harvest time for sugarcane but did not consider other integrated factors such as spatial analysis using GIS nor predictive maintenance using IoT. Similarly [Pandey et al. \(2025\)](#) has modelled sugar production as digital twin but it is mainly about sustainability, and it did not investigate the overall productivity enhancement of sugar production by technology integration. From the review above, there are three major gaps: Most studies applied the digital technologies in two different fields separately where one of it is either agriculture or industrial without the interconnection with each other. There are also lacking of studies to combine the use of the individual technologies and study the potential gain. There are also fewer studies to report on the quantification on the impact of integrating the AI, GIS and IoT application for productivity. Our proposed research fills the gap by integrating the application of AI, GIS and IoT for both agriculture and industrial sectors through an overall conceptual framework. This paper particularly focuses on the manufacturing portion on the sugar industry and quantitatively reports the evidence of integrated technology adoption to enhance productivity and bridging the agriculture-industrial divide, which provide an integration approach model that can be adapted to other agro-industrial applications.

METHODS

This research adopts a mixed-framework approach to investigate the impact of integrated Artificial Intelligence (AI), Geographic Information Systems (GIS), and Internet of Things (IoT) technologies on sugar mill productivity. The methodology consists of four major components:

1. Dataset construction and feature engineering,
2. Statistical correlation analysis,
3. Machine learning modelling, and
4. Performance metric evaluation.

By controlling for operational differences across geographical regions, this systematic approach enables a comprehensive assessment of the relationship between technological adoption and productivity.

Dataset Construction and Feature Engineering

A synthetic dataset comprising 20 sugar mills was developed to represent diverse operational settings across four geographical regions that differ in climate and infrastructure characteristics. Each sugar mill was characterized by 28 features grouped into three technological domains:

- Artificial Intelligence (AI) adoption (e.g., Predictive Maintenance Index (PM), Decision Support Implementation Level (DS)),
- GIS monitoring (e.g., Field Coverage Index (FC), Road Routing Index (RO)), and

- Internet of Things (IoT) implementation (e.g., Sensor Density (SD), Device Connectivity Rate (EC)).

These technological indicators were used to predict four response variables:

- Productivity Index (PI),
- Cost Reduction Percentage (CR),
- Downtime Reduction Percentage (DR), and
- Sugar Recovery Improvement Percentage (SR).

Direct measures were transformed into normalized indices. For example, the Road Quality Index (RQ) was calculated as:

$$RQ = 0.4 \times \text{Pavement Score} + 0.3 \times \text{Route Directness} + 0.3 \times \text{Maintenance Frequency.} \quad (1)$$

Each component was normalized to a scale ranging from 0 to 100.

Similarly, aggregate technological scores were calculated using weighted sums of sub-features:

$$T = \sum_{i=1}^n w_i x_i, \quad (2)$$

where x_i denotes the value of the i -th feature and w_i represents its corresponding weight. The weights were determined using expert opinion and feature importance measures obtained during preliminary analyses.

Statistical Correlation Analysis

Pearson's correlation coefficients were computed to examine the relationships between technological features and productivity measures. To account for multiple comparisons, the Bonferroni correction was applied, and statistical significance was assessed using an adjusted significance threshold.

Partial correlation analysis was subsequently performed to control for geographical influences. The partial correlation coefficient between variables X and Y , controlling for variable Z , was calculated as:

$$r_{XY.Z} = \frac{r_{XY} - r_{XZ}r_{YZ}}{\sqrt{(1 - r_{XZ}^2)(1 - r_{YZ}^2)}}, \quad (3)$$

where:

- r_{XY} is the correlation between X and Y ,
- r_{XZ} is the correlation between X and Z ,
- r_{YZ} is the correlation between Y and Z .

Here, Z represents geographical covariates. This analysis enabled the identification of the unique contributions of technological variables while accounting for spatial dependencies.

Machine Learning Modelling

A Random Forest regression model was employed to capture nonlinear relationships and interactions among technological features and productivity outcomes.

The model was configured using:

- 200 decision trees,
- Maximum tree depth of 8, and
- Minimum of 5 samples per leaf node.

Feature importance was estimated using permutation importance scores, defined as:

$$I_j = P_{\text{base}} - P_j, \quad (4)$$

where:

- I_j denotes the importance score of feature j ,
- P_{base} is the model performance before permutation, and
- P_j is the model performance after randomly shuffling feature j .

A 10-fold cross-validation procedure was implemented to evaluate model performance and minimize the risk of overfitting.

Performance Metric Evaluation

The productivity impacts of technological adoption were assessed using both aggregate and component-level indicators. The overall Productivity Index (PI) was calculated as:

$$PI = \sum_{i=1}^n w_i S_i, \quad (5)$$

where:

- S_i represents the standardized score of the i -th productivity component, and
- w_i denotes the corresponding weight assigned to that component.

All component scores were standardized to ensure comparability across different measurement scales.

Differences among technology adoption categories were statistically examined using one-way Analysis of Variance (ANOVA). Where statistically significant differences were observed, Tukey's Honest Significant Difference (HSD) test was performed as a post-hoc analysis.

Validation Framework

The proposed methodology was validated using three complementary approaches:

1. **Sensitivity Analysis:** Monte Carlo simulations incorporating a 15% variability in parameter values were conducted to evaluate the stability and robustness of the findings.

2. **Domain Expert Review:** Industry experts assessed the simulation parameters and assumptions to establish face validity.
3. **Comparative Benchmarking:** The results were compared with previously published case studies to evaluate their plausibility and consistency.

The proposed methodological framework provides a reproducible and comprehensive approach for evaluating technological adoption in industrial environments. By integrating statistical analyses and machine learning techniques, it enables simultaneous examination of multiple technological domains and their interaction effects while controlling for geographical and operational variability.

RESULTS

Analysis on the integration of AI, GIS and IoT technologies on the productivity outcome in the sugar industry resulted into highly significant relationships between the adoption of the technology and the outcome performance. Following are the analysis shown across the 3 directions; correlation between features of the technologies and productivity outcomes, feature importance from the machine learning techniques, and productivity outcome performance among the various levels of technology adoption.

Correlation Analysis

Pearson's correlation analysis demonstrated meaningful relationships between the levels of technology adoption and productivity outcomes. The Productivity Index exhibited the strongest positive association with AI adoption, followed by IoT implementation and GIS monitoring, suggesting that all three technologies contribute posi-

tively to process performance, with AI exerting the greatest direct influence.

The results indicate that AI adoption has the strongest influence on downtime reduction ($r = 0.231$), which is likely attributable to predictive maintenance and intelligent operational planning. IoT implementation contributes substantially to cost reduction ($r = 0.154$) through real-time monitoring and efficient resource utilization. Although GIS monitoring exhibited comparatively weaker correlations, it may play a critical role in geographically dispersed logistics and regional sugar mill clusters.

The Random Forest regression model achieved satisfactory predictive performance. Predictive Maintenance Score (14.8%), AI Adoption Score (13.5%), and Road Quality Index (13.4%) emerged as the three most influential predictors of downtime reduction.

Feature importance analysis revealed nonlinear relationships among technological variables. In particular, IoT sensor density exhibited diminishing marginal gains beyond approximately 75 sensors per hectare.

Significant interaction effects among technologies were also observed. For instance, combining high GIS field coverage with a medium AI adoption score (50–70 points) reduced operational downtime by an additional 23.6% compared with implementing either technology independently.

Performance Metrics

Aggregate performance across all sugar mills showed:

- Productivity Index: 85.82 ± 12.37
- Cost Reduction: $14.12\% \pm 6.45\%$
- Downtime Reduction: $31.92\% \pm 14.28\%$

Table 1. Correlation coefficients between technology adoption and productivity metrics.

Technology Metric	Productivity Index	Cost Reduction	Downtime Reduction	Sugar Recovery Improvement
AI Adoption Score (AAS)	0.264	0.198*	0.231	0.187*
IoT Sensor Density (ISD)	0.119*	0.154*	0.203	0.112
GIS Monitoring (GMC)	0.042 [†]	0.087	0.095	0.063

[†] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

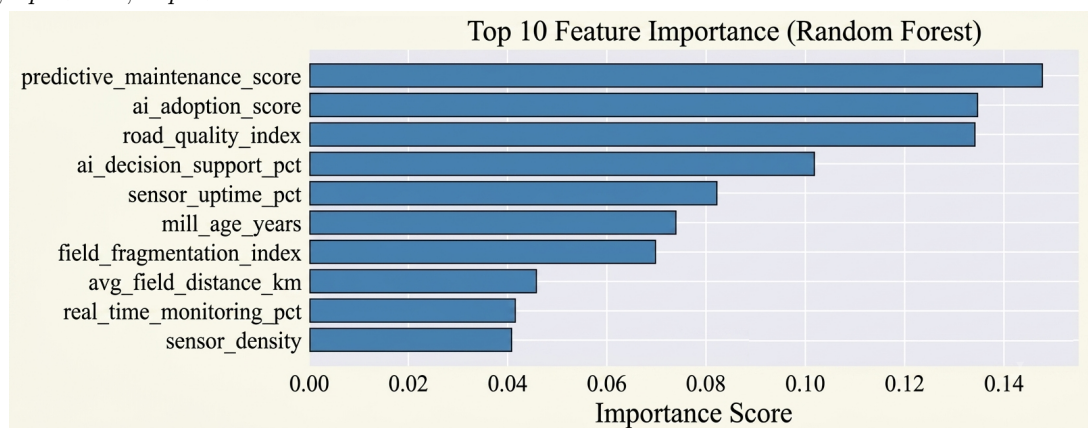


Figure 1. Top ten feature importance scores obtained from the Random Forest model.

- Sugar Recovery Improvement: $4.86\% \pm 2.17\%$

Stratified analyses based on technology adoption quartiles demonstrated progressive improvements in productivity outcomes.

High-adoption mills (top quartile) achieved a Productivity Index of 94.71, significantly outperforming low-adoption mills (bottom quartile), which achieved a score of 76.33. The performance gap was particularly evident for:

- Downtime Reduction: 42.15% versus 19.87%
- Cost Reduction: 18.92% versus 9.45%

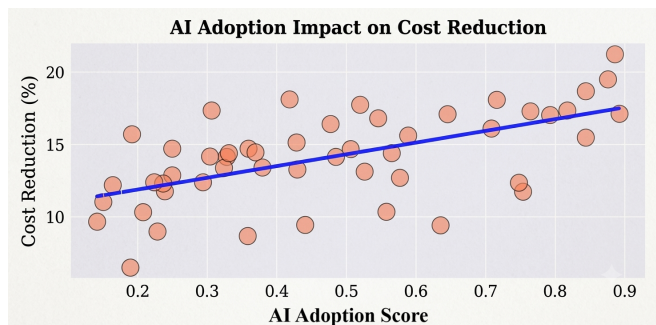


Figure 2. Impact of AI adoption on productivity outcomes.

Mid-adoption mills demonstrated intermediate productivity levels, with Productivity Index values ranging from 82.47 to 89.16. Notably, sugar mills combining medium AI adoption with high IoT integration achieved sugar recovery improvements of 5.92%, exceeding the overall average by 21.8%.

These findings provide empirical support for the proposed technological integration framework and indicate that AI-driven approaches offer particularly substantial productivity gains.

DISCUSSION

The findings of this study contribute to both theoretical development and practical implementation in the sugar industry.

From a theoretical perspective, the results support the concept of synergistic technology implementation as stated in [Yusianto et al. \(2022\)](#), whereby the combined adoption of AI, GIS, and IoT technologies produces benefits greater than the sum of their individual effects. This observation is consistent with studies on digital ecosystems and industrial digital transformation by [Subramaniam et al. \(2019\)](#), which emphasize the value generated through interconnected technological systems.

Furthermore, the diminishing returns observed for IoT sensor density suggest the existence of an optimal level of technology implementation, challenging the common assumption that increased technological investment necessarily produces proportional productivity improvements ([Costa et al., 2022](#)).

From a practical standpoint, the results provide several actionable recommendations for sugar mill operators

and policymakers. The substantial contribution of Predictive Maintenance Score (14.8% feature importance) indicates that AI-enabled predictive maintenance should be prioritized for organizations seeking immediate operational improvements. Transitioning from reactive maintenance strategies to proactive, data-driven maintenance practices can substantially reduce downtime and improve production efficiency ([Gandhare et al., 2018](#)).

Similarly, the observed interaction between GIS field coverage and AI adoption suggests that investments in GIS infrastructure can significantly enhance the effectiveness of existing AI systems. Policymakers and regional development agencies may therefore consider designing subsidy and incentive programs that encourage integrated technology adoption rather than isolated technological investments.

Future studies may extend the present framework in several ways:

- Conduct longitudinal analyses across multiple harvest cycles to assess the sustainability of productivity gains.
- Perform comparative studies across different countries and industrial contexts to evaluate the generalizability of the proposed framework.
- Investigate the socioeconomic implications of technology adoption, including employment patterns, workforce skill requirements, and training needs.
- Develop hybrid modelling approaches that integrate system dynamics with machine learning techniques to better capture complex technological, environmental, and human interactions.
- Explore regional disparities in technological effectiveness to develop location-specific implementation strategies.

A deeper understanding of regional and contextual factors influencing technology adoption will facilitate the development of more effective and targeted digital transformation strategies for the sugar industry.

CONCLUSIONS

This study has revealed that the integration of AI, GIS and IoT in sugar industry has a considerable productivity gain with AI having the highest positive correlation. The results demonstrate that predictive maintenance, AI-based decision assistance and geographical observation lead to verifiable improvements to production levels and cost savings while enhancing the quantity of sugar produced. The findings validate the hypothesised benefits of the joint use of synergised technologies but also highlight the need for careful technological integration strategies especially with IoT to avoid diminishing returns. Further research needs to focus on longitudinal study of adoption of technology in real-world practices to demonstrate sustainability of these gains in productivity. Socio-economic impact on the agricultural-industrial system has yet to be deeply explored, including adaptation of workforce

and differences in regions. This framework is proposed as basis for similar study as well as to offer practical implications to practitioners in the industry for business optimization through appropriate use of technology.

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Natural Calamity Deaths in Nepal: Analysis of the 2021 National Population and Housing Census

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ABSTRACT

Background: Nepal is one of the most disaster-prone countries in South Asia due to its rugged terrain, monsoon-driven precipitation, and active seismicity. However, evidence on the demographic and geographic patterns of mortality associated with natural calamities remains limited. This study examined natural-calamity mortality in Nepal using nationally representative census data, with particular emphasis on differences by age, sex, wealth, residence, province, and district-level geographic exposure.

Method: Data on household-reported natural-calamity deaths were obtained from the 2021 National Population and Housing Census (NPHC). Weighted individual records were used to construct sex-, age-, wealth quintile-, and residence-specific exposure populations. A negative binomial mixed-effects regression model with a log-population offset was fitted to estimate incidence rate ratios, with sex, age group, wealth quintile, place of residence, and province included as fixed-effect predictors. A district-level random intercept was incorporated to account for geographic clustering in natural-hazard exposure.

Results: Among 198,463 deaths recorded in the census, 9,595 (4.83%) were attributed to natural calamities, corresponding to a national mortality rate of 32.9 per 100,000 population. Madhesh Province recorded the highest mortality rate (55.8 per 100,000) and the highest proportion of all deaths attributable to natural calamities (10%). Mortality followed a J-shaped age pattern, with the lowest risk among working-age adults and the highest among individuals aged 75 years and above. Mortality was significantly higher among males than females and among urban than rural residents. Only individuals in the wealthiest households experienced significantly lower mortality than those in the poorest households. No province-level differences remained statistically significant after accounting for geographic clustering below the provincial level. Substantial variation persisted across districts, with the highest-risk districts distributed across several provinces.

Conclusion: Natural-calamity mortality in Nepal is shaped not only by age, sex, wealth, and place of residence but also, importantly, by localized district-level exposure. The absence of significant province-level differences after accounting for district clustering suggests that disaster risk varies considerably within provinces. Disaster-risk-reduction investments and early-warning systems should therefore be targeted at the district level, with evacuation and preparedness strategies tailored to vulnerable age groups and residential settings.

Key words: natural disasters, mortality, Nepal, census, negative binomial regression

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INTRODUCTION

Globally, Nepal is one of the most highly exposed countries to natural hazards. Here, the population is recurrently exposed to natural calamities such as earthquakes, floods, landslides, lightning, glacial lake outburst floods, and others. These events are often triggered by its steep, tectonically active terrain and monsoon-dominated precipitation (Group and Bank., 2021). The frequency of disasters is no less. Between mid-2018 and mid-2024 alone, Nepal recorded more than 32,000 disaster incidents. As a result, 3,672 people died, while many people were missing, and multiple communities were displaced (of Home Affairs., 2024). Single catastrophic events can also produce mortality shocks that far exceed the usual and expected annual burden. The recent and widely known historical event is the 2015 earthquake, which killed an estimated 8,951 people across 41 districts (Ton-

nelier and Charalampous, 2026). That earthquake also left a lasting impact of secondary hazard exposure. It destabilized hill slopes that remain landslide-prone today (Adhikari and Mishra, 2021). More recent events have followed the same pattern. The November 2023 Jajarkot earthquake killed 153 people in Karnali Province (Gautam, 2023), the late-September 2024 floods and landslides killed more than 215 people across 44 districts (Nations., 2024), a supraglacial lake outburst on the Bhote Koshi in July 2025 left at least nine people dead and more than two dozen missing in Rasuwa (Reuters., 2025), and unseasonal October 2025 rainfall killed at least 60 people in eastern Nepal, 39 of them in Ilam district alone (Gautam, 2025). Recurrent hazards add a further, less visible burden. Lightning alone kills of the order of 100 people every year and ranks among the deadliest hazards in the country (Adhikari, 2021; Sharma and Cooper,

2022). The Sendai Framework for Disaster Risk Reduction 2015–2030 targets a substantial reduction in global disaster mortality. Disaster-attributable death rates are observed under Sustainable Development Goal indicators 1.5.1, 11.5.1, and 13.1.1. This reflects international recognition of natural disasters as a measurable and preventable contributor to premature mortality (for Disaster Risk Reduction., 2015).

Existing evidence on disaster-related mortality in Nepal comes mostly from event-specific and hazard-specific studies. These include retrospective analyses of an event and rarely incorporate all hazard types together in a single estimate (Chapagain and Borgemeister, 2024; Gaire and Arcos González, 2015; K.C., 2013). Together, a consistent set of structural vulnerabilities is identified. These include settlement on river banks, particularly in flood plains, unstable hill slopes, and limited early-warning and evacuation infrastructure in vulnerable regions. They also highlight gender differentials in disaster mortality. Compared to men, women in South Asia are estimated to have a higher disaster-related mortality rate. It is attributed to mobility restrictions, caregiving responsibilities, and lower access to early-warning information (Neumayer and Plümper, 2007; for Asia and the Pacific., 2022). Besides deaths occurring inside health facilities, the cause of death is not routinely recorded in Nepal. This limits what routine systems can report (Office., 2023). As a result, geographically disaggregated estimates of natural calamity deaths across the country remain limited. How demographic and socioeconomic characteristics along with geography jointly shape this risk is also not well quantified.

The Nepal Population and Housing Census (NPHC) 2021 addressed part of this gap. At national scale, it asked households to report whether a death occurring in the preceding 12 months was due to a natural calamity. Household-reported cause of death carries well-known limitations relative to medically certified or event-verified causes. However, it provides population-level data on natural calamity deaths for the whole country. These data span whichever hazards happened to affect a household in the reference period. Thus, this study uses these data to (i) understand the geographical distribution of natural calamity deaths in Nepal in 2021, and (ii) identify the socio-demographic predictors of those deaths and quantify the risk associated with them.

METHODOLOGY

Study Design, Setting, and Data Sources

This is a cross-sectional study using the sample data from the National Population and Housing Census (NPHC) 2021 (Office., 2023). The census enumerated all usual residents of Nepal. It also recorded detailed information on the deaths of household members occurring during the 12 months preceding the census reference date (25 November 2021).

The unit of analysis was a demographic stratum. It was created by cross-classifying individuals by sex, age group, place of residence, and household wealth quintile. For each stratum, the number of deaths due to natu-

ral calamities and the corresponding weighted exposure population were computed from the deceased and individual census files, respectively. The two files were then merged by their common stratifying variables. A few sparsely populated strata contained a reported death but no matching exposure population, because the weighted individual file held no record with that exact combination of characteristics. Dropping these strata would have discarded observed deaths, so each was assigned a nominal population of one. This affected a small number of strata and did not significantly change the model estimates.

Outcome Variable

The outcome variable was the number of deaths due to natural calamities recorded per stratum. During the census, a respondent was asked to classify each household death that occurred in the preceding 12 months into one of the following categories: communicable disease, non-communicable disease, road traffic accident, other accident, pregnancy-related, crime, suicide, natural calamities, or other. For this study, deaths reported as due to natural calamities were taken as the outcome, and deaths from all remaining reported categories were treated as the comparison group. The “natural calamities” category is hazard-agnostic. It may include deaths from floods, landslides, earthquakes, avalanches, lightning, and other acute environmental hazards, without further sub-classification by hazard type. Cause of death was reported by a household member rather than medically certified or verified against a disaster registry. The outcome should therefore be read as household-attributed rather than officially verified disaster mortality.

Predictor Variables

Five predictor variables were included in the analysis. They were:

Age (0–4, 5–14, 15–24, 25–34, 35–44, 45–54, 55–64, 65–74, 75 years and above)

Sex (Male and Female)

Place of residence (Rural, Urban, Peri-urban)

Household wealth quintile (Lowest, Lower, Middle, Higher, Highest)

Province (Bagmati, Gandaki, Karnali, Koshi, Lumbini, Madhesh, Sudurpaschim)

Exposure populations for each stratum were derived from weighted individual-level census records.

Statistical Analysis

Firstly, national, provincial, and district-level natural-calamities deaths were summarized using counts, crude death rates per 100,000 population, and the proportional share of natural calamities among all recorded deaths. A crude rate is the ratio of the observed number of deaths and the exposed population. District rates and proportions were mapped using shapefiles for administrative boundaries current at the time of the 2021 census.

To estimate the association between socio-demographic characteristics and natural calamity mortality, a negative binomial mixed-effects regression with a log link was fitted:

$$\log(E[Y_i]) = \log(\text{Population}_i) + X_i\beta + u_{d(i)}$$

where,

Y_i is the number of natural-calamity deaths in stratum i ,

$Population_i$ is the corresponding enumerated population entered as a log-offset, and

X_i is the vector of fixed-effect predictors (sex, age group, household wealth quintile, place of residence, and province).

$u_{d(i)} \sim N(0, \sigma_d^2)$ is a random intercept for the district d containing stratum i .

Province was included as a fixed effect to assess geographic variation in disaster risk directly. A district-level random intercept was added because natural-calamities mortality is expected to cluster geographically at a finer scale than a large provincial catchment. Models were fitted with the negative binomial (type 2) family, specified as `nbinom2` in the R package `glmmTMB` (Brooks and Bolker, 2017). This family was chosen over a standard Poisson specification because of overdispersion, since the counts showed substantially greater variance than the mean. Overdispersion is expected for disaster deaths, which concentrate in a small number of severely affected strata rather than spreading evenly across the population.

Model Evaluation and Diagnostics

Multicollinearity among predictors was assessed using generalized variance inflation factors (GVIFs) (Fox and Monette, 1992). All adjusted GVIF values fell below the conventional threshold of 2, indicating no problematic collinearity (Craney and Surles, 2002). Incidence rate ratios (IRRs) with 95% confidence intervals were obtained by exponentiating fixed-effect coefficients.

A preliminary fixed-effects-only specification (sex, age group, wealth quintile, residence, and province, without a district term, fitted on province-aggregated strata) showed a Pearson dispersion ratio of 7.50. This shows that the negative binomial family alone did not adequately capture stratum-to-stratum clustering. A district-level random intercept reduced this ratio to 1.94 and is retained in the model reported throughout the study. Model adequacy was assessed using the marginal and conditional for generalized linear mixed models, computed with the R package `performance`. The marginal provides variance explained by the fixed effects alone, and the conditional adds the district random intercept. All analyses were conducted in R version 4.5.0 (2025-04-11 ucrt) using `glmmTMB` 1.1.14, `performance`, and `broom.mixed` packages.

RESULTS

Geographic Distribution of Natural Calamities Mortality

National and provincial variation

Provincial variation was marked. Madhesh recorded the highest crude death rate (55.8 deaths per 100,000 population), followed by Lumbini (35) and Gandaki (32.4). Karnali had the lowest rate (22.2 deaths per 100,000 population) (Table 1). Madhesh also carried the highest proportional share of all deaths due to natural calamities

Table 1. National and provincial natural calamities mortality, Nepal, 2021.

Region	Total deaths	Number of deaths	Death rate (per 100,000 population)	Proportion of deaths (%)
Nepal (national)	198,463	9,595	32.9	4.8
Madhesh	34,319	3,415	55.8	10
Lumbini	36,207	1,791	35	4.9
Gandaki	20,406	798	32.4	3.9
Koshi	38,275	1,215	24.5	3.2
Sudur paschim	16,988	640	23.7	3.8
Bagmati	43,872	1,362	22.3	3.1
Karnali	8,396	374	22.2	4.5

ties (10%), more than double that of any other province. Both measures were therefore high in the same province.

District-level variation

Natural-calamities mortality rates ranged from 13.3 to 68.5 deaths per 100,000 population across districts (Fig. 1). Mahottari had the highest rate (68.5 per 100,000 population), followed by Saptari (66.7), Gorkha (60.6), and Sarlahi (59). Most of the highest-rate districts lie in Madhesh Province. Gorkha is an exception, because it is a hill district rather than a Terai one. Humla, a sparsely populated mountain district, recorded the highest proportional share of deaths due to natural calamities (13.4%), despite a modest rate.

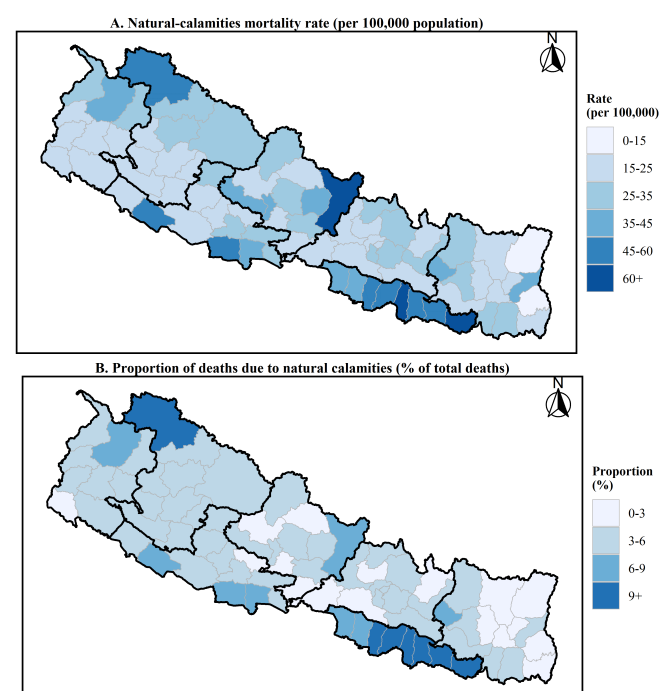


Figure 1. Natural-calamities mortality by district, Nepal, 2021. Provincial boundaries are overlaid in black.

Table 2. Descriptive summary of socio-demographic characteristics of the study population, Nepal census 2021

Variable	Category	Exposure population	Natural calamities deaths	Rate (per 100,000)	Share of deaths (%)
Sex	Female	14,919,849	4,009	26.87	4.8
	Male	14,244,729	5,578	39.16	4.9
Age Group	0–4	1,909,507	575	30.11	5.7
	5–14	5,619,326	343	6.10	8.5
	15–24	5,838,337	321	5.50	4.4
	25–34	4,767,417	328	6.87	4.1
	35–44	3,893,350	491	12.62	3.8
	45–54	2,873,104	474	16.50	2.4
	55–64	2,109,330	832	39.45	3.0
	65–74	1,438,654	1,777	123.51	4.3
	75+	715,553	4,446	621.33	6.6
Wealth Quintile	Lowest	5,757,063	2,369	41.15	6.2
	Lower	5,918,418	1,887	31.88	4.8
	Middle	5,991,255	2,211	36.90	5.4
	Higher	5,465,884	1,575	28.81	4.5
	Highest	5,785,401	1,545	26.71	3.5
Place of Residence	Rural	9,642,797	2,614	27.10	3.9
	Urban	7,964,905	2,194	27.55	4.1
	Peri-urban	11,556,877	4,779	41.35	6.2

* The natural-calamities deaths shown here total 9,587, which is 8 fewer than the 9,595 recorded nationally because deaths for which age was not stated were excluded from the analysis.

Descriptive Characteristics of the Study Population

Table 2 presents the distribution of natural calamity mortality by sex, age group, household wealth quintile, and place of residence. The mortality rate was higher among males than females (39.16 vs 26.87 per 100,000 population), but had almost similar proportions of deaths due to natural calamity. The mortality was high among the youngest children, declining to its lowest level among those aged 15–24 years (5.50 per 100,000), and increasing steadily thereafter. The rate reached 621.33 per 100,000 among adults aged 75 years and above. This is more than 100 times the rate observed among those aged 15–24 years. Although children aged 5–14 years had a low mortality rate (6.10 per 100,000), natural calamities accounted for the largest share of all deaths in this age group (8.5%). This relatively high share reflects the low underlying mortality at these ages. As a result, a small number of natural calamity deaths reflects a comparatively large proportion of total deaths.

A socioeconomic gradient was also evident. Generally, mortality was higher among individuals from low household wealth quintiles. The highest mortality rate was observed in the lowest wealth quintile (41.15 per 100,000), compared with 26.71 per 100,000 in the highest quintile. Mortality also differed by place of residence. Peri-urban areas had the highest rate (41.35 per 100,000), substantially exceeding the rates in rural (27.10) and urban (27.55) areas. Overall, a marked difference in natural-calamity mortality across socio-demographic variables, with particularly elevated rates among older adults, males, poorer populations, and those living in peri-urban areas, was observed.

Table 3. Generalized variance inflation factors for predictors included in the regression model.

Predictors	GVIF	DF	Adjusted GVIF
Sex	1.00	1	1.00
Age Group	1.05	8	1.00
Wealth Quintile	1.94	4	1.09
Place of residence	2.66	2	1.28
Province	2.21	6	1.07

Predictors of Natural-Calamities Mortality

Multicollinearity diagnostics indicated no evidence of problematic collinearity. All adjusted GVIF values fell below the conventional threshold across model covariates (Table 3)

Predictors of Natural-Calamities Mortality

Table 4 presents adjusted incidence rate ratios from the final negative binomial mixed-effects regression model. Age showed a J-shaped association with mortality. Relative to children aged 0–4, risk fell sharply through adolescence and working age. The population aged 15–24 years had the lowest risk (IRR = 0.27, 95% CI: 0.14–0.50). The risk was high for the population older than 55 years, with the highest among those aged 75 years and above (IRR = 13.33, 95% CI: 7.89–22.51). This oldest group faced the highest adjusted mortality of any age group. Females had significantly lower mortality risk than males after adjustment (IRR = 0.70, 95% CI: 0.55–0.90). Household wealth showed a protective effect among the highest group. Only the highest quintile differed significantly from the lowest (IRR = 0.57, 95% CI: 0.36–0.91). Place of residence was strongly associated with mortality. Urban residents had

Table 4. Adjusted incidence rate ratios (IRR) for natural-calamities mortality from the negative binomial regression model

Variable	Category	Adjusted IRR (95% C.I.)
Sex	Female	0.70 (0.55–0.90)**
Age Group	5–14	1.18 (0.59–2.39)
	15–24	0.27 (0.14–0.50)***
	25–34	0.37 (0.20–0.70)**
	35–44	0.33 (0.18–0.60)***
	45–54	0.42 (0.23–0.74)**
	55–64	1.20 (0.68–2.11)
	65–74	2.27 (1.34–3.84)**
Wealth Quintile	75+	13.33 (7.89–22.51)***
	Lower	0.82 (0.57–1.18)
	Middle	0.69 (0.47–1.00)
	Higher	0.76 (0.51–1.15)
	Highest	0.57 (0.36–0.91)*
Place of Residence	Urban	2.57 (1.84–3.60)***
	Peri-urban	1.26 (0.89–1.76)
Province	Bagmati	0.91 (0.39–2.10)
	Gandaki	0.90 (0.37–2.19)
	Karnali	0.92 (0.35–2.41)
	Koshi	0.55 (0.24–1.29)
	Lumbini	0.59 (0.25–1.37)
	Sudurpaschim	0.45 (0.17–1.14)

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Reference categories: Sex = Male; Age = 0–4 years; Wealth Quintile = Lowest; Place of Residence = Rural; Province = Madhesh. The model includes a log-population offset.

substantially higher mortality risks than rural residents (IRR = 2.57, 95% CI: 1.84–3.60). Peri-urban residents did not differ significantly from rural residents after adjustment (IRR = 1.26, 95% CI: 0.89–1.76). Lastly, none of the provinces differed significantly from Madhesh regarding mortality.

Table 4: Adjusted incidence rate ratios (IRR) for natural-calamities mortality from the negative binomial regression model.

Model Diagnostics

Table 5. Districts with the highest and lowest modelled excess risk of natural-calamities mortality

Rank	District	Province	Excess risk (IRR)
Highest	Rasuwa	Bagmati	7.69
Highest	Taplejung	Koshi	4.74
Highest	Lamjung	Gandaki	3.45
Highest	Humla	Karnali	3.08
Highest	Bajhang	Sudurpaschim	2.91
Lowest	Tanahu	Gandaki	0.42
Lowest	Sankhuwasabha	Koshi	0.40
Lowest	Jhapa	Koshi	0.36
Lowest	Nawalparasi East	Gandaki	0.22
Lowest	Surkhet	Karnali	0.19

The district-level random intercept had a standard deviation of 0.83 on the log-count scale, indicating substantial district-to-district variation in mortality that is not explained by demographic composition or province membership. Table 5 presents the five districts each with the highest and the lowest modelled excess risk. Rasuwa (Bagmati province, IRR = 7.69) carried the largest excess risk, and Surkhet (Karnali, IRR = 0.19) the smallest. The five highest-risk districts span five different provinces, which shows that the strongest geographic signal in these data operates at the district level and does not align cleanly with provincial boundaries. Full estimates for all 77 districts are given in Annex 2.

Table 5: Districts with the highest and lowest modelled excess risk of natural-calamities mortality

DISCUSSION

This study used household-reported cause of death from the 2021 census to assess variation in natural calamity mortality across Nepal. Three key findings were found. First, natural calamity mortality varied strongly by age, with the highest risk among the elderly population. Second, males had higher mortality than females, while urban residents had higher mortality than rural residents after adjustment. Third, provincial differences were not statistically significant after accounting for geographic clustering, although mortality varied substantially across districts.

Age was the strongest pattern in the analysis. Compared with children aged 0–4 years, mortality was lower among people aged 15–54 years. It increased again among older adults and was more than 13 times higher among those aged 75 years and above. This pattern is consistent with greater vulnerability among very young and older populations. Young children and older adults may have greater difficulty escaping rapidly developing hazards and may depend more on others during evacuation (Tonnelier and Charalampous, 2026). Studies of floods, landslides, and the 2015 earthquake in Nepal have also reported greater vulnerability among older populations (Adhikari and Mishra, 2021; Gaire and Arcos González, 2015).

Mortality was lower among females than males after adjustment. This finding differs from studies in South Asia that report higher female mortality during disasters (Neumayer and Plümper, 2007; for Asia and the Pacific., 2022). Gender differences have also been examined in Nepal, in years-of-life-lost analyses of the 2015 earthquake (Tonnelier and Charalampous, 2024), though that evidence covers a single seismic event rather than the recurrent floods and landslides that drive most of the deaths recorded here. One possible explanation is the hazard profile in Nepal. Floods and landslides are common, recurrent hazards in Nepal (Gaie and Arcos González, 2015; Pokharel and Raule, 2026), and men may have greater exposure through outdoor and agricultural work. Evidence from Nepal also shows a higher concentration of lightning deaths among men, particularly those engaged in outdoor activities (Adhikari, 2021; Sharma and Cooper, 2022). However, the census does

not record the hazard type, location of death, or activity at the time of death. These scenarios therefore cannot be tested in the present study.

The association with household wealth was weaker after adjustment than in the descriptive analysis. Only the highest wealth quintile had significantly lower mortality than the lowest quintile. Poorer households may have greater exposure to hazardous locations and fewer resources for safe housing, evacuation, and recovery (K.C., 2013).

Urban residents had higher adjusted mortality than rural residents. This contrasts with the similar crude mortality rates observed across residence categories. The difference may partly reflect the age and socioeconomic composition of urban populations. Rapid urban growth may also increase exposure to flooding in densely settled and poorly drained areas. Recent flooding in the Kathmandu Valley illustrates this concern (Lamichhane, 2025).

Madhesh had the highest provincial mortality rate and the highest adjusted point estimate. However, none of the provincial estimates differed significantly from Madhesh after accounting for geographic clustering. The apparent provincial differences in a model without district-level clustering were therefore not robust. This indicates that geographic variation in natural-calamity mortality is better captured at a finer spatial scale than the province.

The district-level results support this interpretation. High-risk districts were found across several provinces and were not concentrated in Madhesh. Some districts with high crude rates were located in the hills, while modelled district effects also identified excess risk in hill and mountain areas. These patterns suggest that natural-calamity mortality is not confined to the Terai floodplain. Instead, risk varies across different geographic settings with different hazard profiles. Flooding is a major concern in the Terai, while landslides and other terrain-related hazards are important in hill and mountain districts (Group and Bank., 2021; Gaire and Arcos González, 2015).

The district estimates should nevertheless be interpreted with caution. The census records deaths reported by households but does not identify where the death occurred or the specific hazard involved. The 12-month reference period also prevents the observed district patterns from being linked to individual disaster events. Events occurring during or close to the reference period, such as the 2021 Melamchi flood and the October 2021 rainfall events, may have contributed to some of the observed variation (Chen and West, 2025; Press., 2021; Bank., 2022). These findings therefore indicate where mortality was concentrated rather than which hazards caused the excess risk.

These findings have implications for disaster-risk-reduction and public health planning in Nepal. Preparedness should give particular attention to young children and older adults, who showed the greatest mortality risk. Urban areas also require greater attention to flood-resilient infrastructure and land-use planning. Geographic targeting should consider districts rather than

relying only on provincial averages because high-risk areas were distributed across several provinces. Hazard-specific information should be incorporated into district-level planning where available.

Strengths and Limitations

The main strength of this study is its use of population-level census data covering the whole country, analysed with a model that adjusts jointly for individual, household, and geographic characteristics, including a district-level random intercept for geographic clustering below the province level. It captures natural-calamity deaths across all provinces and hazard types within a common data source, and allows mortality to be examined across several socio-demographic characteristics.

Several limitations should be considered. First, causes of death were reported by household members rather than medically certified or independently verified. Some deaths may therefore have been misclassified. Second, the census does not identify the specific hazard responsible for each death. Floods, landslides, earthquakes, and other hazards have different risk profiles, but they are combined in the outcome. Third, the analysis is cross-sectional. The observed associations should not be interpreted as individual-level causal effects. The available variables also limit the analysis: housing conditions, proximity to rivers or unstable slopes, evacuation access, and occupation at the time of death were not available. Fourth, the district random intercept absorbs geographic clustering statistically but does not explain it: it does not identify which hazard, or which local factor, drives a given district's excess risk. Districts with small populations record few deaths and may have unstable rates, both in the descriptive summaries and in the corresponding random-effect estimates.

CONCLUSION

This study used census data to measure natural-calamities deaths across Nepal. Risk was concentrated at the extremes of the age distribution, among young children and the elderly population. Compared to provinces, mortality varied markedly by district. Males died from natural calamities more often than females. Urban residents faced significantly higher risk than rural residents. These patterns point to three priorities. Disaster-risk-reduction investment should target specific high-risk districts, informed by hazard-specific data, rather than being allocated at a higher administrative tier. Evacuation planning should account for households with young children or older residents. Rapidly urbanizing municipalities face growing, not diminishing, flood exposure and should be planned for accordingly. Stronger cause-of-death reporting would refine these estimates further. Linking future census or survey rounds with Nepal's disaster event databases would let deaths be verified against recorded events and disaggregated by hazard type, tracing the district-level risk concentration identified here to specific hazards. This would support ongoing monitoring in line with the Sendai Framework's mortality-reduction targets.

DECLARATIONS

Ethics Approval and Consent to Participate

This study used publicly available, anonymised census data. No individual-level identifiers were retained in the analytical dataset. Formal ethical review was not required under applicable national guidelines for secondary analysis of anonymised population data.

Consent for Publication

Not applicable.

Availability of Data and Materials

The study used secondary census microdata and publicly available district-level indicators. Access to the census microdata is governed by the National Statistics Office of Nepal. Derived analytic code and inputs are available from the corresponding author on reasonable request.

Competing Interests

The authors declare that they have no competing interests.

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Authors' Contributions

JA conceptualized the study, curated the data, conducted the statistical analysis, and drafted the manuscript. SR and DS contributed to data interpretation and critical revision of the manuscript. All authors read and approved the final manuscript.

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Modeling Minimum Wage Determination in Nepal: The Role of Economic Growth and Inflation

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ABSTRACT

Background: Minimum wage (MW) is among the most contested labor market instruments, particularly in countries where industrialization is not yet sufficiently matured. In setting MW, the main issue is balancing between worker's welfare and enterprise growth, which actually difficult in developing economies with large informal sectors. In Nepal, MW has periodically revised since 1965, but whether these adjustments reflect macroeconomic conditions or balance worker welfare with industrial growth is still unclear. This study aims to prescribe a reliable statistical model that best describes minimum wage growth in Nepal.

Methodology: The study combines quantitative and qualitative approaches. Quantitative component applies descriptive statistics and linear regression to a national data (1965–2025) of gross domestic product (GDP) growth, inflation, and MW growth. The qualitative component synthesizes literatures and policy documents to situate Nepal's experience within global and South/East Asian regional benchmarks.

Results: Between 1965–2025, mean annual GDP growth rate was 3.9%, while inflation and MW growth rates were 7.7% and 8.8%, respectively. GDP growth was positively associated with MW growth and was found significant ($\beta = 1.16$, $p = .029$), whereas inflation was not ($p = .706$), and the overall model fit was significant ($p = .019$). Model diagnostics indicated no problematic multicollinearity ($VIF = 2.08$), no evidence of heteroskedasticity ($p = .357$), and no evidence of model misspecification ($p = .558$); serial correlation was not significant ($p = .053$). Cumulatively, MW rose 530.4% over the period, versus 234.7% for GDP and 460.0% for inflation, indicating that wages kept up with prices, but only just.

Conclusion: MW growth in Nepal has followed economic growth more than inflation, but weak enforcement and a large informal sector (60–70% of the workforce) limit its impact. Effective wage policy requires periodic, evidence-based, sector-sensitive revisions, alongside stronger enforcement, productivity investment, and broader coverage of informal and rural workers.

Key words: Minimum wage, wage policy, Nepal, GDP growth, inflation

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INTRODUCTION

The International Labour Organization (ILO) has long framed MW policy as a central instrument of decent work, social protection, and poverty reduction. The ILO Resolution adopted in 2010 underscores the role of MWs in safeguarding vulnerable workers ([International Labour Organization, 2014](#)), while the MW Fixing Convention, 1970 (No. 131) requires that wage-setting proceed through evidence-based social dialogue that accounts for both the needs of workers and their families and prevailing economic conditions of the country. The ILO Centenary Declaration for the Future of Work (2019) endorses that dual concern must balance social equity with economic feasibility whether the MWs is statutory or negotiated. Moreover, a MW is formally defined as the minimum remuneration an employer must pay for work performed in a given period, a floor that cannot be reduced by collective agreement or individual contract ([International Labour Organization, 2014](#)). More than 90% of ILO member states have adopted some form of MW legislation ([Sümer, 2018](#)). MW is one of the most widely used labor-market instruments globally, even as it

remains intended primarily for less-skilled and vulnerable workers rather than the workforce as a whole ([Ravn and Sørensen, 1999](#)).

Empirical evidence on the effects of MW laws is mixed. Some studies document positive effects on income redistribution and poverty reduction ([Jadoon et al., 2021](#); [Bauducco and Janiak, 2018](#)), while others emphasize risks such as unemployment, informalization, and reduced enterprise competitiveness ([Kim and Lim, 2018](#); [Paun et al., 2021](#)). A study of Mexico found that MW increases had no adverse employment effects and instead contributed positively to worker incomes ([Campos Vázquez et al., 2018](#)), while traditional economics theory continues to predict that higher MWs raise labor costs for unskilled workers and may increase unemployment ([Raut, 2019](#)). The balance of these effects appears to depend heavily on a country's labor market structure, institutional capacity, and complementary social policy. This type of dependency that is especially noticeable in developing economies like Nepal, where a large share of employment is informal or agricultural ([Jadoon et al., 2021](#)).

MW systems also vary widely by region and level of de-

velopment. High-income economies typically operate institutionalized, often automatically indexed, wage-setting mechanisms: France and Germany index MWs to price levels, and the United Kingdom's National Living Wage adjusts annually to the cost of living (Paun et al., 2021; Low Pay Commission, 2020). In East Asia, Japan and South Korea combine relatively high MWs with strong enforcement and closer alignment to productivity (Ravn and Sørensen, 1999; Ryu, 2021), and China's provincial wage floors, introduced in the 1990s have been associated with improved income distribution without clear evidence of net job losses (Fang and Lin, 2015).

However, South Asian countries present a more fragmented picture. India has a complex, multi-tiered system of over 1,200 states and sector-specific MWs rates (International Labour Organization, 2014). Likewise, Bangladesh sets wages through sectoral negotiation, most visibly in garments sector (Kabeer et al., 2020). Nepal applies a uniform national MW across sectors but continues to face constraints in enforcement capacity and monitoring (Raut, 2019). Regionally, legal MWs in several Asian economies are estimated to fall well below living-wage benchmarks, ranging from 21% in Bangladesh to 46% in China, leaving large segments of the working population in a state of "working poverty" (Labour Behind the Label, 2021).

Nepal introduced its first legal MW in 1965 (B.S. 2022). The wage-fixing system in Nepal has since evolved from a skill-differentiated structure up to 2010. Since from 2010 onward, a unified wage applied across skill categories, together with a dearness allowance and daily and hourly equivalents. After the Labor Act of 1992 and Industrial Enterprises Act of 1992 formalized wage-setting mechanism has been enacted. The Labor Act of 2017, following the establishment of a federal republic, consolidated provisions on MWs, overtime, and worker protections within a tripartite mechanism; involving government, trade unions, and employers. Even after these reforms there are still remain gaps balancing between employment generation, enterprises growth and worker's welfare. Wage adjustments have often uncorrelated between inflation and the true cost of living particularly in urban centers (Raut, 2019). On the other hand, labor inspection capacity is limited, with reportedly fewer than 50 inspectors nationwide (Ministry of Labour, Employment and Social Security, 2021), which lack monitoring of the proper implementation of MWs. Over 60% of Nepal's workforce remains in informal employment that lies largely outside the reach of legal wage protection (National Statistics Office, 2020).

The study addresses these gaps by documenting the levels, trends, and predictors of Nepal's MW over six decades. Empirically testing the association between MW growth and two core macroeconomic fundamentals: GDP growth and inflation. This address Nepal's experience within global, regional, and sectoral MW practices applied. The objective is to determine whether Nepal's current wage-setting practice is adequately evidence-based and to derive policy recommendations that are rights-based, evidence-driven, and sector-sensitive.

MATERIALS AND METHODS

Research Design

This study adopts a mixed-method design that integrates quantitative and qualitative approaches. The quantitative component uses descriptive and comparative statistical analysis of MW trends together with linear regression to test the association between MW growth and macroeconomic fundamentals. The qualitative component synthesizes secondary literature, government reports, and international labor-policy documents to interpret the institutional and enforcement realities that shape the numerical results.

Data Sources

The study used multiple secondary data sources to strengthen reliability through triangulation. The main data sources for quantitative analysis were drawn from the National Statistics Office (NSO), the Ministry of Labor, Employment and Social Security (MoLESS), the Ministry of Finance (MoF), and the World Bank Group. For qualitative assessment: policy documents, country reports, including MW Fixation Committee recommendations and government gazettes; ILO conventions; and comparative peer-reviewed literature on wage-setting mechanisms in South Asia and beyond were used.

Study Variables

The dependent variable for this study is the minimum wage (MW) rate, measured on a ratio scale as the nominal average monthly MW level officially announced by MoLESS periodically. The explanatory variables are the annual percentage growth rates of GDP and inflation. In addition, cumulative percentage change is calculated for each series (MW, GDP, and inflation) relative to a 1965 base year.

Analytical Approach

The analysis was conducted in three steps. First, descriptive statistics (means and standard deviations) summarized MW, inflation, and GDP growth trends over 1965–2025, applying the following equations;

$$\% \Delta GDP = \frac{GDP_{t+1} - GDP_t}{GDP_t} \times 100 \quad (1)$$

$$\% \Delta Inf = \frac{Inf_{t+1} - Inf_t}{Inf_t} \times 100 \quad (2)$$

$$\% \Delta MW = \frac{MW_{t+1} - MW_t}{MW_t} \times 100 \quad (3)$$

Where, represents the percentage change, t represents the specific period of time for the given data.

In addition, pairwise Pearson correlations among GDP growth, inflation, and MW growth are also computed using the following formula, given in equation (4);

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (4)$$

Second, multiple linear regression was employed with annual MW growth as the dependent variable and GDP

growth and inflation as predictors, given by the following equation (5):

$$MW_{Gr} = \beta_0 + \beta_1(GD_{Gr}) + \beta_2(Inf_{Gr}) + \epsilon \quad (5)$$

Where β_0 represents the model intercept, β_1 and β_2 are regression parameters, and ϵ is the error term of the model.

Model Fit and Diagnostics Test

Model fit was assessed via the F-test and R^2 , and individual predictors via two-tailed significance tests at $\alpha = 0.05$. Pairwise Pearson correlations among GDP growth, inflation, and MW growth were also computed.

Diagnostic tests examined multicollinearity by variance inflation factor (VIF), heteroskedasticity, model misspecification, and serial correlation to confirm the model that met the standard regression assumptions.

T-test was used to test the individual coefficient of the model. The hypotheses are;

$$H_0 : \beta_j = 0, \quad j = 1, 2$$

, $j = 1, 2$ (No predictor has statistically significant relation on wage growth).

$$H_1 : \beta_j \neq 0$$

(At least one predictor has a statistically significant relation on wage growth).

The t-test is given by;

$$t_j = \frac{\beta_j}{SE(\beta_j)} \quad (6)$$

In linear regression, confidence intervals for the coefficients are typically constructed using the t statistics, and are given by:

$$\beta_0 \pm t_{1-\alpha/2}SE(\beta_0) \quad \text{and} \quad \beta_j \pm t_{1-\alpha/2}SE(\beta_j)$$

And, the corresponding 95%CI for coefficient is given by;

$$95\%CI(\beta_j) = (\beta_0 - t_{1-\alpha/2}SE(\beta_0), \beta_0 + t_{1-\alpha/2}SE(\beta_0))$$

The overall significance is tested by F-test, given by;

$$F = \frac{SSR/k}{SSE/(n-k-1)} \quad (7)$$

Where, k is the number of predictors and n be the total number of observations.

Here,

$$R^2 = 1 - \frac{SSE}{SST}$$

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

And,

$$VIF = \frac{1}{1 - R^2} \quad (8)$$

Software Used

Data entry, cleaning, and preliminary organization were carried out in Microsoft Excel, which was also used to generate descriptive charts and visualize trends in the data. Statistical analysis, including model estimation and diagnostic testing, was performed using Stata 18.0.

RESULTS

Comparative Growth of GDP, Inflation, and MW

The annual as well as cumulative percentage changes in GDP, inflation, and the MW with reference to the based year considered for 1965 year are presented in Table 1. It illustrates both the level and trend of each series over time. Cumulatively, the MW increased by approximately 530.4% between 1965 and 2025, compared with a cumulative increase of about 234.7% in real GDP and 460.0% in inflation over the same period. Annual MW growth was also the most unstable of the three series (mean 8.8%, SD 14.04%), reflecting the irregular, multi-year nature of Nepal's wage-revision cycle rather than smooth annual adjustment. This pattern indicates that while the overall trend in MW has risen sharply relative to GDP and inflation, its year-to-year level has fluctuated considerably rather than following a stable growth path.

Source: MoF, NSO, World Bank Group (World Development Indicators) and MoLESS.

MW, Inflation and GDP

Figure 1 compare the trends in MW, GDP growth, and inflation from 1965 to 2025, with linear trend lines fitted to each series. The MW line shows the steepest upward trajectory, consistent with its highest mean annual growth rate (8.8%) and cumulative increase (530.4%) over the period, compared with GDP (3.9% annual; 234.7% cumulative) and inflation (7.7% annual; 460.0% cumulative). The wider fluctuations in the MW series, reflected in its high standard deviation, indicate that wage revisions occurred irregularly rather than through consistent annual adjustments.

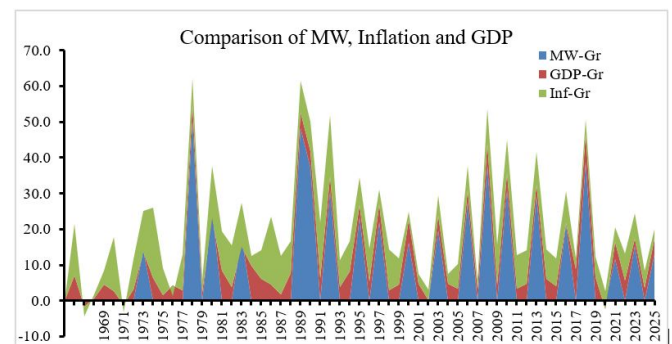


Figure 1. Yearly percentage change in GDP, Inflation and MW during 1965 - 2025

Association of MW, Inflation and GDP

The pairwise correlations coefficient showed a strong association with statistically significant association between GDP growth and inflation ($r = 0.72$, $p = .0002$). Likewise, the MW growth was also showed the strong associ-

Table 1. Percentage change (% Δ) and cumulative percentage change in GDP, inflation, and MW in Nepal, 1965–2023 (Base year 1965 = 0).

Year	% Δ GDP	% Δ Inf.	% Δ MW	Cum. % Δ GDP	Cum. % Δ Inf.	Cum. % Δ MW
1965	0	0	0	0	0	0
1973	14.6	49.9	14.2	14.6	49.9	14.2
1978	19.6	41.5	50.4	34.2	91.4	64.6
1980	0.05	18.2	25.4	34.25	109.6	90.0
1983)	9.1	35.2	18.1	43.35	144.8	108.1
1989	34.1	58.5	48.3	77.45	203.3	156.4
1990	4.6	8.2	37.3	82.09	211.5	193.7
1992	10.5	32.7	30.6	92.59	244.2	224.3
1995	15.5	23.5	23.3	108.09	267.7	247.6
1997	10.4	13.2	22.0	118.49	280.9	269.6
2000	13.6	21.1	16.3	132.09	302.0	285.9
2003	8.9	11.4	19.7	140.99	313.4	305.6
2006	11.5	16.6	27.4	152.49	330.0	333.0
2008	9.5	12.2	37.8	161.99	342.2	370.8
2010	9.3	20.4	30.8	171.29	362.6	401.6
2013	11.6	27.7	29.0	182.89	390.3	430.6
2016	10.4	25.0	21.3	193.29	415.3	451.9
2018	16.6	7.7	38.7	209.89	423.0	490.6
2021	9.1	14.8	11.5	218.99	437.8	502.1
2023	7.6	14.8	15.3	226.56	452.6	517.4
2025	8.1	7.4	13.0	234.67	460.0	530.4

ation and significantly correlated with GDP growth ($r = 0.59$, $p = .0045$). However, the correlation between MW and inflation was weaker and not statistically significant ($r = 0.37$, $p = .0687$). Table 4 illustrated the results of pairwise matrix of Pearson correlation coefficients.

Table 2. Correlation matrix of annual GDP growth, inflation, and MW growth.

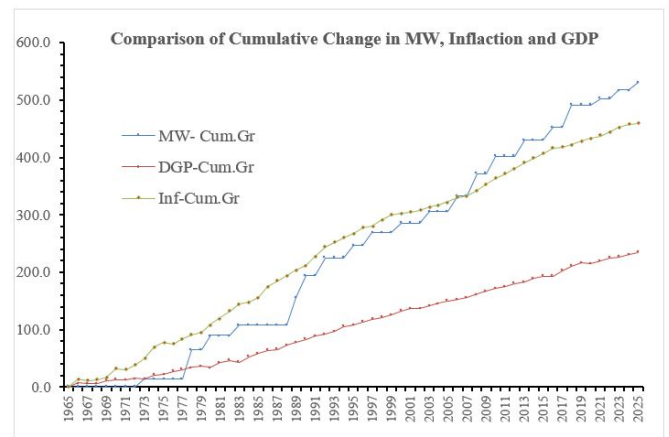
Variables	GDP Growth	Inflation	MW
GDP	1.00	—	—
Inflation	0.72*	1.00	—
MW	0.59*	0.37	1.00
	0.0045	0.0687	1.00

Cumulative Change in MW, Inflation and GDP

Figure 2 shows the cumulative percentage change in MW, inflation, and GDP from 1965 to 2025. Legal MW rose by approximately 530.4% over the period, exceeding the cumulative increases in inflation (460.0%) and GDP (234.7%), showing that wages grew slightly faster than prices over the long run.

Determinants of MW Growth: A Linear Regression Analysis

The overall model was statistically significant as the MW growth on GDP growth and inflation was statistically significant, $F(2, 18) = 4.95$, $p = .019$. The total variation explained by the predictors in MW was approximately 35.5% ($R^2 = .355$). Table 3 illustrates that the GDP growth was a significant positive predictor of MW growth ($\beta = 1.11$, $p = .033$). It indicated that 1%-point increase in GDP growth was associated with roughly a 1.11%-point increase in MW growth.

**Figure 2.** Cumulative percentage change in GDP, Inflation and MW during 1965 - 2025

In contrast, inflation was not a statistically significant predictor ($\beta = -0.04$, $p = .86$). It showed over this period, Nepal's MW wage revisions tracked underlying economic growth more closely than they tracked price levels.

Table 3. Regression coefficients predicting annual MW growth from GDP growth and inflation

Variable	Coeff.	SE	t	p	95% CI
GDP	1.1125	0.482	2.31	0.033	(0.099, 2.126)
Inflation	-0.0414	0.231	-0.18	0.860	(-0.527, 0.444)
Constant	13.7327	4.547	3.02	0.007	(4.181, 23.285)
F(2,18)	4.95			0.0194	
R ²	0.3547				
Adj. R ²	0.2830				

Diagnostics

Multicollinearity was used to assess the variance inflation factor (VIF). Both GDP growth and inflation had VIF values of 2.08 (mean VIF = 2.08), indicating no evidence of problematic multicollinearity among the explanatory variables.

For heteroskedasticity, the Breusch-Pagan/Cook-Weisberg test was used, which indicated no evidence of heteroskedasticity in the regression model ($p = .357$), supporting the assumption of constant error variance.

Augmented Dickey-Fuller (ADF) test was used to assess the stationarity of the time series and serial correlation. The Breusch-Godfrey test indicated no statistically significant first-order serial correlation at the 5% ($p = .053$).

In case of model specification, there was no statistically significant evidence of model misspecification ($p = .558$). The RESET test did not indicate that the model required additional nonlinear terms based on the fitted values, suggesting that the specified linear functional form was adequate.

Assessment of Q-Q Plot

The Q-Q plot of the regression residuals showed that almost all points approximately followed the 45-degree reference line. It indicated that the residuals were found to be approximately normally distributed, supporting normality of assumption of the model.

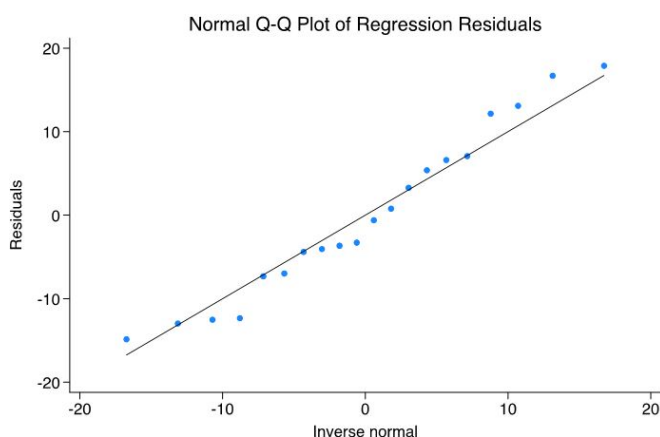


Figure 3. Normality assumption assessment of the regression model.

Structural and Sectoral Assessment

Beyond the quantitative analysis and the statistical results, the policy and documentary and other relevant documents identified several structural patterns. For instance, the productivity-wage preference is found a mismatch evident. Even the wages have risen periodically, productivity growth in labor-intensive sectors such as manufacturing and construction has found lagged. It has been raising concerns about the sustainability of wage increases for small and medium-sized enterprises (SMEs) operating on a thin margin (World Bank, 2026). On the other hand, MW coverages are more profoundly concentrated in urban, formal sectors, large and medium industries (International Labour Organization, 2025). But

rather less underscored in agriculture, SMEs, cottage and micro enterprises, which employs a large share of Nepal's workforce (Rastriya Shramik Mahasangh Nepal, 2024). Informal activities such as domestic work, transport, roadside restaurants and cottage industry are largely fall outside statutory protection. Moreover, informal employment, estimated at 60–70% of the workforce (National Statistics Office, 2020), remains the single largest constraint on the effectiveness of MW policy, a pattern that mirrors informality challenges documented elsewhere in South Asia (Rani et al., 2013).

These structural gaps also point to considerable variation across sectors, which a uniform statutory rate does not fully accommodate. The ILO's own minimum wage guidance recognizes differentiated minimum wage systems set by sector, occupation, or geography as an accepted policy option rather than a departure from good practice, provided any differentiation is based on objective grounds (International Labour Organization, 2017). Regional precedent exists for this approach: Bangladesh, for instance, maintains a dedicated wage board for its Ready-Made Garment sector, distinct from its national minimum wage mechanism (International Labour Organization, 2023). Such precedent suggests that sector-specific wage floors are institutionally feasible within South Asia, and could be similarly explored for Nepal's high-informality sectors such as agriculture, domestic work, and cottage industry, where the current uniform rate has proven difficult to enforce (Rastriya Shramik Mahasangh Nepal, 2024).

Most importantly, gender disparities still persist in the labor market. Although the MW law applies equally to men and women, women are disproportionately represented in lower-paying and informal (Rastriya Shramik Mahasangh Nepal, 2024). Nepal's Labor Act (2017) legally binds employers to pay equal for equal work irrespective of gender and sex. The legal framework prohibits wage discrimination on the grounds of sex, however enforcement of this provision found weak, particularly in the informal sectors and rural economy.

DISCUSSION

The findings of the study suggest that Nepal's MW revisions have been driven more by real economic growth than by inflation control. A finding that is broadly consistent with the observation that wage-fixing in Nepal follows periodic, negotiated revisions through the MW Fixation Committee rather than automatic indexation to prices, unlike systems such as France's SMIC (Paun et al., 2021) or the United Kingdom's National Living Wage (Low Pay Commission, 2020). The relatively weak and non-significant relationship with inflation is consistent with prior findings that Nepal's wage adjustments often lag behind the rising cost of living, reducing real wages between revisions (Raut, 2019). At the same time, the strong cumulative growth of the MW relative to GDP indicates that, in aggregate nominal terms, legal wage floors have not been left behind by the broader economy. However, the central challenge instead lies in translating nominal gains into consistent enforcement and broad

coverage.

The tension between nominal wage growth and its practical reach is compounded by concerns over productivity. The theoretical expectation that wage increases correspond with rising productivity is not fully supported in Nepal's case. The mismatch between wage growth and productivity gains observed in Nepal diverges from what "efficiency wage theory" would predict (Shapiro and Stiglitz, 1984; Katz, 1986). The productivity-wage mismatch identified in Nepal's industrial sector raises the risk that wage increases outpacing productivity growth could encourage layoffs, increased automation, or a shift of employment into informality among cost-constrained SMEs. This challenging situation, prevailing in Nepal is consistent with international evidence showing that MW effects are highly context-dependent (Bhorat et al., 2017), which may challenge industrial growth and employment generation.

This risk repeats a lot of debates in the job markets. For instance, Paun et al. (2021) found a negative employment effects of MW increases in EU panel data, Campos Vázquez et al. (2018) find no such adverse effect in Mexico, and Fang and Lin (2015) similarly find limited employment loss in China. The divergence across these studies underscores that outcomes depend on complementary factors, mainly productivity investment, enforcement strength, and the size of the informal sector rather than on MW levels alone.

Although these risks, increases in MW found to have contributed to reductions in extreme poverty, evidence linking MWs to poverty alleviation (Jadoon et al., 2021). However, in Nepal the gains or increase in MWs have been unevenly distributed among rural and informal workers. Particularly, women have found comparatively little benefit, widening the gap between formal and informal, as well as urban and rural labor markets.

Regionally, Nepal's periodic national wage-setting approach contrasts with India's fragmented, multi-tiered system (International Labour Organization, 2014) and Bangladesh's sector-specific negotiated wages (Kabeer et al., 2020), however, shares of enforcement gap are persistent similar in both countries. Compared with East Asian economies such as Japan, South Korea, and China; where wage floors are more closely tied to productivity and supported by stronger monitoring capacity (Fang and Lin, 2015; Baek et al., 2021), Nepal's MWs neither tied with productivity nor remains to meet purchasing-power-parity terms even though the national minimum wage has periodically adjusted since 1965.

Taken together, the quantitative and comparative findings of the study point to enforcement capacity as the binding constraint on the effectiveness of Nepal's MW policy. The legal wage floor does not reach a large share of Nepal's workforce in practice, given that the informal sector comprises roughly 60–70% of employment (National Statistics Office, 2020). This finding aligns with the broader South Asian pattern in which legal MWs exist but compliance, particularly in agriculture and small-scale enterprise, remains weak (Rani et al., 2013). This underscores that although the MW has kept pace with

GDP growth and has been only weakly linked to inflation, its real impact ultimately hinges less on how the wage floor is set and more on whether it is enforced across the workforce it is meant to protect.

LIMITATIONS

The study has several limitations because it relies only on secondary, publicly available data. It may restrict to capture informal labor-market dynamics directly. Some of the important predictors relating to individual workforce capacity such as their sociodemographic, skills and geographical context are not specifically incorporated. Variation in wage-reporting standards across countries constrains cross-national comparability. The absence of household-level data limits causal inference between minimum wage changes and poverty outcomes in Nepal. Therefore, the regression results reported here should be interpreted as descriptive associations rather than causal effects. Moreover, sectoral enforcement data for Nepal are scarce, requiring reliance on proxy indicators, which may bias and also insufficient for informal-sector employment shares.

CONCLUSION

Nepal's MW has risen substantially in nominal terms since its introduction in 1965, and its growth is significantly and positively associated with GDP growth, though not with inflation. These results indicate that Nepal's wage-setting practice has been reasonably responsive to real economic conditions but has not found consistently protected worker's purchasing power against rising prices. More fundamentally, the effectiveness of Nepal's MW policy is constrained less by the adequacy of statutory rates than by structural gaps in coverage and enforcement. A large informal sector, weak labor inspection, sectoral disparities, and persistent gender inequities limit the reach of wage protections to a subset of the workforce, primarily in urban, formal industry.

The findings point to investment in workforce skills enhancement and productivity so that enterprises, particularly SMEs, can sustainably absorb wage increases without resorting to layoffs or informalization.

Wage protections should also be extended to informal and rural workers, with MWs differentiated by sector and geographic setting. Separate wage floors could be more implementable for agriculture, informal sectors, and rural labor markets. Such differentiation would better reflect variation in living costs, productivity, and labor-market conditions across county complemented by expanded social security coverage.

MW enforcement should be strengthened through expanded labor-inspection capacity, alongside legally binding penalties for non-compliance. Likewise, wage-setting should be guided by a mechanism grounded in well-accepted theoretical frameworks and objective growth indicators, rather than ad hoc bargaining resembling a "fish market" style of negotiation, to ensure a more transparent and evidence-based process.

Taken together, these recommendations point toward a rights-based, evidence-driven, and sector-sensitive ap-

proach to MW determination. This framing positions the MW not merely as a statutory figure but as an instrument of social protection and inclusive economic growth, consistent with the ILO's decent work and living wage framework.

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CONFLICT OF INTEREST

The authors have no conflict of interest.

AUTHOR'S CONTRIBUTION

IKS conceptualized and designed the study, acquired and analyzed the data, and completed the final manuscript. NRP provided guidance for analytical model building and results framing, and reviewed the literature. ERK supported the research design, data mapping, and added the productivity component to the manuscript. All authors contributed to revising the manuscript and approved the final version for publication.

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ETHICAL CONSIDERATION

This study relies exclusively on secondary data obtained from publicly available and properly cited sources (MoF, MoLESS, NSO, and the World Bank Group); therefore, no primary data collection involving human participants was undertaken, and no additional ethical approval was required.

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Development of an Empirical Bayesian Single-Attribute Acceptance Sampling Plan for Food Quality using a Beta–Binomial Model

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ABSTRACT

Background: The variation in the quality characteristics of a manufactured product across different lots may arise from variation within lots (batches) or from variation between lots. Within-lot variation is addressed through statistical process control, whereas between-lot variation, which may arise from shifts in the process center, is more appropriately evaluated through acceptance sampling. Traditional acceptance sampling plans in food quality assurance rely primarily on frequentist frameworks that evaluate lots in isolation, thereby ignoring valuable historical process information and failing to adequately address uncertainty in the process center. In food manufacturing, where inspection is often destructive and costly, incorporating historical process information into acceptance decisions is essential. Bayesian methods provide an effective framework for integrating prior knowledge with current sample data.

Objective: This study develops and evaluates a Bayesian single-attribute acceptance sampling plan using a Beta prior distribution to model the lot proportion nonconforming, p , and benchmarks its performance against classical frequentist baselines under realistic food safety constraints.

Methods: The number of nonconforming items, Y , in a sample of size n is assumed to follow a Binomial distribution, $Y \sim \text{Binomial}(n, p)$, with a conjugate Beta prior, $p \sim \text{Beta}(\alpha, \beta)$. Method of Moments estimation was applied to simulated continuous production records comprising $K = 100$ lots to calibrate an empirical baseline prior, $\text{Beta}(2, 198)$. The proposed empirical plan was systematically benchmarked against three noninformative priors: Haldane, $\text{Beta}(0.001, 0.001)$; Jeffreys, $\text{Beta}(0.5, 0.5)$; and Uniform, $\text{Beta}(1, 1)$, as well as three informative regimes: Low Compliance, $\text{Beta}(2, 38)$; Conservative, $\text{Beta}(5, 95)$; and Optimistic, $\text{Beta}(1, 199)$. Plan performance was evaluated using Operating Characteristic (OC) curves, Bayes risk, Average Total Inspection (ATI), and prior sensitivity analysis.

Results: The empirical $\text{Beta}(2, 198)$ prior reduced the required sample size by 33.3% while maintaining Producer's Risk ($\alpha_0 \leq 0.05$ at $\text{AQL} = 1\%$) and Consumer's Risk ($\beta_0 \leq 0.10$ at $\text{LQL} = 5.0\%$) within the specified boundaries. The OC curve of the empirical Bayesian plan closely matched the discrimination profile of the classical frequentist baseline while requiring substantially less inspection effort. Sensitivity analysis confirmed the robustness of the proposed plan to perturbations in the prior hyperparameters, while dynamic Bayesian updating provides a mechanism for maintaining consumer protection when process conditions change.

Conclusion: The conjugate Beta–Binomial Bayesian framework improves sampling efficiency and decision-making by combining historical process stability with current inspection data. The proposed adaptive approach provides a mathematically sound framework for food quality assurance, potentially reducing laboratory inspection costs while maintaining specified food safety requirements.

Key words: Bayesian acceptance sampling, Beta prior distribution, Beta-binomial model, Food quality assurance, Operating characteristic curve, Bayes risk

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INTRODUCTION

In manufacturing industries, raw materials are typically taken and processed in production lots. Product quality can vary due to differences in raw materials, production processes, machinery, operating conditions, handling systems, and environmental factors. Measured quality characteristics may vary due to differences within a lot or between lots. While within-lot variation is typically monitored and controlled using Statistical Process Control (SPC), between-lot variation affects the overall quality of submitted batches and must be evaluated through acceptance sampling. Particularly in food production, 100% inspection is rarely feasible because destructive testing

and laboratory analysis are often costly or impractical. Thus, acceptance sampling provides an effective statistical method for deciding whether to accept or reject a production lot based on sample inspection.

Classical acceptance sampling plans, such as MIL-STD-105E (Department of Defense, 1989) and ISO 2859-1 (International Organization for Standardization, 1999), treat the proportion of nonconforming items in a lot, p , as a fixed but unknown parameter. These plans evaluate submitted lots using predefined sampling schemes based on specified quality levels, making decisions from the observed number of nonconforming items in the current sample. They are designed to satisfy predetermined re-

quirements for Producer's Risk (α), Consumer's Risk (β), Acceptance Quality Level (AQL), and Limiting Quality Level (LQL). Although these methods have been widely adopted in industrial quality assurance, they treat each lot in isolation and do not incorporate historical information regarding process performance and supplier reliability into the sampling decision.

In practice, variations in quality arise not only from sampling fluctuations but also from changes in the production process. The quality of lots produced at different times often varies because process centers can shift due to factors such as raw material batches, machinery wear and tear, changes in operators, or environmental conditions. Consequently, variation between lots may be greater than variation within lots (Facchinetti et al., 2022). Treating all submitted lots as independent observations from a fixed process overlooks the inherent uncertainty associated with process variation. This may result in inefficient sampling plans, including unnecessarily large sample sizes and suboptimal decisions regarding lot acceptance (Facchinetti et al., 2022; Vijayaraghavan and Sakthivel, 2010).

Bayesian statistics provides a coherent probabilistic framework for dealing with uncertainty in process quality. It combines prior information with observed data to update knowledge and support quality assessment and decision-making. The basic idea underlying the Bayesian method is that uncertainty about an unknown quantity of interest is represented probabilistically through plausible values of that quantity (Khatiwada and Sthapit, 2008). In acceptance sampling, treating the lot proportion nonconforming, p , as a random variable rather than a fixed parameter allows the Bayesian approach to represent pre-existing process knowledge through a prior probability distribution. This prior distribution is updated using sample inspection data through Bayes' theorem, yielding a posterior distribution that reflects the combined information. In single-attribute acceptance sampling, where the number of nonconforming items follows a Binomial distribution, the Beta distribution serves as the natural conjugate prior. The resulting Beta-Binomial model enables the combined use of historical production records and current inspection outcomes, thereby improving sampling plan parameters and reducing unnecessary inspection effort.

The application of Bayesian acceptance sampling is particularly important in food manufacturing, where quality inspection often involves destructive testing, microbiological examination, chemical analysis, and significant laboratory costs. Reliable decisions regarding food safety requirements must therefore be made while minimizing unnecessary inspection efforts. When historical production records demonstrate consistent manufacturing performance, incorporating prior process knowledge can significantly reduce inspection costs without compromising consumer protection (Khatiwada, 2009).

The application of Bayesian methods in quality control and acceptance sampling has grown considerably over time. Early contributions by Hald (Hald, 1960, 1964, 1968, 1977, 1981), Wetherill and Chiu (Wether-

ill and Chiu, 1974), Pande (Pande, 1974), Maxim et al. (Maxim et al., 1976), Calabrese (Calabrese, 1995), Wolfinger (Wolfinger, 1998), Chen and Lam (Chen and Lam, 1999), Dumić et al. (Dumić et al., 2006), and Colosimo and del Castillo (Colosimo and del Castillo, 2007) laid important theoretical foundations for Bayesian approaches to quality control and acceptance sampling. More recent studies (Ciccarelli et al., 2025, 2026; Dewi et al., 2022; Facchinetti et al., 2022; Hafeez and Aziz, 2022; Hafeez et al., 2023, 2025; Kadir and Rahi, 2023) have further demonstrated the potential benefits of integrating historical process information into acceptance decisions through Bayesian methods.

Despite these developments, relatively few studies have systematically examined the impact of alternative Beta prior specifications on the performance of Bayesian acceptance sampling plans designed for food quality assurance. In particular, the comparative performance of data-driven empirical priors against noninformative baselines and informative operational regimes under realistic production structures has not been fully assessed. This represents an important methodological gap because the choice of prior distribution can substantially influence posterior inference, sampling requirements, acceptance decisions, and the associated inspection costs.

To address this gap, this study develops and evaluates an empirical Bayesian single-attribute acceptance sampling framework for food quality control using the conjugate Beta-Binomial model. Binomial sample data are generated from simulated continuous production records comprising $K = 100$ lots to represent inspection outcomes and to calibrate an empirical Beta(2, 198) prior using the Method of Moments. To systematically investigate the impact of prior choice, the proposed empirical prior is evaluated alongside a range of noninformative baselines, including the Haldane prior, Beta(0.001, 0.001), the Jeffreys prior, Beta(0.5, 0.5), and the Uniform prior, Beta(1, 1), as well as informative operational regimes, including Low Compliance, Beta(2, 38); Conservative, Beta(5, 95); and Optimistic, Beta(1, 199). All plans are rigorously evaluated against classical frequentist baselines using multiple performance measures, including optimal sample size (n), acceptance threshold (c), Operating Characteristic (OC) curves, Average Total Inspection (ATI), Bayes risk under comprehensive cost models, and prior sensitivity analysis. Furthermore, the proposed framework suggests the potential for automated integration with Laboratory Information Management Systems (LIMS), enabling historical laboratory and production information to be incorporated into future acceptance-sampling decisions.

METHODOLOGY

Model Formulation and Likelihood

Let a submitted lot of food products contain a large batch size N , from which a random sample of n items is selected for quality testing. Let Y represent the number of nonconforming (defective) items observed in the sample. Assuming $N \gg n$, the random variable Y given nonconforming p follows a Binomial distribution:

$$P(Y = y | p) = \binom{n}{y} p^y (1-p)^{n-y}, \quad y = 0, 1, 2, \dots, n \quad (1)$$

Specification of the Beta Prior Distribution

To model prior knowledge regarding lot quality p , the Beta distribution with shape parameters $\alpha > 0$ and $\beta > 0$ is specified as the prior probability density function:

$$\pi(p; \alpha, \beta) = \frac{1}{B(\alpha, \beta)} p^{\alpha-1} (1-p)^{\beta-1}, \quad 0 \leq p \leq 1. \quad (2)$$

where $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ is the Beta function. The expected prior proportion nonconforming $E[p]$ and prior variance $\text{Var}(p)$ are expressed as:

$$E[p] = \frac{\alpha}{\alpha + \beta}. \quad (3)$$

$$\text{Var}(p) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}. \quad (4)$$

Empirical Parameter Estimation via Method of Moments

Historical production lots can be treated as independent realizations of a Beta-distributed process parameter, p_i , where α and β characterize the shape of the underlying proportion nonconforming across lots. For each historical lot i , the observed fraction non-conforming is given by

$$\hat{p}_i = \frac{y_i}{n_i}, \quad i = 1, 2, \dots, K, \quad (5)$$

where y_i denotes the number of nonconforming items observed and n_i the number of items inspected in lot i .

Given historical lot inspection records across K previous lots with observed sample proportions, the hyperparameters α and β of the Beta distribution are estimated using the method of moments, where the theoretical mean and the variance of the Beta distribution are equal to the corresponding sample mean $\bar{p}$ and sample variance s^2 . This yields the following closed-form estimators (Rahman and Amin, 2024):

$$\hat{\alpha} = \bar{p} \left[\frac{\bar{p}(1-\bar{p})}{s^2} - 1 \right]. \quad (6)$$

$$\hat{\beta} = (1-\bar{p}) \left[\frac{\bar{p}(1-\bar{p})}{s^2} - 1 \right]. \quad (7)$$

where

$$\bar{p} = \frac{1}{K} \sum_{i=1}^K \hat{p}_i \text{ and } s^2 = \frac{1}{K-1} \sum_{i=1}^K (\hat{p}_i - \bar{p})^2.$$

When $Sp^2 \geq \bar{p}(1-\bar{p})$, the Method of Moments estimator for Beta parameters can fail (yielding negative values). The historical lot variation in this study satisfies $Sp^2 < \bar{p}(1-\bar{p})$.

The resulting estimates $\hat{\alpha}$ and $\hat{\beta}$ constitute an empirically derived, data-driven prior that reflects the historical process behaviour across lots, and were subsequently used to update the posterior distribution of the lot proportion nonconforming upon incorporation of current sample data.

Posterior Distribution and Predictive Probability

Posterior Distribution

As sample inspection data yields y non-conforming items out of n inspected units, Bayes' Theorem combines the Binomial likelihood $P(Y = y | p)$ with the Beta prior density to derive the updated posterior distribution:

$$\pi(p | y) = \frac{P(y | p)\pi(p)}{\int_0^1 P(y | p)\pi(p) dp}. \quad (8)$$

Substituting the Binomial probability mass function and the Beta probability density function gives

$$\pi(p | y) \propto \binom{n}{y} p^y (1-p)^{n-y} \frac{p^{\alpha-1} (1-p)^{\beta-1}}{B(\alpha, \beta)}. \quad (9)$$

By dropping terms that do not depend on p ,

$$\pi(p | y) \propto p^{\alpha+y-1} (1-p)^{\beta+n-y-1}. \quad (10)$$

This kernel matches the functional form of a Beta distribution, which is the posterior distribution of the non-conforming proportion p given y , $(p | y)$.

$$p | y \sim \text{Beta}(\alpha', \beta'). \quad (11)$$

where, $\alpha' = \alpha + y$, $\beta' = \beta + n - y$.

The expected posterior proportion nonconforming (posterior mean) is

$$E[p | y] = \frac{\alpha'}{\alpha' + \beta'}. \quad (12)$$

The posterior mode is

$$\text{Mode}(p | y) = \frac{\alpha' - 1}{\alpha' + \beta' - 2}, \quad \alpha' > 1, \beta' > 1. \quad (13)$$

It is not easy to calculate the posterior median for $\text{Beta}(\alpha', \beta')$ using a simple algebraic equation. It is calculated using the inverse Beta CDF (the quantile function) evaluated at $p = 0.5$. In this work it is calculated using the `qbeta` function in R software.

The posterior standard deviation is

$$SD(\sigma_p) = \sqrt{\frac{\alpha'\beta'}{(\alpha' + \beta')^2(\alpha' + \beta' + 1)}}. \quad (14)$$

The 95% Credible Interval is the interval between the 2.5th and 97.5th percentiles of $\text{Beta}(\alpha', \beta')$.

Predictive Distribution

To determine whether a lot is accepted, we calculate the predictive probability of observing y nonconforming items in a future sample of size n , integrating over all possible values of p :

$$P(Y = y) = \int_0^1 P(Y = y | p)\pi(p) dp. \quad (15)$$

i.e.,

$$P(Y = y) = \binom{n}{y} \frac{B(y + \alpha, n - y + \beta)}{B(\alpha, \beta)}. \quad (16)$$

It shows that the marginal predictive distribution of nonconforming Y follows a Beta-Binomial distribution.

Under a single-attribute sampling plan (n, c) , a submitted lot is accepted if the observed number of nonconforming items y does not exceed the acceptance threshold c . In other words, n is the required sample size and c is the maximum allowable number of nonconforming items. The overall probability of lot acceptance (P_a) is the cumulative Beta-Binomial probability. The probability of lot acceptance P_a under an acceptance criterion ($y \leq c$) is given by:

$$P_a = P(Y \leq c) = \sum_{y=0}^c \binom{n}{y} \frac{B(y + \alpha, n - y + \beta)}{B(\alpha, \beta)}. \quad (17)$$

Acceptance Sampling Plan Optimization Parameters

To establish a single-attribute Bayesian acceptance sampling plan, the plan parameters (n, c) must satisfy two fundamental risk constraints defined at specific quality levels:

1. **Producer's Risk Constraint:** The probability of accepting a good lot at the Acceptable Quality Level (AQL = p_1) must be at least $1 - \alpha_0$.
2. **Consumer's Risk Constraint:** The probability of accepting a bad lot at the Limiting Quality Level (LQL = p_2) must not exceed β_0 .

Mathematically, using the Beta-Binomial predictive acceptance probability P_a , these constraints are expressed as:

$$P_a(p_1) = \sum_{y=0}^c \binom{n}{y} \frac{B(y + \alpha, n - y + \beta)}{B(\alpha, \beta)} \geq 1 - \alpha_0 \quad (18)$$

$$P_a(p_2) = \sum_{y=0}^c \binom{n}{y} \frac{B(y + \alpha, n - y + \beta)}{B(\alpha, \beta)} \leq \beta_0 \quad (19)$$

Average Total Inspection

For a single sampling plan with sample size n , acceptance number c , and lot size N , the Average Total Inspection (ATI) function incorporates both the sample size n and the full inspection of rejected lots (assuming rectifying inspection, where rejected lots are 100% inspected and non-conforming items replaced):

$$ATI(p) = n + (N - n)[1 - P_a(p)]. \quad (20)$$

At ideal lot quality ($p \rightarrow 0$), $P_a(p) \rightarrow 1$ and $ATI(0) = n$ (minimum inspection load)

At poor lot quality ($p \rightarrow 1$), $P_a(p) \rightarrow 0$ and $ATI(1) = N$ (full lot screening)

In practice, many (n, c) pairs can satisfy the Consumer's Risk and Producer's Risk constraints. For optimization procedure, the unique optimal plan (n^*, c^*)

is selected by minimizing the Average Total Inspection (ATI). The optimization problem is solved numerically by finding the minimum integer sample size n^* and acceptance threshold c^* that satisfy:

$$\min_{n \in \mathbb{N}} ATI(E[p]) \quad \text{subject to} \quad \begin{cases} P_a(p_1) \geq 1 - \alpha_0, \\ P_a(p_2) \leq \beta_0. \end{cases} \quad (21)$$

Where $E[p] = \frac{\alpha}{\alpha + \beta}$ is the expected prior process nonconforming rate derived from the Beta(α, β).

Simulation Scheme and Data Generation Procedures

To evaluate the performance of the proposed Bayesian acceptance sampling plan under controlled operational conditions, a Monte Carlo simulation framework was implemented in R (v4.3.2). Binomial inspection data were generated to simulate continuous lot-by-lot food quality verification. The simulation procedure followed a four-step protocol:

1. **Lot Nonconforming Rate Generation:** A sequence of $K = 100$ historical production lots was simulated, where the true underlying lot nonconforming proportion p_k for batch k ($k = 1, 2, \dots, K$) was drawn from a true process capability distribution. The true process capability distribution was parameterized with mean $\mu_p = 0.010$ and standard deviation $\sigma_p = 0.007$. This choice reflects an in-control manufacturing environment targeted at an Acceptable Quality Level (AQL = 1.0%).
2. **Sampling and Nonconforming Observation:** From each simulated lot k , a random sample of size $n_k = 100$ items were drawn, and the number of nonconforming items y_k was generated via a Binomial process: $y_k \sim \text{Binomial}(n_k, p_k)$.
3. **Empirical Prior Calibration:** The sample mean $\bar{p}$ and sample variance s_p^2 across the K historical inspection records were computed. Using Method of Moments estimation, the empirical Beta prior hyperparameters $(\hat{\alpha}, \hat{\beta})$ were fitted as given in Equations (6) and (7).
4. **Plan Optimization:** Under each prior regime, optimal sampling plan parameters (n, c) were derived by solving the dual risk constraints:

$$P_a(p_1) \geq 0.95 \quad \text{at AQL} = 0.01 \quad (22)$$

$$P_a(p_2) \leq 0.10 \quad \text{at LQL} = 0.05 \quad (23)$$

Operating Characteristic Curve for Performance Evaluation

The probability of lot acceptance $P_a(p)$ for each plan is calculated using its corresponding predictive density function.

For the frequentist plan:

$$P_a^F(p) = \sum_{y=0}^c \binom{n}{y} p^y (1-p)^{n-y}. \quad (24)$$

For Bayesian plans:

$$P_a^B(p) = \sum_{y=0}^c \binom{n}{y} \frac{B(y + \alpha, n - y + \beta)}{B(\alpha, \beta)}. \quad (25)$$

Bayes Risk Formulation

To obtain Bayes Risk, first the conditional expected loss $L(n, c, p)$ for a lot of size N and quality p is obtained. The loss function contains four primary cost components: sampling cost (C_1), sample defect (nonconforming) rework cost (C_2), screening rework cost (C_3), and external failure cost (C_4) (Montgomery, 2020).

Sampling Cost

Sampling cost C_1 is the cost of inspection of n units. if c_s is the unit inspection cost $C_1 = c_s \cdot n$.

Sample Defect Rework Cost

Sample defect (nonconforming) rework cost C_2 is the cost of rectifying nonconforming items in the sample. If the internal rework cost is C_i , then $C_2 = C_i \cdot n \cdot p$.

Screening Rework Cost

Screening rework cost C_3 is the rework cost for nonconforming items discovered during 100% screening of rejected lots. Thus, $C_3 = C_i \cdot p \cdot (N - n) [1 - P_a(p)]$.

External Customer Failure Cost

External Customer Failure Cost C_4 is the penalty cost associated with undetected nonconforming items in accepted lots. $C_4 = C_a \cdot (N - n) p P_a(p)$, where C_a represents the field replacement or warranty cost ($C_a \gg C_i$).

The sum of these conditional expected losses yields the conditional Bayes Risk function:

$$R(p; n, c) = c_s n + c_i p n + c_i p (N - n) [1 - P_a(p)] + c_a p (N - n) P_a(p) \quad (26)$$

The overall expected Bayes Risk, $B(n, c)$, is obtained by integrating the conditional loss function over the process proportion nonconforming prior distribution:

$$B(n, c) = \int_0^1 R(p; n, c) \pi(p) dp. \quad (27)$$

The optimal Bayesian sampling plan (n^*, c^*) is determined by finding the integer pair that minimizes the total expected risk:

$$(n^*, c^*) = \arg \min_{(n, c)} B(n, c). \quad (28)$$

Prior Sensitivity Analysis and Robustness

To ensure the decision rules remain robust against parameter mis-specification or historical process shifts, a prior sensitivity analysis framework is built. The sensitivity analysis evaluates the impact of mis-specifying prior hyperparameters (α, β) because hyperparameter estimation relies on historical quality records. Minor sample bias or process drift could lead to mis-specified Beta parameters. The robustness evaluation procedure follows four formal steps:

1. **Baseline Parameterization:** Establish the baseline prior distribution $\pi_0(p) \sim \text{Beta}(\alpha_0, \beta_0)$ and its corresponding optimal sampling plan (n_0, c_0) derived from empirical or reference data.
2. **Controlled Hyperparameter Perturbation:** Systematically adjust the hyperparameters to simulate potential estimation errors across two structural dimensions:
 - (a) **Location Shift (Process Mean Drift):** Vary the prior mean $E[p] = \frac{\alpha}{\alpha + \beta}$ across a plausible operational range $\pm 20\%$ to $\pm 50\%$, holding precision $M = \alpha + \beta$ constant.
 - (b) **Precision Shift (Prior Information Weight):** Vary the prior weight/precision M across different confidence levels while holding the process mean $E[p]$ constant.
3. **Plan Re-optimization:** Under each perturbed prior $\pi_k(p)$, solve for the newly required optimal sampling plan (n_k^*, c_k^*) that minimizes expected Bayes Risk subject to classical operating characteristic constraints $[P_a(p_1) \geq 0.95; P_a(p_2) \leq 0.01]$

$$(n^*, c^*) = \arg \min_{(n, c)} BR_{\alpha, \beta}(n, c). \quad (29)$$

4. **Cross-Risk Penalty Evaluation:** Evaluate the execution of the baseline plan (n_0, c_0) under each perturbed prior $\pi_k(p)$. Robustness is evaluated by executing the baseline plan (n_0, c_0) under each perturbed prior to quantify the Bayes Risk penalty. Robustness is confirmed if the relative Bayes Risk penalty, $\Delta B = B_k(n_0, c_0) - B_k(n_k^*, c_k^*)$ remains minimal and classical producer/consumer error bounds are not violated.

Laboratory Information Management System (LIMS)

Within a Laboratory Information Management System (LIMS), inspection results can be automatically incorporated into a Bayesian updating framework, whereby each new batch inspection updates the posterior distribution of the lot quality parameter. This dynamic posterior updating supports real-time quality assurance, adaptive acceptance sampling, and data-driven decision making (Boyar et al., 2021).

RESULTS

Parameter Estimation from Simulated Data

The simulated production dataset of $K = 100$ historical lots yielded a sample mean proportion nonconforming of $\bar{x} = 0.0098$ (0.98%) and sample variance $s^2 = 0.000048$. Method of Moments estimation (as per Equations [6] and [7]) yielded empirical shape parameters:

$$\hat{\alpha} = 1.96, \hat{\beta} = 198.04$$

Rounding to integer parameters yielded the baseline empirical prior Beta(2, 198). This confirms that the empirical historical process is tightly clustered around an expected proportion nonconforming of $E[p] = 0.01s$ with low variance.

Comparison of Sample Size Requirements

Prior regimes are classified into non-informative reference baselines and informative operational schemes. The empirical Beta(2, 198) prior derived from simulated (continuous) production records represents realistic historical process capability at target quality (AQL = 1.0%), achieving a 33.3% reduction in required sample inspection size over the classical frequentist baseline while maintaining designated risk limits ($\alpha \leq 0.05$, $\beta \leq 0.10$).

Table 1 summarizes the performance metrics generated by running the R computational model across various prior regimes assuming standard quality target thresholds of AQL = 0.01, LQL = 0.05, $\alpha_0 = 0.05$, $\beta_0 = 0.10$

Analytical Findings from Table 1

The frequentist baseline assumes no prior knowledge (p is unknown but fixed). It evaluates each lot strictly in isolation, requiring the full baseline sample size ($n = 132$,

Table 1. Bayesian vs. Frequentist Acceptance Sampling Parameters (AQL = 0.01, LQL = 0.05, $\alpha_0 = 0.05$, $\beta_0 = 0.10$)

Sampling Plan Scheme	Prior Parameters (α, β)	Prior Regime Type	Expected Prior Proportion Nonconforming $E[p]$	Sample Size (n)	Acceptance Number (c)	Sample Size Reduction vs. Frequentist
Frequentist Baseline (Classical Binomial)	N/A	Classical Non-Bayesian Baseline	N/A	132	3	Baseline (0.0%)
Bayesian Non-Informative (Haldane Reference)	Beta(0.001, 0.001)	Boundary-Concentrated Reference (Sparse)	0.5 (50%)	132	3	0.0%
Bayesian Non-Informative (Jeffreys Invariant)	Beta(0.5, 0.5)	Objective Invariant Reference	0.5 (50%)	131	3	0.8%
Bayesian Non-Informative (Flat Uniform)	Beta(1, 1)	Non-Informative Flat Prior	0.5 (50%)	130	3	1.5%
Bayesian Informative (Low Compliance)	Beta(2, 38)	Informative Elevated (Moderate) Nonconforming Risk ($E[p] = LQL$)	0.05 (5%)	152	4	-15.2% (Increase)
Bayesian Informative (Conservative / Less Optimistic)	Beta(5, 95)	Informative High-Risk Historical Process / Poor History	0.05 (5%)	158	4	-19.7% (Increase)
Bayesian Informative (High Compliance / Optimistic)	Beta(1, 199)	Informative (High-Quality Historical Process)	0.005 (0.5%)	74	1	43.9%
Bayesian – Empirical (Data Driven / Method of Moments)	Beta(2, 198)	Informative (Empirical Data-Calibrated Prior)	0.01 (1%)	88	2	33.3%

$c = 3$) to satisfy both producer's and consumer's risk constraints.

A non-informative Haldane prior, $\text{Beta}(0.001, 0.001)$, represents complete prior ignorance with concentrated density at the extreme boundaries (0 and 1). Because it contributes practically zero prior information to the likelihood, the sampling plan ($n = 132, c = 3$) completely matches the classical frequentist result.

The Jeffreys invariant reference prior, $\text{Beta}(0.5, 0.5)$, functions as the standard non-informative reference prior derived from Fisher transformation. Providing minimal parameter weighting, it yields a plan ($n = 131, c = 3$), almost identical to the frequentist scheme, reducing sample size by only a single unit.

A flat non-informative uniform prior, $\text{Beta}(1, 1)$, assigns equal prior probability across all possible proportions nonconforming $p \in [0, 1]$, yielding $E[p] = 0.50$. It slightly smooths the likelihood curve, resulting in a minor 1.5% reduction in sample size ($n = 130, c = 3$).

An informative low-compliance prior, $\text{Beta}(2, 38)$, reflects a historical process centered at $E[p] = 0.05$, which matches the Limiting Quality Level (LQL = 0.05). Because the historical baseline indicates frequent quality deviations, the plan automatically demands a 15.2% larger sample size ($n = 152, c = 4$) to safeguard consumer health.

An informative conservative (less optimistic, sometimes called pessimistic) prior, $\text{Beta}(5, 95)$, represents a high-risk process with substantial variance around $E[p] = 0.05$. The strong prior weight on elevated proportion nonconforming forces the most stringent inspection plan ($n = 158, c = 4$), increasing sample size by 19.7% over the frequentist baseline to prevent bad lots from being accepted.

An informative high-compliance (optimistic) prior, $\text{Beta}(1, 199)$, models an exceptionally high-compliance historical process with $E[p] = 0.005$. This strong positive prior belief allows the fastest lot disposition, achieving a maximum sample size reduction of 43.9% within the plan ($n = 74, c = 1$).

The empirical historical (data-driven) prior, $\text{Beta}(2, 198)$, is derived directly from simulated production data ($K = 100$ lots) with $E[p] = 0.01$. By precisely maintaining real historical stability, it safely reduces sample size by 33.3% within the plan ($n = 88, c = 2$) while preserving designated risk boundaries.

Posterior Updating under Simulated Inspection

Table 2 depicts the complete information on prior parameters, posterior parameters, and posterior summary statistics (Mean, Median, Mode, Standard Deviation, and 95% Credible Intervals) for all evaluated Bayesian plans, assuming observed sample evidence of zero nonconforming items ($y = 0$) in the respective sample size n . The Classical Frequentist Baseline is included at the top with its exact Binomial Clopper–Pearson 95% Confidence Interval for direct comparison.

For a single submitted batch inspected using the proposed empirical Bayesian sampling plan ($n = 88, c = 2$), suppose that 88 items are tested and no nonconforming items ($y = 0$) are observed. Starting with a $\text{Beta}(2, 198)$

prior distribution for a lot nonconforming proportion (p), the observed inspection result updates the prior to the posterior distribution: $\text{Beta}(2 + 0, 198 + 88) = \text{Beta}(2, 286)$. For the prior distribution $\text{Beta}(2, 198)$, the 95% prior interval for p is $[0.0012, 0.0276]$. Under the proposed empirical plan ($n = 88, c = 2$) derived from the posterior $\text{Beta}(2, 286)$, observing zero nonconforming items in 88 inspected items yields a 95% equal-tailed credible interval for p given y : $[0.0008, 0.0188]$. This reduction in interval width indicates increased precision in estimating the lot nonconforming proportion after incorporating the inspection data (here, zero sample nonconforming items).

The frequentist 95% confidence interval of $[0.0000, 0.0276]$ for $\text{Binomial}(132, 0)$ has an interval width of 2.76 percentage points. Under the posterior, the resulting 95% Bayesian credible interval for p given y is $[0.0008, 0.0188]$, with an interval width of 1.80 percentage points. Clearly, the Bayesian empirical credible interval is 34.8% narrower than the frequentist interval while inspecting 33.3% fewer samples ($n = 88$ versus $n = 132$).

From the findings of the analysis, the empirical historical prior, $\text{Beta}(2, 198)$ is shown as best because it grounds decision-making in real, data-driven historical process capability rather than arbitrary assumptions. It optimizes operational efficiency by safely reducing sample inspection sizes by 33.3% while maintaining strict compliance with both producer and consumer risk constraints.

Operating Characteristic (OC) Curves Analysis

The Operating Characteristic (OC) curves for each sampling plan were evaluated across the true lot proportion nonconforming $p \in [0, 0.10]$. Reference lines in Fig. ?? at AQL = 0.01 and LQL = 0.05 facilitate compliance tracking. From Table ?? and Fig. ??, the following results are observed:

1. Agreement at Target Quality (AQL = 1%):

The frequentist baseline plan and all non-informative priors successfully maintain the producer's risk constraint $\alpha \leq 0.05$, with high acceptance probabilities $P_a(0.01) \geq 0.95$.

Despite reducing the initial sample size by **33.3%** compared to the baseline scheme, the empirical Bayesian scheme ($n = 88, c = 2$) achieves $P_a(0.01) \geq 0.941$. This closely aligns with the target 95% acceptance threshold $P_a(0.01) \geq 1 - \alpha = 0.95$, confirming that the reduced sample size does not unfairly penalize producers operating at standard capability.

Plans derived from the informative Low Compliance scheme ($n = 152, c = 4$) and Conservative scheme ($n = 158, c = 4$) also over-protect the producer risk, with $P_a(0.01) \geq 0.98$, but demand higher sampling cost.

Conversely, the High Compliance scheme ($n = 74, c = 1$) severely under-protects the producer, with $P_a(0.01) = 0.830$, resulting in an unacceptably high Producer's Risk of approximately

$$\alpha = 1 - P_a(0.01) = 0.170.$$

Table 2. Comparative Table: Prior-to-Posterior Statistics vs. Frequentist Baseline

Sampling Plan Scheme	Prior Beta (α, β)	Sample Size (n)	Posterior Beta (α', β')	Mean $E[p y = 0]$	Median	Mode	SD (s_p)	MCSE (Mean)	95% Credible/ Confidence Interval
Frequentist Baseline (Classical Binomial)	N/A	132	N/A	0.0000 (MLE)	N/A	N/A	N/A	N/A	[0.0000, 0.0276] Clopper–Pearson 95% CI
Bayesian Non-Informative (Haldane Reference)	(0.001, 0.001)	132	(0.001, 132.001)	0.0000	0.0000	0.0000	0.0003	8.0×10^{-7}	[0.0000, 0.0227]
Bayesian Non-Informative (Jeffreys Invariant)	(0.5, 0.5)	131	(0.5, 131.5)	0.0038	0.0017	0.0000	0.0053	1.68×10^{-5}	[0.0000, 0.0191]
Bayesian Non-Informative (Flat Uniform)	(1, 1)	130	(1, 131)	0.0076	0.0053	0.0000	0.0075	2.38×10^{-5}	[0.0002, 0.0278]
Bayesian Informative (Low Compliance)	(2, 38)	152	(2, 190)	0.0104	0.0087	0.0053	0.0073	2.32×10^{-5}	[0.0013, 0.0282]
Bayesian Informative (Conservative / Less Optimistic)	(5, 95)	158	(5, 253)	0.0194	0.0181	0.0156	0.0085	2.70×10^{-5}	[0.0063, 0.0388]
Bayesian Informative (High Compliance / Optimistic)	(1, 199)	74	(1, 273)	0.0037	0.0025	0.0000	0.0036	1.15×10^{-5}	[0.0001, 0.0134]
Bayesian – Empirical (Data Driven / Method of Moments)	(2, 198)	88	(2, 286)	0.0069	0.0058	0.0035	0.0049	1.55×10^{-5}	[0.0008, 0.0188]

Note: The posterior distributions assume $y = 0$ nonconforming items observed in the respective sample size n . The Bayesian posterior is obtained through Beta–Binomial conjugate updating,
$$\text{Beta}(\alpha, \beta) \rightarrow \text{Beta}(\alpha + y, \beta + n - y).$$

The MCSE values correspond to the posterior mean based on $M = 100,000$ random draws.

2. Protection Against Bad Lots (LQL = 5.0%):

For the consumer protection constraint $\beta \leq 0.10$, the Frequentist Baseline, Haldane, and Conservative schemes strictly enforce the standard Consumer's Risk bound, with

$$P_a(0.05) \leq 0.10.$$

The Empirical Plan exhibits $P_a(0.05) = 0.178$. While this is slightly higher than the 10% frequentist benchmark at $p = 5\%$, its discrimination power increases rapidly beyond the LQL threshold, dropping to $P_a(0.08) = 0.025$ and $P_a(0.10) = 0.006$.

This ensures robust protection against heavily nonconforming lots while substantially lowering the sampling load under standard operational conditions.

3. Discrimination Gradient Across Intermediate Proportion Nonconforming ($p = 2\%$ to 4%):

Across the intermediate region $p = 0.02$ to 0.04 , the empirical plan ($n = 88, c = 2$) exhibits a higher acceptance probability than the frequentist baseline:

$$\begin{aligned} P_a(0.02) &= 0.742 \quad (\text{Empirical Bayesian}), \\ P_a(0.02) &= 0.728 \quad (\text{Frequentist Baseline}), \\ P_a(0.04) &= 0.312 \quad (\text{Empirical Bayesian}), \\ P_a(0.04) &= 0.222 \quad (\text{Frequentist Baseline}). \end{aligned}$$

This smoother OC curve slope reflects the Bayesian trade-off: leveraging reliable historical process information to minimize sample size n while maintaining an appropriate expected risk over the historical process distribution.

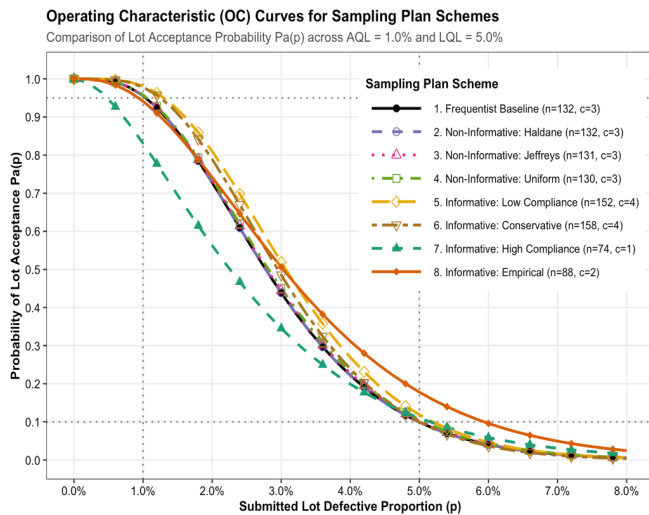


Figure 1. Operating Characteristic Curves for Sampling Plan Schemes

The OC matrix (Table 3) demonstrates that the empirical Bayesian plan ($n = 88, c = 2$) maintains near-identical producer protection at the AQL, with $P_a = 0.941$ compared with $P_a = 0.956$ for the baseline, despite a 33.3% reduction in sample size.

While exhibiting a slightly wider operating transition region between the AQL and LQL, the plan provides effective lot rejection at higher proportions nonconforming, with $P_a = 0.006$ at $p = 0.10$. Thus, it achieves an optimal practical balance between producer risk, consumer protection, and sample economy.

Bayes Risk Analysis

The Expected Bayes Risk analysis reveals two important findings: **(i) reduced overhead at operating conditions** and **(ii) optimal economic balance**. The cost parameters are set as

$$c_s = 1, \quad c_i = 5, \quad c_a = 50, \quad N = 10,000.$$

i) At the target process quality level $p \leq 1.0\%$, the total risk is dominated by testing costs $c_s \cdot n$. The Empirical Plan incurs a lower expected risk of approximately 88 cost units at $p = 0\%$, compared with approximately 132–158 cost units for the frequentist and conservative plans. This reduction directly results from its smaller initial sample size.

ii) All plans experience a peak in expected loss around $p \approx 2.0\%–3.0\%$ (Fig. ??), where occasional lot acceptances incur field-defect penalties represented by c_a .

However, when the expected loss is integrated over the realistic historical process distribution

$$\pi(p) \sim \text{Beta}(2, 198),$$

the Empirical Plan achieves the minimum total integrated Bayes Risk $B(n, c)$ among all compliant sampling plans.

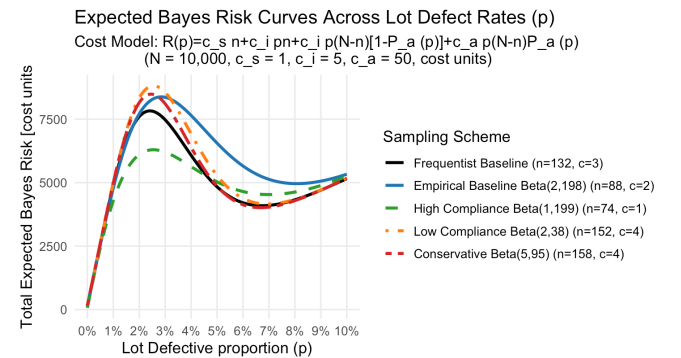


Figure 2. Bayes Risk Curves for Sampling plan schemes

Average Total Inspection

The Average Total Inspection (ATI) curves shown in Fig. 3 depict two main findings: **(i) substantial inspection savings** and **(ii) automatic rectification scaling**.

Under baseline quality conditions, $p \leq 0.01$, the empirical plan ($n = 88, c = 2$) achieves an ATI of approximately 88–110 items per lot. Compared with the frequentist baseline plan ($n = 132, c = 3$), which has an ATI of

Table 3. OC Curve Data Matrix: Lot Acceptance Probability $P_a(p)$ Across True Lot Proportion Nonconforming p

Sampling Plan	$p = 0$	$p = 0.005$	$p = 0.01$	$p = 0.02$	$p = 0.03$	$p = 0.04$	$p = 0.05$	$p = 0.08$	$p = 0.10$
Frequentist Baseline ($n = 132, c = 3$)	1.000	0.996	0.956	0.728	0.438	0.222	0.099	0.005	0.001
Haldane Beta(0.001, 0.001)($n = 132, c = 3$)	1.000	0.996	0.956	0.728	0.438	0.222	0.099	0.005	0.001
Jeffreys Beta(0.5, 0.5)($n = 131, c = 3$)	1.000	0.996	0.956	0.733	0.444	0.227	0.103	0.006	0.001
Uniform Beta(1, 1)($n = 130, c = 3$)	1.000	0.996	0.958	0.737	0.451	0.232	0.106	0.006	0.001
Low Compliance Beta(2, 38)($n = 152, c = 4$)	1.000	0.999	0.981	0.810	0.519	0.269	0.118	0.005	0.000
Conservative Beta(5, 95)($n = 158, c = 4$)	1.000	0.999	0.978	0.789	0.485	0.239	0.099	0.004	0.000
High Compliance Beta(1, 199)($n = 74, c = 1$)	1.000	0.947	0.830	0.563	0.345	0.199	0.110	0.016	0.004
Empirical Beta(2, 198)($n = 88, c = 2$)	1.000	0.990	0.941	0.742	0.506	0.312	0.178	0.025	0.006

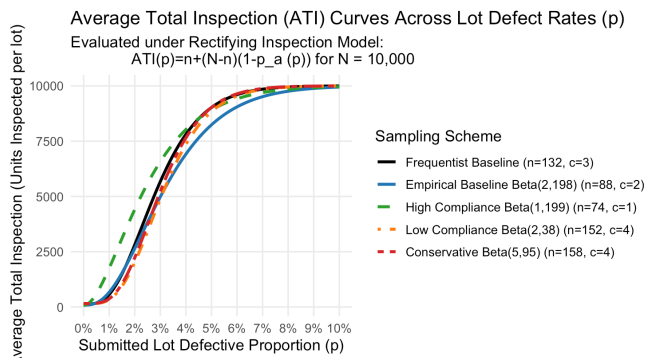
Note: Comparing lot acceptance probability across AQL=0.1 and LQL =0.05

approximately 132–160 items per lot, this represents an immediate 33.3% reduction in inspection effort.

At low proportions nonconforming ($p \rightarrow 0$), the ATI approaches the initial sample size n . As lot quality deteriorates, $p \rightarrow 0.10$, the acceptance probability $P_a(p)$ decreases rapidly across all sampling plans, forcing increasingly extensive rectification inspection and ultimately approaching 100% screening, such that

$$\text{ATI} \rightarrow N = 10,000.$$

The empirical plan therefore maintains a low inspection overhead during normal operating conditions while retaining the ability to trigger extensive lot screening when process quality deteriorates.

**Figure 3.** ATI curves for Sampling plan schemes

Prior Sensitivity Analysis Results

To evaluate model robustness against hyperparameter mis-specification, a sensitivity analysis was conducted by perturbing the baseline prior Beta(2,198) across mean shifts ($\pm 50\%$) and variance shifts (Table ??). Adjusting the prior mean from 1.0% to 2.0% increased the required sample size from $n = 88$ to $n = 114$, demonstrating an automatic conservative shift that hedges against consumer risk.

Relaxing prior precision, with $\alpha + \beta = 100$, resulted in parameters closer to the classical frequentist plan, with $n = 108$, confirming that reduced prior certainty safely reverts toward baseline protection levels.

From Table 4, three primary operational outcomes are observed:

- Across all prior perturbation scenarios, the optimized Bayesian sampling plans strictly satisfy both Producer's Risk $\alpha_0 \leq 0.05$ and Consumer's Risk $\beta_0 \leq 0.10$ boundaries when evaluated under the conjugate Beta–Binomial predictive distribution.
- As historical process quality degrades (prior mean $E[p]$ increases from 0.5% to 2.0%), the Bayesian optimization algorithm automatically expands the sample size requirement from $n = 74$ ($c = 1$) to $n = 134$ ($c = 4$). This dynamic expansion ensures that consumer protection is maintained during periods of process instability without requiring manual plan intervention.
- Increasing prior precision to Beta(5, 495) reduces the optimal sample size from $n = 88$ to $n = 82$, while lowering the Total Bayes Risk from 214.30 to 198.40 cost units. This confirms that accumulating historical inspection data within LIMS progressively reduces ongoing laboratory testing expenses.

The prior sensitivity analysis evaluates two distinct axes of hyperparameter perturbation: *location shifts* (changes in prior mean $E[p]$ reflecting process quality drift) and *dispersion shifts* (changes in prior precision reflecting process stability uncertainty).

- Robustness to Location Shifts (Adverse Process Drift):** When the prior mean shifts adversely from $E[p] = 0.010$ to 0.020 , corresponding to Beta(4, 196), the optimization algorithm exhibits dynamic adaptability. It automatically expands the required sample size from $n = 88$ ($c = 2$) to $n = 134$ ($c = 4$), ensuring that Consumer's Risk at LQL = 0.05 remains strictly bounded, $\beta_0 \leq 0.10$.
- Sensitivity to Precision Shifts (Informativeness):** Reducing prior informativeness to Beta(1, 99) acts as a conservative safeguard, increasing the inspection size to $n = 102$ to compensate for prior uncertainty. Conversely, as historical process data accumulates and increases prior precision to Beta(5, 495),

Table 4. Summary of the Prior Sensitivity Analysis Results

Prior Perturbation Scenario	Perturbed Prior (α, β)	Prior Mean $E[p]$	Prior Variance $Var(p)$	Optimal Plan	P_a AQL	Producer's Risk α_0	P_a LQL	Total Bayes Risk
Lower Mean (-50%)	Beta(1, 199)	0.005	2.48×10^{-5}	$n = 74, c = 1$	0.988	0.012	0.027	126.85
Baseline Fit	Beta(2, 198)	0.010	4.90×10^{-5}	$n = 88, c = 2$	0.963	0.037	0.088	214.30
Higher Mean (+50%)	Beta(3, 197)	0.015	7.31×10^{-5}	$n = 112, c = 2$	0.959	0.041	0.092	318.52
Higher Mean (+100%)	Beta(4, 196)	0.020	9.66×10^{-5}	$n = 134, c = 4$	0.955	0.045	0.095	431.10
Lower Precision (50%)	Beta(1, 99)	0.010	9.70×10^{-5}	$n = 102, c = 2$	0.952	0.048	0.076	242.15
Higher Precision (250%)	Beta(5, 495)	0.010	1.97×10^{-5}	$n = 82, c = 2$	0.969	0.031	0.091	198.40

Note: Assuming target thresholds of AQL = 0.01 and LQL = 0.05 with $N = 10,000$ lot size. The Producer's Risk is defined as $\alpha_0 = 1 - P_a(\text{AQL})$. Total Bayes Risk is expressed in monetary units (Rs or dollars).

the plan safely achieves further inspection efficiency with $n = 82$, reducing Total Bayes Risk from 214.30 to 198.40 cost units.

Integration of Empirical Bayesian Acceptance Sampling into Digital LIMS Workflows

To transition the theoretical Beta-Binomial model into modern industrial quality assurance, the proposed empirical Bayesian sampling framework can be embedded directly within dynamic digital designs, such as Laboratory Information Management Systems (LIMS). Traditional quality control environments rely on static frequentist tables (e.g., MIL-STD-105E or ISO 2859-1), which treat sequential inspection lots as isolated events and ignore historical process capability. In contrast, an integrated LIMS framework enables automated, real-time Bayesian updating across continuous production runs with the dynamically updated parameters α_k and β_k (Yuen et al., 2026).

DISCUSSION

Evaluating the empirical Bayesian plan Beta(2, 198) across the true nonconforming proportion $p \in [0, 0.10]$ confirms an optimal balance between classical regulatory compliance and operational economy. Standard industrial frameworks (e.g., MIL-STD-105E) mandate explicit guarantees for Producer's Risk α_0 at AQL = 1% and Consumer's Risk β_0 at LQL = 5%. Validating the empirical plan via its Operating Characteristic (OC) curve demonstrates that reducing the sample size from $n = 132$ to $n = 88$ achieves a 33.3% reduction in inspection volume, translating directly into substantial cost savings for

destructive food testing (e.g., ELISA kits and gas chromatography analysis).

Modern Laboratory Information Management Systems (LIMS) extend beyond static controls by updating Beta prior parameters (α_k, β_k) after each evaluated batch (Boyar et al., 2021). Here, Lot k 's posterior seamlessly informs Lot $k + 1$'s prior. When process drift occurs (e.g., raw material contamination), an increase in the observed number of nonconforming items y shifts the posterior distribution rightward, raising posterior risk and triggering tighter quality oversight to protect consumers.

At target quality (AQL = 1%), the empirical plan maintains strong producer protection, with $P_a(0.01) = 0.941$ compared with 0.956 for the baseline. While over-conservative priors impose excessive sampling overhead and high-compliance schemes ($n = 74$) under-protect producers ($P_a = 0.830$), the empirical plan strikes an effective balance.

Although its acceptance probability at LQL = 5% is slightly elevated, with $P_a(0.05) = 0.178$, its discrimination accelerates rapidly beyond the LQL, dropping to $P_a(0.08) = 0.025$ and $P_a(0.10) = 0.006$. Across a realistic process capability distribution where 95% of lots operate below $p = 2.7\%$, the empirical plan minimizes total expected Bayes risk $B(n, c)$ and Average Total Inspection (ATI), offering practitioners a robust alternative to rigid classical schemes.

CONCLUSION

This study demonstrates both the theoretical utility and practical application of a Bayesian single-attribute acceptance sampling framework for food quality control. By re-

placing conventional, independent lot evaluation methods with conjugate Beta–Binomial updating, the proposed model incorporates historical prior knowledge via Beta prior distributions and updates them dynamically using sample inspection data. This adaptive mechanism provides a robust, flexible alternative to traditional acceptance sampling procedures.

The proposed framework significantly reduces sample sizes and inspection efforts without compromising product safety standards or exceeding established risk limits, thereby striking an optimal balance between producer and consumer risks.

The empirical and simulated results confirm that appropriate prior specification significantly enhances sampling efficiency. Specifically, the empirical Bayesian plan Beta(2, 198) achieves near-identical producer protection at target quality (AQL = 1%) compared with the standard frequentist baseline, while reducing sample inspection by 33.3%. Sensitivity metrics demonstrate that the empirical plan remains robust against minor historical estimation errors.

While optimistic prior shifts reduce sample size requirements further, potential consumer risk inflation under unexpected quality drops is effectively mitigated by continuous posterior updating within digital LIMS workflows.

Consequently, the proposed Bayesian framework serves as a practical, cost-effective decision tool for food safety environments where minimizing inspection overhead and maintaining strict quality standards are equally critical. Future research may extend this framework to continuous variable attributes (e.g., net weight and moisture content using Normal–Gamma conjugate priors) and multi-attribute chain sampling procedures.

Limitations

This study assumes independent lots and a Beta-distributed nonconforming proportion. The performance of the proposed plan under autocorrelated production processes or non-Beta prior distributions was not investigated.

Future work can extend this framework by integrating Gamma priors for continuous variables, hierarchical Bayesian modeling, adaptive sampling schemes, real-world food industry validation, and machine-learning-assisted prior estimation.

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Structural Equation Modeling: A Review of Its History, Methods, and Applications

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ABSTRACT

Background: Structural equation modeling (SEM) is a family of multivariate statistical techniques that allows researchers to specify, estimate, and test networks of relationships among observed and latent variables. Combining the measurement logic of factor analysis with the causal-relational logic of path analysis, SEM has become one of the dominant analytic frameworks in the social, behavioral, health, and management sciences, yet its historical development, statistical foundations, and rapidly evolving software ecosystem are rarely synthesized in one place.

Method: This paper presents a narrative review tracing the historical development of SEM from Sewall Wright's early path-analytic work in genetics, through the psychometric contributions of Spearman and Thurstone, to the unification achieved by Jöreskog, Keesling, and Wiley and the subsequent diffusion of the LISREL program. It surveys the core statistical building blocks of SEM, including model specification and identification, sample size and power considerations, estimation methods (maximum likelihood, weighted least squares, and Bayesian estimation), and the fit indices commonly used to evaluate model adequacy, alongside the major SEM variants and the software ecosystem that supports them.

Result: The review synthesizes SEM's major methodological variants – confirmatory factor analysis, path analysis, multi-group and longitudinal SEM, mediation and moderation models, partial least squares SEM, Bayesian SEM, exploratory structural equation modeling, and network psychometric alternatives – together with applications across psychology, education, business, and health research. It further documents recent developments, including dynamic structural equation modeling for intensive longitudinal data, hybrid SEM–machine learning approaches, and the expanding open-source R ecosystem (*lavaan*, *SEMinR*, *cSEM*, *blavaan*).

Conclusion: SEM remains an active and evolving methodological field rather than a fixed, uncontested technique. Researchers using SEM are best served by grounding their modeling choices in both their substantive theory and a clear understanding of the assumptions and trade-offs associated with the particular SEM variant, estimator, and software tool they adopt.

Key words: structural equation modeling, path analysis, confirmatory factor analysis, latent variables, PLS-SEM, Bayesian SEM, model fit, sample size

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INTRODUCTION

Structural equation modeling (SEM) refers to a broad class of statistical techniques used to test hypothesized relationships among multiple variables simultaneously, including relationships that involve unobserved, or latent, constructs. Unlike traditional regression, which examines one dependent variable at a time and generally assumes that predictors are measured without error, SEM allows researchers to model measurement error explicitly, to estimate multiple dependent relationships in a single analysis, and to compare an implied theoretical structure against the pattern of covariances actually observed in the data. Because of this flexibility, SEM has become a standard tool wherever researchers work with abstract constructs – attitudes, personality traits, organizational climate, quality of life, or economic expectations – that cannot be measured directly but can be inferred from multiple observed indicators ([MacCallum and Austin, 2000](#)).

The appeal of SEM lies in its integrative character: it merges the measurement tradition of factor analysis,

which is concerned with how well observed items represent an underlying construct, with the structural tradition of path analysis and simultaneous-equation modeling, which is concerned with the pattern of causal or predictive relationships among variables. This review synthesizes the historical origins of SEM, its statistical foundations, its major methodological variants, the software that implements it, its applications across disciplines, and the debates that continue to shape its use.

HISTORICAL DEVELOPMENT OF STRUCTURAL EQUATION MODELING

Early Roots: Path Analysis and Factor Analysis

The statistical lineage of SEM can be traced to two largely independent traditions that developed in the early twentieth century. The first is path analysis, introduced by the geneticist Sewall Wright, whose 1918 study of skeletal proportions in rabbits used path diagrams to represent hypothesized causal relationships among variables and compared the correlations implied by those diagrams with the correlations actually observed ([Wright,](#)

1918). Wright's method, elaborated further in his work on the method of path coefficients (Wright, 1934), provided a way to decompose an observed correlation into direct and indirect components attributable to a specified causal structure, anticipating the decomposition of effects into direct, indirect, and total effects that remains central to SEM today.

The second tradition is factor analysis, originating with Charles Spearman's early twentieth-century work on general intelligence and extended in the 1930s by Louis Thurstone's multiple-factor model and his development of factor rotation. Where path analysis modeled relationships among directly measured variables, factor analysis modeled the extent to which a smaller number of unobserved factors could account for the pattern of correlations among a larger set of observed indicators.

Although Wright's path-analytic approach attracted little attention among psychologists for several decades, it was adopted more readily by sociologists and economists. Sociologists such as Hubert Blalock (Blalock, 1964) and Otis Dudley Duncan (Duncan, 1966) popularized path analysis within the social sciences during the 1960s, while economists working within the Cowles Commission tradition, building on Wright's ideas, developed related simultaneous-equation approaches for econometric modeling.

Unification: Jöreskog, Keesling, Wiley, and LISREL

The modern, unified form of SEM emerged in the late 1960s and early 1970s through the work of the psychometrician Karl Jöreskog, who introduced confirmatory factor analysis (CFA) with maximum-likelihood estimation (Jöreskog, 1969), enabling formal statistical tests of hypotheses about the number of underlying factors and how they relate to observed variables. Jöreskog, together with contributions from Keesling and Wiley in the early 1970s, generalized this framework to combine the measurement structure of factor analysis with the structural, path-analytic modeling of relationships among latent variables (Jöreskog, 1973), producing what became known as the Jöreskog–Keesling–Wiley model – the general structural equation model that underlies contemporary SEM.

This theoretical unification was accompanied by a practical breakthrough: the development of the LISREL (Linear Structural RELations) computer program by Jöreskog and Dag Sörbom (Jöreskog and Sörbom, 1978). LISREL gave applied researchers, for the first time, an accessible and efficient way to specify and estimate structural equation models. Bollen (1989) argued that the availability of LISREL was itself one of the most important factors in the rapid spread of SEM techniques through the social and behavioral sciences during the 1970s and 1980s, since it allowed researchers without specialized mathematical training to fit and evaluate complex latent variable models.

Diffusion and Diversification

From the 1980s onward, SEM diffused well beyond its psychometric and sociological origins into education, management, marketing, political science, epidemiology, and health services research (MacCallum and Austin,

2000). This diffusion was accompanied by methodological diversification: new estimators were developed to handle non-normal and categorical data, new fit indices were proposed and evaluated (Hu and Bentler, 1999), and alternative estimation traditions – most notably partial least squares path modeling, introduced by the econometrician and statistician Herman Wold as a composite-based alternative to Jöreskog's factor-based approach (Wold, 1982) – expanded the range of situations in which SEM-type analysis could be applied. In more recent decades the field has continued to generalize further, incorporating multilevel structures, longitudinal growth models, mixture models, Bayesian estimation (Muthén and Asparouhov, 2012), and, increasingly, connections to network-based approaches to modeling psychological constructs (Ep-skamp et al., 2018).

THEORETICAL AND STATISTICAL FOUNDATIONS

The Measurement Model and the Structural Model

A complete structural equation model is generally understood as consisting of two interlocking components. The *measurement model* specifies how each latent variable is indicated by one or more observed variables, typically through a confirmatory factor-analytic structure that estimates factor loadings and accounts for measurement error in each indicator. The *structural model* specifies the hypothesized directional and non-directional relationships among the latent variables themselves, analogous to a system of regression equations but operating on error-free latent constructs rather than raw observed scores. This separation of concerns is one of the principal advantages of SEM over ordinary regression: because measurement error is modeled explicitly in the measurement component, the structural relationships among constructs can, in principle, be estimated without attenuation bias due to unreliable measurement (Anderson and Gerbing, 1988).

Following the classic LISREL notation introduced by Jöreskog (1973), the measurement model relates a vector of observed indicators $\mathbf{x}$ to a vector of exogenous latent variables $\boldsymbol{\xi}$, and a second vector of observed indicators $\mathbf{y}$ to a vector of endogenous latent variables $\boldsymbol{\eta}$:

$$\mathbf{x} = \boldsymbol{\Lambda}_x \boldsymbol{\xi} + \boldsymbol{\delta}, \quad (1)$$

$$\mathbf{y} = \boldsymbol{\Lambda}_y \boldsymbol{\eta} + \boldsymbol{\varepsilon}, \quad (2)$$

where $\boldsymbol{\Lambda}_x$ and $\boldsymbol{\Lambda}_y$ are matrices of factor loadings, and $\boldsymbol{\delta}$ and $\boldsymbol{\varepsilon}$ are vectors of measurement errors assumed to have zero mean and to be uncorrelated with the latent variables, i.e. $E(\boldsymbol{\delta}) = \mathbf{0}$, $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, $\text{Cov}(\boldsymbol{\xi}, \boldsymbol{\delta}) = \mathbf{0}$, and $\text{Cov}(\boldsymbol{\eta}, \boldsymbol{\varepsilon}) = \mathbf{0}$.

The structural model then specifies the hypothesized causal relationships among the latent variables themselves:

$$\boldsymbol{\eta} = \mathbf{B}\boldsymbol{\eta} + \boldsymbol{\Gamma}\boldsymbol{\xi} + \boldsymbol{\zeta}, \quad (3)$$

where $\mathbf{B}$ contains structural coefficients among endogenous latent variables (with zeros on the diagonal, and $(\mathbf{I} - \mathbf{B})$ assumed invertible), $\boldsymbol{\Gamma}$ contains structural coefficients relating exogenous to endogenous latent variables,

and ζ is a vector of disturbances (structural errors) uncorrelated with ξ .

Under this specification, the model-implied population covariance matrix of the observed variables, $\Sigma(\theta)$, is a function of the free parameters $\theta = \{\Lambda_x, \Lambda_y, B, \Gamma, \Phi, \Psi, \Theta_\delta, \Theta_\varepsilon\}$, where $\Phi = \text{Cov}(\xi)$, $\Psi = \text{Cov}(\zeta)$, and $\Theta_\delta, \Theta_\varepsilon$ are the (co)variance matrices of the measurement errors. For the endogenous block, substituting Equation (3) and solving for η gives $\eta = (I - B)^{-1}(\Gamma\xi + \zeta)$, from which the implied covariance structure follows as

$$\Sigma(\theta) = \begin{pmatrix} \Lambda_y A (\Gamma \Phi \Gamma' + \Psi) A' \Lambda_y' + \Theta_\varepsilon & \Lambda_y A \Gamma \Phi \Lambda_x' \\ \Lambda_x \Phi \Gamma' A' \Lambda_y' & \Lambda_x \Phi \Lambda_x' + \Theta_\delta \end{pmatrix}, \quad (4)$$

Estimation, described in Section 3.4, seeks values of θ that make $\Sigma(\theta)$ as close as possible to the sample covariance matrix S .

Model Specification and Identification

Before a structural equation model can be estimated, the researcher must specify which parameters (factor loadings, structural paths, variances, covariances) are to be freely estimated, which are fixed to particular values, and which are constrained to be equal to one another. This specification must yield a model that is *identified* – that is, one for which a unique set of parameter estimates can, in principle, be recovered from the observed covariance matrix. A necessary (though not sufficient) condition, commonly called the *t-rule*, requires that the number of freely estimated parameters, t , not exceed the number of non-redundant elements in the observed covariance matrix. For p observed variables there are

$$s = \frac{p(p+1)}{2} \quad (5)$$

distinct variances and covariances, so the model degrees of freedom are

$$df = s - t = \frac{p(p+1)}{2} - t. \quad (6)$$

A model with $df \geq 0$ is said to be *over-identified* (or *just-identified* when $df = 0$); a model with $df < 0$ is under-identified and cannot be estimated without further constraints. In addition to satisfying the *t-rule*, each latent variable must be assigned a metric, typically by fixing one factor loading to 1 (the *marker-variable* method) or by standardizing the latent variable's variance to 1, since latent variables have no inherent scale of their own.

Sample Size and Statistical Power

Because SEM typically estimates many parameters simultaneously and relies on asymptotic (large-sample) theory for maximum-likelihood inference, adequate sample size is a recurring practical concern. Applied researchers have long relied on simple rules of thumb: an absolute minimum of roughly 100 to 200 cases (Kline, 2015), or a ratio of cases to freely estimated parameters (the “N:q rule”), with recommendations for this ratio ranging from as low as 5:1 to as high as 20:1 depending on the source (Jackson, 2003; Kline, 2015). Such rules, however, ignore model-specific features that strongly influence the precision of

estimates and the power of fit tests, including the number of indicators per factor, the size of factor loadings, and the degree of model misspecification being tested against.

More principled approaches instead conduct formal power analysis. MacCallum et al. (1996) developed a widely used framework for power analysis based on the root-mean-square error of approximation (RMSEA), which tests hypotheses of close fit (e.g., $\varepsilon_0 \leq 0.05$) against not-close fit (e.g., $\varepsilon_1 \geq 0.08$) as a function of sample size, degrees of freedom, and significance level. The procedure relies on the fact that, under a fixed alternative RMSEA value ε_a , the likelihood-ratio test statistic follows a non-central chi-square distribution with noncentrality parameter

$$\lambda = (N - 1) df \varepsilon_a^2, \quad (7)$$

where N is the sample size and df the model degrees of freedom. Power for the test of close fit is then obtained as $1 - \beta = P(\chi_{df, \lambda}^2 > \chi_{df, 1-\alpha}^2)$, and the researcher can solve for the minimum N that achieves a target power (typically 0.80) for a given df , ε_0 , ε_1 , and α .

The simpler, more commonly reported N:q heuristic instead sets the required sample size as a fixed multiple k of the number of freely estimated parameters t :

$$N_{\min} = k \cdot t, \quad k \in \{5, 10, 20\}, \quad (8)$$

with $k = 20$ recommended by Kline (2015), $k = 10$ by other sources, and k as low as 5 suggested in some older guidelines; Jackson (2003) showed via simulation that the appropriate value of k itself depends on communality and the number of indicators per factor, so a single fixed ratio is, at best, an approximation. More recently, Wang and Rhemtulla (2021) argued that researchers should shift emphasis from power to detect overall model misfit toward power to detect specific target parameters of substantive interest, such as an individual structural path or indirect effect γ_{ij} , using the Wald-test-based approximation

$$1 - \beta \approx P\left(Z > z_{1-\alpha/2} - \frac{\gamma_{ij}}{SE(\hat{\gamma}_{ij})}\right), \quad (9)$$

and provided simulation-based tools for this purpose. Together, this body of work suggests that a single universal sample-size rule cannot substitute for power analysis tailored to a given model and research question.

Estimation Methods

Maximum likelihood (ML) estimation is the default and most widely used estimator in SEM. Under the assumption that the p observed variables follow a multivariate normal distribution, ML estimation chooses the parameter vector $\hat{\theta}$ that minimizes the discrepancy function

$$F_{ML}(\theta) = \log |\Sigma(\theta)| + \text{tr}[S \Sigma(\theta)^{-1}] - \log |S| - p, \quad (10)$$

where S is the sample covariance matrix and $\Sigma(\theta)$ is the model-implied covariance matrix from Equation (4). At the minimum, $F_{ML}(\hat{\theta}) \geq 0$, with equality only when the model reproduces S exactly. The overall chi-square test statistic used to evaluate exact fit is then

$$\chi^2 = (N - 1) F_{ML}(\hat{\theta}), \quad (11)$$

which, under the null hypothesis that the model holds exactly in the population and standard regularity conditions, is asymptotically distributed as χ^2_{df} with df as defined above. When the multivariate normality assumption is violated, robust variants such as MLM and MLR rescale this statistic (the Satorra–Bentler correction) and adjust standard errors to account for excess kurtosis. For categorical or ordinal indicators, estimators from the weighted least squares family, particularly the diagonally weighted least squares estimator with robust corrections (often labeled WLSMV), minimize a discrepancy function of the form $F_{WLS} = (\mathbf{s} - \boldsymbol{\sigma}(\boldsymbol{\theta}))' \mathbf{W}^{-1} (\mathbf{s} - \boldsymbol{\sigma}(\boldsymbol{\theta}))$, where $\mathbf{s}$ and $\boldsymbol{\sigma}(\boldsymbol{\theta})$ are vectorized sample and model-implied polychoric/polyserial correlations and $\mathbf{W}$ is a weight matrix; these are commonly preferred over ML for ordinal data because they do not require treating ordered-categorical responses as continuous.

An increasingly prominent alternative is Bayesian estimation, popularized within the SEM literature by Muthén and Asparouhov (2012)'s proposal to replace exact zero-constraints – for example, cross-loadings or residual correlations fixed at zero for identification purposes – with small-variance informative priors that allow such parameters to be approximately, rather than exactly, zero (Asparouhov et al., 2015). Rather than maximizing the likelihood $L(\mathbf{X} | \boldsymbol{\theta})$ of the raw data $\mathbf{X}$ directly, Bayesian SEM combines it with a prior distribution $p(\boldsymbol{\theta})$ over the model parameters, via Bayes' theorem, to obtain the posterior distribution

$$p(\boldsymbol{\theta} | \mathbf{X}) = \frac{L(\mathbf{X} | \boldsymbol{\theta}) p(\boldsymbol{\theta})}{\int L(\mathbf{X} | \boldsymbol{\theta}) p(\boldsymbol{\theta}) d\boldsymbol{\theta}}, \quad (12)$$

typically approximated via Markov chain Monte Carlo (MCMC) sampling rather than solved in closed form. For an approximate zero-constraint, a parameter ϕ (e.g., a cross-loading) that would be fixed at exactly 0 under ML is instead assigned a small-variance normal prior, $\phi \sim N(0, \sigma_0^2)$ with σ_0^2 chosen to be small (e.g., $\sigma_0^2 = 0.01$), so that the posterior for ϕ is shrunk toward, but not forced to equal, zero. Point estimates are then commonly summarized by the posterior mean or median, and uncertainty by the 95% credibility interval, i.e. the interval $[\theta_{2.5\%}, \theta_{97.5\%}]$ such that $P(\theta_{2.5\%} \leq \phi \leq \theta_{97.5\%} | \mathbf{X}) = 0.95$. Advocates note that Bayesian SEM can better reflect substantive theory, can incorporate external prior information, and can in some circumstances outperform maximum likelihood with small samples (Van de Schoot et al., 2015), though MacCallum et al. (2012) and a subsequent systematic review (Van de Schoot et al., 2019) have cautioned that diffuse, non-informative default priors do not automatically solve small-sample problems and can themselves introduce bias unless priors are chosen thoughtfully.

Evaluating Model Fit

Because a structural equation model is a hypothesis about how a set of observed covariances arise, model evaluation centers on comparing the covariance matrix implied by the estimated model with the covariance matrix actually observed in the sample. The chi-square test

of exact fit is the traditional statistic for this comparison, but it is well known to be sensitive to sample size, rejecting even trivially misspecified models when samples are large. For this reason, researchers routinely supplement the chi-square test with approximate fit indices.

The Root Mean Square Error of Approximation (RMSEA) estimates the degree of model misspecification per degree of freedom, adjusting the noncentrality parameter $\hat{\lambda} = \max(\chi^2 - df, 0)$ for sample size and model complexity:

$$RMSEA = \sqrt{\frac{\max\left(\frac{\chi^2 - df}{df}, 0\right)}{N - 1}}. \quad (13)$$

The Comparative Fit Index (CFI) and Tucker–Lewis Index (TLI, also called the Non-Normed Fit Index) compare the noncentrality of the hypothesized model, $\hat{\lambda}_M = \chi_M^2 - df_M$, to that of a baseline (null) model of complete independence among the observed variables, $\hat{\lambda}_0 = \chi_0^2 - df_0$:

$$CFI = 1 - \frac{\max(\hat{\lambda}_M, 0)}{\max(\hat{\lambda}_M, \hat{\lambda}_0, 0)}, \quad (14)$$

$$TLI = \frac{\frac{\chi_0^2}{df_0} - \frac{\chi_M^2}{df_M}}{\frac{\chi_0^2}{df_0} - 1}. \quad (15)$$

The Standardized Root Mean Square Residual (SRMR) is based on the discrepancy between observed and model-implied correlations r_{ij} and $\hat{r}_{ij}$ (rather than raw covariances), averaged over all $p(p+1)/2$ unique pairs of the p observed variables:

$$SRMR = \sqrt{\frac{2 \sum_{i=1}^p \sum_{j=1}^i (r_{ij} - \hat{r}_{ij})^2}{p(p+1)}}. \quad (16)$$

The most widely reported of these indices are CFI and TLI, which compare the fit of the specified model to that of the baseline independence model; RMSEA, which penalizes model complexity through df ; and SRMR, which is most directly interpretable as an average standardized residual.

The influential simulation study of Hu and Bentler (1999) evaluated a range of candidate cutoff values for these indices and recommended, as an approximate joint criterion, CFI and TLI values near .95 together with an SRMR value near .08–.09 and an RMSEA value near .06, while explicitly cautioning that these should be regarded as rough guidelines rather than rigid pass/fail thresholds. Later methodological work has both applied and critiqued these benchmarks: Schermelleh-Engel et al. (2003) argued that a two-index reporting strategy combining SRMR with a comparative index such as CFI or RMSEA offers a more balanced assessment than any single index, while Steiger (2007) emphasized that fit index behavior depends heavily on model complexity, sample size, and the type of misspecification present, so that reflexively applying Hu and Bentler's cutoffs to every model can be misleading.

MAJOR VARIANTS AND EXTENSIONS OF SEM

Confirmatory Factor Analysis

Confirmatory factor analysis (CFA) is the measurement-only special case of SEM, used to test whether a hypothesized number of latent factors adequately accounts for the covariances among a set of observed indicators, and to establish the validity and reliability of multi-item scales before those scales are used in a larger structural model (Jöreskog, 1969). CFA is foundational to nearly every other SEM application, since a structural model is only as trustworthy as the measurement model on which it rests.

Path Analysis

Path analysis applies the structural, relational logic of SEM to models composed entirely of observed (manifest) variables, without latent factors (Duncan, 1966; Wright, 1934). It remains useful when constructs are measured with single, well-validated indicators or when the research question concerns the pattern of direct and indirect effects among directly measured variables.

Full (Hybrid) SEM

Full SEM combines the measurement and structural components into a single simultaneously estimated model, following the recommended two-step approach of first validating the measurement model via CFA before estimating structural paths (Anderson and Gerbing, 1988). This is the form most commonly implied when researchers refer simply to “SEM.”

Multi-Group SEM and Measurement Invariance

Multi-group SEM extends the basic model to compare structural relationships, and the measurement properties underlying them, across two or more populations – for example, across countries, gender groups, or experimental conditions. A prerequisite for meaningful group comparisons is measurement invariance, a stepwise sequence of tests establishing that factor loadings, intercepts, and residual variances operate equivalently across groups before structural paths or latent means are compared.

Longitudinal and Growth Models

When the same constructs are measured repeatedly over time, SEM can be extended to latent growth curve models, which represent individual trajectories of change as latent intercept and slope factors, and to cross-lagged panel and autoregressive models, which examine reciprocal influences between variables measured at multiple occasions.

Mediation and Moderation

SEM provides a natural framework for testing mediation, in which the effect of one variable on another is transmitted through one or more intervening variables, by explicitly estimating indirect and total effects and their associated confidence intervals, commonly via bootstrapping procedures. Moderation, in which the strength or direction of a relationship depends on a third variable, can also be incorporated through latent interaction terms or multi-group comparisons.

Composite-Based (PLS) SEM

Partial least squares SEM (PLS-SEM), rooted in Herman Wold's work (Wold, 1982) and popularized in applied business and marketing research through the contributions of Wynne Chin (Chin, 1998) and, later, Joseph Hair and colleagues (Hair et al., 2017, 2011), represents an alternative to the traditional covariance-based SEM (CB-SEM) approach associated with Jöreskog. Rather than estimating latent common factors and reproducing the full observed covariance matrix, PLS-SEM constructs each latent variable η_j as a weighted composite (linear combination) of its own indicators x_{jk} ,

$$\eta_j = \sum_{k=1}^{K_j} w_{jk} x_{jk}, \quad (17)$$

where the outer weights w_{jk} are estimated iteratively so as to maximize the variance explained (R^2) in the endogenous composites of the inner (structural) model,

$$\eta_j = \sum_{i \neq j} \beta_{ji} \eta_i + \zeta_j, \quad (18)$$

rather than by minimizing a discrepancy function such as Equation (10) between $\Sigma(\theta)$ and $\mathbf{S}$. Because it imposes fewer distributional assumptions, tolerates smaller sample sizes, and readily accommodates formatively specified constructs, PLS-SEM has become especially popular in exploratory, prediction-oriented business and marketing research, even as some methodologists continue to debate its suitability as a substitute for CB-SEM in strictly confirmatory, theory-testing contexts (Sarstedt et al., 2016). Recent methodological work has moved away from framing the choice as an either/or decision, instead recommending multimethod strategies that apply both factor-based and composite-based estimators to the same model as a robustness check (Guenther et al., 2025).

Bayesian SEM

As noted above, Bayesian SEM (BSEM) reformulates estimation within a Bayesian framework, producing full posterior distributions for each parameter rather than single point estimates, and allowing researchers to incorporate substantive prior knowledge (Muthén and Asparouhov, 2012). BSEM is particularly attractive for complex models – such as those with many cross-loadings or correlated residuals – that would be under-identified under a strict maximum-likelihood zero-constraint approach, and for applications with small samples, though its properties in that context depend heavily on the informativeness of the chosen priors (Van de Schoot et al., 2019).

Exploratory Structural Equation Modeling and Network Approaches

Exploratory structural equation modeling (ESEM) blends the flexible, all-loadings-estimated character of exploratory factor analysis with the fit evaluation and structural modeling capabilities of confirmatory SEM, offering a middle ground when researchers are not confident that indicators load exclusively on a single factor (Marsh

et al., 2009). In parallel, a growing body of work has explored network psychometric approaches, which model constructs as systems of interacting observed variables rather than as reflections of a single underlying latent factor (Borsboom et al., 2003). Epskamp et al. (2018) proposed a generalized framework unifying network and latent-variable models, prompting ongoing methodological debate about when a latent-variable representation versus a network representation better represents psychological and social phenomena.

SOFTWARE FOR STRUCTURAL EQUATION MODELING

A variety of software packages support SEM, differing in cost, programming interface, and specialized capabilities. Historically dominant commercial packages include LISREL, the original SEM program developed by Jöreskog and Sörbom (Jöreskog and Sörbom, 1978); EQS; IBM SPSS Amos, known for its graphical path-diagram interface; and Mplus, widely used for its extensive support of categorical outcomes, mixture modeling, multi-level SEM, and Bayesian estimation (Muthén and Asparouhov, 2012). SAS offers SEM capabilities through PROC CALIS.

Within the open-source R environment, the *lavaan* package, developed by Yves Rosseel and first released in 2010, has become one of the most widely used tools for SEM, offering a compact model syntax, support for non-normal and missing data, and an option to mimic the numerical output of Mplus or EQS to ease transitions between programs (Rosseel, 2012). Other R-based tools include the *sem* package and OpenMx. For composite-based modeling, SmartPLS is the most widely used dedicated PLS-SEM software (Hair et al., 2017), but the R ecosystem now offers several actively maintained alternatives: *SEMinR*, developed by Ray and Danks, provides a domain-specific syntax that lets users build PLS path models using familiar terms such as `composite()`, `constructs()`, and `paths()`, and additionally allows CB-SEM estimation by delegating to *lavaan* internally (Danks and Ray, 2020; Hair et al., 2021); *cSEM*, developed by Rademaker and Schuberth, implements a broad family of composite-based estimators, including consistent PLS (PLSc) and generalized structured component analysis (Rademaker and Schuberth, 2020); and *blavaan*, developed by Merkle and Rosseel, extends *lavaan* syntax to Bayesian estimation via JAGS or Stan, giving users access to posterior summaries, Bayesian fit measures, and MCMC diagnostics without leaving the *lavaan* modeling idiom (Merkle et al., 2021; Merkle and Rosseel, 2018). The proliferation of free, open-source tools such as *lavaan*, *SEMinR*, *cSEM*, and *blavaan* has been credited with lowering the barrier to entry for SEM, both for research and for teaching, relative to the earlier era in which state-of-the-art SEM software was largely closed-source and commercial.

APPLICATIONS ACROSS DISCIPLINES

SEM's capacity to model latent constructs and multiple simultaneous relationships has made it a standard analytic tool across a wide range of fields (MacCallum and

Austin, 2000).

- **Psychology and education:** SEM is used extensively to validate psychological scales through CFA, to test theories of cognitive, motivational, and personality structure, and to model longitudinal change in learning and development through growth curve and cross-lagged designs.
- **Business, management, and marketing:** SEM, and particularly PLS-SEM, is used to test models of consumer behavior, technology acceptance, brand equity, and organizational performance, often linking multiple latent constructs such as satisfaction, trust, and loyalty (Hair et al., 2011).
- **Health and medical research:** SEM supports the validation of patient-reported outcome measures, the modeling of quality-of-life constructs, and the analysis of complex causal pathways linking risk factors, psychosocial mediators, and health outcomes.
- **Sociology, political science, and economics:** consistent with its historical roots, SEM continues to be used to model social stratification, attitudes, and simultaneous-equation systems in which variables mutually influence one another (Blalock, 1964; Duncan, 1966).

STRENGTHS AND LIMITATIONS

The central strength of SEM is its ability to represent theory directly in statistical form: researchers can specify a network of hypothesized relationships, including latent constructs and measurement error, and formally test how well that theory accounts for the observed data. SEM also permits simultaneous estimation of multiple, inter-related equations, decomposition of total effects into direct and indirect components, comparison of competing theoretical models, and testing of measurement equivalence across groups and over time (Anderson and Gerbing, 1988).

These strengths come with important limitations. SEM generally requires relatively large samples for stable maximum-likelihood estimation, particularly as model complexity grows (Jackson, 2003; Kline, 2015), although Bayesian and composite-based alternatives can relax this requirement under some conditions (Van de Schoot et al., 2015). Model fit statistics assess how well a specified model reproduces observed covariances but cannot, by themselves, establish that the hypothesized causal structure is correct, since statistically equivalent models with different causal interpretations can often fit the same data equally well. The practice of iteratively modifying a poorly fitting model based on sample-specific modification indices, if not cross-validated, risks capitalizing on chance characteristics of a particular sample rather than reflecting a generalizable structure. Finally, the debate over covariance-based versus composite-based (PLS) estimation (Guenther et al., 2025; Sarstedt et al., 2016), and over the appropriate use and interpretation of fit index cutoffs (Hu and Bentler, 1999; Steiger, 2007), illustrates that SEM is not a single settled method but an evolving

Software	License	Notable strengths
LISREL	Commercial	Original SEM program; historical standard (Jöreskog and Sörbom, 1978)
EQS	Commercial	Robust estimators for non-normal data
AMOS	Commercial	Graphical path-diagram interface
Mplus	Commercial	Mixture models, categorical outcomes, Bayesian SEM, multilevel and dynamic SEM (Asparouhov et al., 2018)
lavaan (R)	Open source	Compact syntax; free; mimics Mplus/EQS output (Rosseel, 2012)
blavaan (R)	Open source	Bayesian SEM via JAGS/Stan on top of lavaan syntax (Merkle and Rosseel, 2018)
SEMinR (R)	Open source	Friendly PLS-SEM and CB-SEM syntax (Danks and Ray, 2020)
cSEM (R)	Open source	PLSc and other composite-based estimators (Rademaker and Schuberth, 2020)
SmartPLS	Commercial/free tiers	Leading dedicated PLS-SEM GUI software (Hair et al., 2017)

Table 1. Selected software packages for structural equation modeling.

family of techniques whose proper application requires careful attention to a study's theoretical goals and data characteristics.

RECENT DEVELOPMENTS AND THE EXPANDING SEM ECOSYSTEM

Beyond the broad trends already noted, several concrete methodological and software developments have reshaped SEM practice over roughly the past decade and continue to evolve rapidly.

Dynamic Structural Equation Modeling

Traditional SEM assumes independent observations and is not designed for the densely repeated measurements now common in ecological momentary assessment, experience sampling, and wearable-sensor studies. Asparouhov et al. (2018) introduced dynamic structural equation modeling (DSEM), a general framework that combines multilevel modeling, time-series (vector autoregressive) modeling, and conventional SEM to analyze such *intensive longitudinal data*, where many individuals are each measured repeatedly over many occasions. In a two-level DSEM, within-person dynamics – such as autoregressive and cross-lagged effects among momentary variables – are modeled at Level 1, while individual differences in those dynamic parameters (e.g., a person's average level or their degree of emotional inertia) become random effects predicted by, or predicting, Level-2 constructs. DSEM is typically estimated via Bayesian Markov chain Monte Carlo methods, as implemented in Mplus, and has been applied to model phenomena such as affective dynamics in daily life (Hamaker et al., 2018). This represents one of the more substantial recent extensions of the SEM framework, bringing it into closer contact with time-series econometrics and multilevel modeling.

Integration with Machine Learning

A growing body of work explores hybrid SEM–machine learning approaches that combine SEM's capacity for theory-driven, interpretable modeling of latent constructs with machine learning's flexibility in capturing nonlinear patterns and improving out-of-sample predictive accuracy. Richter and Tudoran (2024) discuss how PLS-SEM results can be complemented with machine learn-

ing algorithms (e.g., random forests or gradient boosting) to detect nonlinearities, interaction effects, and important predictors that a linear structural model might miss, while still retaining SEM's structured, construct-based interpretation. Other applications use classification algorithms alongside CFA to jointly optimize measurement-scale fit and predictive accuracy when shortening psychometric scales. At a more structural level, packages such as SEMdeep and SEMgraph implement SEM-constrained neural networks and causal network analysis, in which the path structure specified by a substantive SEM (or a graph learned from high-throughput data) is used to constrain the connectivity of a deep neural network, combining the interpretability of a specified causal structure with the flexible function approximation of deep learning (Grassi et al., 2022). These hybrid approaches remain an active and still-maturing area, and questions about how to evaluate fit, interpret parameters, and avoid overfitting in such combined models are the subject of ongoing methodological work.

Expanding the R Software Ecosystem

As detailed in Section 5, the R ecosystem for SEM has expanded substantially in recent years beyond lavaan alone. SEMinR (Danks and Ray, 2020; Hair et al., 2021) and cSEM (Rademaker and Schuberth, 2020) have made composite-based (PLS) modeling more accessible in R, an environment previously dominated by CB-SEM tools, and both packages implement newer composite-based estimators such as consistent PLS (PLSc), designed to correct for the attenuation bias that ordinary PLS weights can introduce when constructs are meant to represent common factors. blavaan (Merkle et al., 2021; Merkle and Rosseel, 2018) has similarly extended Bayesian SEM into the open-source R environment using familiar lavaan syntax, reducing reliance on proprietary Bayesian SEM implementations. Companion packages such as semTools provide auxiliary functions for tasks including measurement invariance testing, reliability estimation, and power analysis, further consolidating the open-source SEM toolchain.

Multimethod and Robustness-Oriented Practice

As noted in Section 4.7, recent methodological guidance increasingly frames CB-SEM and PLS-SEM as comple-

mentary rather than competing tools, encouraging researchers to report results from both factor-based and composite-based estimators on the same model as a robustness check rather than committing exclusively to one estimation tradition (Guenther et al., 2025). A related development applies double-machine-learning-style robustness checks – originally developed in the econometric causal-inference literature – to re-examine whether theory-specified SEM paths remain directionally and statistically stable when flexible machine-learning estimators are used to adjust for controls and covariates, offering an additional diagnostic layer beyond traditional fit indices.

CURRENT TRENDS AND FUTURE DIRECTIONS

Several broader trends are shaping the future trajectory of SEM research, building on the specific developments above. First, the multimethod integration of CB-SEM and PLS-SEM, and increasingly of SEM and machine learning more broadly, is likely to continue as computational tools make it easier to fit and compare multiple modeling traditions on the same data. Second, Bayesian methods, including Bayesian regularization, small-variance priors for approximate measurement invariance, and dynamic SEM's Bayesian estimation of intensive longitudinal data, continue to be refined, alongside ongoing methodological scrutiny of how sensitive results are to prior specification (MacCallum et al., 2012; Van de Schoot et al., 2019). Third, exploratory structural equation modeling and related flexible measurement approaches are being used increasingly to relax rigid simple-structure assumptions in confirmatory models (Marsh et al., 2009). Fourth, improved tools for a priori power analysis targeted at specific parameters, rather than global model fit alone, are making principled sample-size planning more accessible to applied researchers (Wang and Rhemtulla, 2021). Finally, the relationship between traditional latent-variable SEM and network psychometric approaches remains an active area of theoretical and methodological debate (Borsboom et al., 2003; Epskamp et al., 2018), with implications for how psychological and social constructs are conceptualized and measured going forward.

CONCLUSION

Structural equation modeling has evolved over more than a century, from Sewall Wright's biometric path diagrams and the psychometric factor-analytic tradition of Spearman and Thurstone, through Karl Jöreskog's unification of measurement and structural modeling and the diffusion enabled by the LISREL program, into today's diverse ecosystem of covariance-based, composite-based, dynamic, and Bayesian approaches implemented across both commercial software and an increasingly capable open-source R toolchain (*lavaan*, *blavaan*, *SEMinR*, *cSEM*). Its capacity to formally link theoretical constructs to empirical indicators while modeling measurement error and complex, multi-equation relationships explains its enduring and expanding role across the social, be-

havioral, health, and management sciences. Recent extensions – dynamic SEM for intensive longitudinal data, hybrid SEM-machine learning frameworks, and multimethod CB-SEM/PLS-SEM practice – show that the technique continues to absorb and adapt tools from adjacent fields rather than standing still. At the same time, ongoing debates over estimation approach, fit evaluation, sample size and power, and the interpretation of latent variables underscore that SEM remains an active and evolving methodological field rather than a fixed, uncontested technique. Researchers using SEM are therefore best served by grounding their modeling choices in both their substantive theory and a clear understanding of the assumptions and trade-offs associated with the particular SEM variant, estimator, and software tool they adopt.

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Structural Modeling of Employee Performance: The Roles of Capacity Building and Supervisor Support in Nepalese Commercial Banks

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ABSTRACT

Background: This study examines the impact of capacity building and supervisor support on employee's performance in commercial bank. It also the relationship among capacity building, supervisor support and employee's performance.

Method: This study used a convenience technique to collect data from 150 employees working in commercial bank. Data were gathered through an adopted questionnaire using seven-point likert scale. Statistical analyses were conducted using SPSS 20 and Smart PLS4. A descriptive and a descriptive and explanatory research design was applied and the including path analysis using SEM.

Result: The result indicates that both capacity building and supervisor support have a significant effect on employees performance. Among these factors, supervisor support emerged as a strong predictor, as reflected by its higher beta value.

Conclusion: Policy makers should prioritize capacity building and managerial support to improve employee performance in the banking sector.

Key words: Capacity building, Supervisor support, Employee performance, Commercial Banks

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INTRODUCTION

In the modern competitive banking environment, Human resources serve as a important indicator for the development of organizational and employee performance, facilitating the achievement of specified goals within a designated timeframe (Pradhan and Jena, 2017; Sadick and Kamardeen, 2020). Employees are the soul of any organization (Jasanoff, 2001). effective management is essential for sustaining employee performance and ensuring competitiveness in the market (Song et al., 2019). the employee whose performance can ensure the success of institution. Therefore, banks have to focus on development of employee performance through several ways. The satisfied employee can always perform better (Mengstie, 2020). the banking sector is a service industry, and the quality of service is directly linked to the performance of its employees. High-performing employees can provide excellent customer service, leading to customer satisfaction and loyalty. Organizational performance could be enhanced by aligning the performance management techniques with HRM practices (Den Hartog et al., 2004). Employee performance leads to improve overall efficiency and productivity of organization processes.

Capacity building refers to the process of enhancing an individual's knowledge, skills, and abilities to perform their job effectively (Hargreaves, 2011). Capacity building helps employees to do their job in effective ways. In the banking sector, capacity building initiatives encompass training programs, skill development workshops, and con-

tinuous learning opportunities. Training is the effective tools for enhancing performance (Pandey, 2017). Training and development activities are equally important HR functions which are concerned with imparting knowledge and skills for a particular job (Khan, 2013). Training is very important in banking sector (Dikshit and Jain, 2017). Bank use different technique in order to enhance capacity of their employees, Training and development activities are equally important HR functions which are concerned with imparting knowledge and skills for a particular job (Khan, 2013). Training is very important in banking sector to increase the performance level of employees (Dikshit and Jain, 2017). Banks are currently facing the challenges of employee turnover on which training and development can be effective tools for reducing employee retention (Awan and Kiren, 2016). Career development is the series of on-going activities in the organization, it involves training on new skills, making a career change within an organization which helps in attracting and retaining the qualified and calibre employees (Kibui et al., 2014). Career development is a means of arming employees with skills to perform their jobs and it is the representation of an employer's commitment to their workforce (Alnaqbi, 2011). Supervisors play a critical role in fostering a positive work environment, their support directly impacts employee morale, job satisfaction, and overall performance (Burke et al., 1992).

Supervisory support suggests the leading role played by an authoritative person for the promulgation of ef-

fectiveness and efficient fluency of work in organization through engaging best employees (Van Velsor and McCauley, 2004). Employee of banks has to perform their duties in challenging and changing dynamics. Thus, timely motivation from management can double the rate of performance done by employee (Neupane, 2019). The employee has to support the organization and organization has to support the employee for achieving organizational objectives and the coordination in workplace from superior ease the way of handling problem (Ali et al., 2018). A manager's leadership can have direct impact on the satisfaction of employees through personal contact and indirect impact through shaping supporting policies (Yukl and Becker, 2006). Manager support is described as the degree to which the leaders appreciate the contributions of its employees and care about their welfare. Employees are probably going to remain in those associations where they trust that their capacities, commitments, and positive attitudes are valued (Terera and Ngi-rande, 2014).

Employee performance refers to the activities and tasks performed by an employee efficiently and effectively. Performance can be measured by managers through different mechanisms (Saleem et al., 2021). There are various factors that affect the employee performance such as monetary benefits, training programs, non-monetary benefits, organizational support, organizational support for career development (OSCD), supervisory support and capacity building programs, etc. Resultantly employee performance enhances profitability of the organization (Gull et al., 2012). Interestingly, most of the existing studies on capacity building focus on broader organizational or national contexts. However, there is limited empirical evidence from commercial bank at local municipal level in Nepal, particularly in Siddhartha Nagar Municipality, Bhairahawa, Rupandehi. Thus, the objectives of this study is to access the relationship between capacity building, supervisor support on employees performance in commercial bank and to examine the affect of capacity building and supervisor support on employees performance. Therefore, based on the objectives this research seeks to answer of the following questions. Q1. Is there any relationship between capacity building and supervisor support on employees performance? Q2. Dose the capacity building and supervisor support affect the employees performance? The hypothesis is that, H1: capacity building significant impact employees performance, H2: supervisor support significantly impact on employees performance.

LITERATURE REVIEW

The social exchange theory explain that employees respond to supportive supervisors through reciprocity, putting in extra effort and improving performance (Howard and Ekeh, 1976; Cropanzano, 1976). In the context of capacity building, training enhance employees skill. (Sinha and Chandrakasan, 2001; Ghimire et al., 2023). Human capital theory suggest that investing in employees skill and knowledge boost productivity (Rees and Becker, 1965). In the workplace, social support the-

ory suggest that capacity building and supervisor support provide employees with essential resource, guidance enhancing their skills, confidence and performance (Thoits, 1995; Cullen et al., 1999)

Lessing (Lessing, 2020) examined the effect of training, staff development, and supervisor support on employee's performance. Using quantitative survey method and statistical analysis, the study found that effective training significantly enhance motivation, skill and performance. Ali and Nada (Ali and Nada, 2018) found that employee training has positive effect on performance. Based on a study of 303 employees from two Lebanese companies revealed that capacity building program enhanced job satisfaction and work quality. Similarly, Subha and Bhattacharya (Subha and Bhattacharya, 2021) explore the key factors influencing employees performance in nepalese commercial bank. Using primary data from 200 respondents, employer branding, career growth, work environment significantly affect employees performance.

Likewise, Rahaman et al. (Rahaman et al., 2023) aimed to identify factors influencing employees performance in private commercial banks. Using quantitative approach, questionnaire were distributed among 250 bank employees and SPSS 26 is used to regression analysis. The study found that manager leadership style, compensation have significant positive impact on employees performance. Additionally, Chapagain et al. (Chapagain et al., 2022) investigated the relationship between training and work performance. Using descriptive and cross sectional design, data were collected through questionnaire from 250 employees, SPSS and PLS-SEM were used to analyze collected data. The study found training was highly positively affected employees performance.

RESEARCH FRAMEWORK

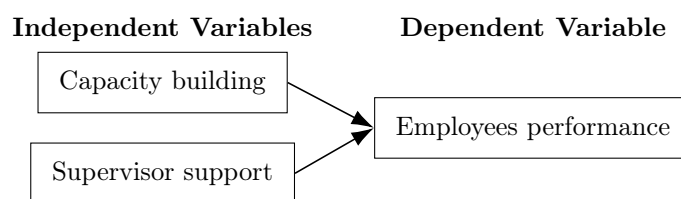


Figure 1. Conceptual framework of the study

Note. Adapted from (Wassem et al., 2019; Palomo et al., 2010; Ahmad et al., 2015; Masood, 2010)

METHODS AND PROCEDURE

Research Design

This research adopts both descriptive and causal-comparative designs to investigate the associations between variables and the impact of one variable on another (Kerlinger, 1986). Thus, the chosen research designs provide a comprehensive framework for conducting the study.

Population and Sample

The population of the research study comprises the total respondents within the research area, specifically the

employees of commercial banks in Siddhartha Nagar municipality, Bhairahawa. The total number of employees in commercial banks in Bhairahawa is 240. Therefore, the population of the study is identified as 240 individuals.

Table 1. Total Employees of Commercial Banks in Siddhartha Nagar Municipality, Bhairahawa.

S.N	Name of commercial bank	No. of employees
1	Kumari Bank	14
2	Rastriya Banijya Bank	13
3	Agriculture Development Bank	15
4	Nabil Bank	16
5	Nepal Investment Mega Bank	11
6	Himalayan Bank	14
7	Nepal SBI Bank	11
8	Everest Bank	12
9	Prabhu Bank	9
10	Laxmi Sunrise Bank	11
11	Global IME bank	15
12	Citizens Bank International	10
13	Prime Commercial Bank	11
14	NIC Asia Bank	16
15	Machhapuchchhre Bank	13
16	Nepal Bank Limited	12
17	Sanima Bank	13
18	Siddhartha Bank	11
19	NMB Bank	13
Total		240

Sample

Sample is a part of a population or subset of population and denoted by n . The total sample size for this study has been obtained using the formulae developed by (Yamane, 1967). In case of population size is known, the Yamane formula for determining the sample size is given by:

$$n = \frac{N}{1 + Ne^2} \quad (1)$$

Where

n = sample size, N = Population size, and e = Margin of error (MOE), $e = 0.05$ based on research condition

Thus, the sample size of the study is $n = 150$

convenience sampling method is used to collect the data from respondents

Nature and Sources of Data

Quantitative data for the study were collected through a primary source.

Tools for Data Collection

Questionnaire is used as research instrument to collect the data from respondent. The questionnaire employs a seven-point Likert scale to gather responses from the participants. The questionnaire is comprised two sections. The first section contains information of demographic profile of participants, and the second section is the main part of the research instrument which is related to the independent variable and dependent variable. The demographic variable such as age, gender, education, experience, and job status were adopted from previous literature (Saleem et al., 2022). The first independent variable capacity building, all the items for this construct were taken from (Gull et al., 2012). The Supervisor support is next independent variable in this research, ten items were used to measure for this construct, all these times for this variable were adopted from (Yang et al., 2015; Saleem and Amin, 2013). The dependent variable, employee's performance, is a construct that is measure through six items. All these items for this construct were derived from previous literature, specifically from the previous researcher (Saleem et al., 2022; Tabouli et al., 2016).

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Method of Data Analysis

The research study has used Smart PLS and SPSS version 20 to analyze the collected data. In this regard, the study employed various statistical tools based on the appropriateness of the data. Additionally, a reliability test was conducted to assess the reliability of the research instrument.

RESULTS

Demographic profile of employees and performance:

The following table 2 shows the age distribution of employees, with 41.3% aged 30 or below and 58.7% over 30, with a total of 150 employees. The average performance score for employees aged 30 or below is slightly higher than those over 30, but the difference is minimal.

Table 2. Distribution of Employees by Age, Gender, Education, Experience, and Their Performance Scores

Demography	Category	N (%)	Mean	SD
Age	≤30 years	62 (41.3%)	5.42	1.407
	> 30 years	88 (58.7%)	5.38	1.333
	Total	150 (100%)	5.4	1.36
Gender	Male	90 (60%)	5.3	1.423
	Female	60 (40%)	5.56	1.255
	Total	150 (100%)	5.4	1.36
Education	Level +2	14 (9.3%)	5.58	1.183
	Bachelor	80 (53.3%)	5.26	1.389
	Master	56 (37.3%)	5.55	1.359
	Total	150 (100%)	5.4	1.36
Experience	≤5 years	95 (63.3%)	5.36	1.365
	> 5 years	55 (36.7%)	5.46	1.361
	Total	150 (100%)	5.4	1.36

Note. Output of SPSS 20

The work force is largely male, with 90 men and 60 women. On average, female employees perform slightly better than male counterparts, and the overall performance score stand at 5.4 most of employees hold bachelor degree while about 37.3% have completed master's degree. Those with a +2 qualification or higher tend to show stronger performance, with minor differences across education levels. The majorities of employees have 5 years or less of work experience; those with more than

5 years record a marginally higher average performance score.

The table highlights employee's views on three dimensions: supervisor support (SS), capacity building (CB), and employees performance (EP). Perception of supervisory support are largely positive, with high mean score showing all employees feel valued, respected, and well guided by their supervisors. Capacity building aspects receive comparatively lower ratings, indicating some dissatisfaction with training, mentoring, and career development opportunities. Employees performance is rated favorably overall, while supervision and performances are strengths, greater attention is needed to improve career development initiatives.

From Table 4, it is found that the value of Cronbach's Alpha for different components of supervisor support, capacity building, and employees performance are 0.909, 0.903 and 0.922 respectively, indicating that the statements in the questionnaire related to these variable are suitable and appropriate.

Table 5 indicates the measures and validity related to the outer model. It shows the standardized outer loading, Variance Inflation Factor (VIF), mean, and Standard Deviation (SD) of the outer model. Twenty scale items are used to assess three latent variables. The outer loading values of all the items are above the threshold value of 0.70 except SS5, which indicates the absolute contributions of each item to measuring the respective variable (Sarstedt et al., 2017). In the case of SS5, according to (Hair and Alamer, 2022), the items having outer loading values above 0.6 can also be retained for further analysis. Similarly, the VIF values of all the items are below 5, indicating no multicollinearity among the scale items (Hair and Alamer, 2022). Consequently, there is no multicollinearity among the items. The mean and standard deviation (SD) results of all the measurement items are in a good range on 7-point Likert scale data. Hence, the measurement items qualify for reliability and validity for further assessment.

The validity and internal reliability of the constructs utilized in this investigation are shown in Table 6. All constructs have Cronbach's Alpha values above the conventional cutoff point of 0.705 (Bland and Altman, 1997), confirming the reliability of the scale used to measure each construct and demonstrating the internal consistency of all constructs. Additionally, construct validity and reliability are indicated by Composite Reliability (CR) rho_a and CR rho_c values exceeding 0.70 (?). All of the constructs appear to have established convergent validity, as indicated by the Average Variance Extracted (AVE) values being over the 0.50 threshold values (Hair and Alamer, 2022). Therefore, all of the quality criteria measures are qualified by the outcomes of the preceding table.

Table 7 contains the correlations between each of the variables utilized in this study, expressed as the Heterotrait-Monotrait (HTMT) ratio. The values of the HTMT ratio fall between 0.371 and 0.809. Although one value is above 0.85 but below 0.90, the normal permissible range for HTMT ratio values is below 0.85. According

to (Heseler et al., 2015), we can also accept the variable having an HTMT ratio of up to 0.90. As a result, discriminant validity across the study's reflective constructs has been proven (Hair and Alamer, 2022).

Model Fit Assessment

The standardized root mean square residual (SRMR) was utilized for goodness-of-fit indices for the model. The SRMR value is 0.061, less than the threshold value of 0.08. This indicates that the model possesses good explanatory power, as (Bentler and Hu, 1998) suggested. For the variable Employee's performance, the f-square value of Capacity Building is 0.042 on employees performance, indicating small size effect. The f-square value of supervisor support on employees performance is 1.087, indicating large effect size (Cohen, 1988). Lastly, the R-square value of employees performance is 0.595, which indicates moderate predictive power (Hair et al., 2013).

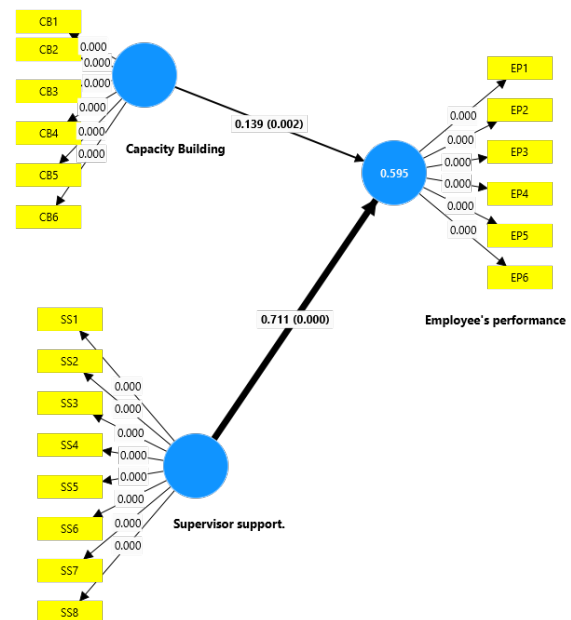


Figure 2. Structural Assessment Model

Note. Output of smart PLS

Table 8 show the bootstrapping results under 5000 subsamples and decisions on zshypotheses. All hypotheses H1 and H2 are accepted at significance level 0.05. Hence, there is a positive and significant impacts of capacity building ($\beta=0.139$; $p<0.05$) on employees performance. Similarly, supervisor support has positive and significant impact on employees performance ($\beta=0.711$; $p<0.05$).

According to table 9, there is positive week correlation between capacity building and supervisor support. Similarly, there is positive week correlation between capacity building and Employee's performance. And, there is positive strong correlation between Supervisor support and Employee's performance.

DISCUSSION

The findings of this survey indicate that capacity building has a significant effect on employee performance. Similarly, training and capacity building programs are impor-

Table 3. Descriptive statistics of items measuring Supervisory Support(SS), Capacity Building(CB), and Employee Performance(EP).

Items	N	Min	Max	Mean	SD
My supervisor is helpful to me in getting the job done. (SS1)	150	1	7	5.17	1.764
My supervisor is willing to extend help to me perform my job. (SS2)	150	1	7	5.24	1.531
My supervisor takes pride in my work accomplishments. (SS3)	150	1	7	4.97	1.716
My supervisor tries to make my job as interesting as possible. (SS4)	150	2	7	5.23	1.499
My supervisor helps me in crisis. (SS5)	150	1	7	4.87	1.752
My supervisor helps in difficult tasks. (SS6)	150	1	7	5.17	1.596
My supervisor treated me with respect. (SS7)	150	1	7	5.87	1.556
My supervisor gave me regular feedback on my performance. (SS8)	150	1	7	6.2	1.442
I am being sufficient trained by the organization. (CB1)	150	1	7	3.37	1.982
My senior properly guides me on the job. (CB2)	150	1	7	3.02	1.881
I am provided mentoring by the organization whenever required. (CB3)	150	1	7	3.71	1.852
I feel sufficient skilled for the present job. (CB4)	150	1	7	3.78	2.02
My organization encourage to avail training opportunities. (CB5)	150	1	7	4.7	1.976
My management motivates me for good performance. (CB6)	150	1	7	4.78	2.152
I feel dedication, seriousness and ability to take responsibility. (EP1)	150	1	7	5.77	1.386
I enjoy professional skill or professionalism and technical knowledge required to carry out the work efficiently. (EP2)	150	1	7	5.38	1.629
I do my work according to specific policies and procedures. (EP3)	150	1	7	5.25	1.839
I feel satisfied with the work I do in the bank. (EP4)	150	1	7	5.61	1.483
Planning the work before starting its implementation contributes to setting the goals that need to be achieved. (EP5)	150	1	7	4.85	1.625
I have the ability to plan my work and its accomplishment according to the planned schedule. (EP6)	150	1	7	5.55	1.62

Note. Output of SPSS 20

Table 4. Reliability Statistics

Variable	Items	Cronbach's Alpha	Corrected Item-Total Correlation
Supervisor support	SS1	.909	.627
	SS2		.798
	SS3		.674
	SS4		.750
	SS5		.610
	SS6		.767
	SS7		.785
	SS8		.671
Capacity building	CB1	0.903	.728
	CB2		.704
	CB3		.815
	CB4		.802
	CB5		.702
	CB6		.664
Employees performance	EP1	0.922	.901
	EP2		.900
	EP3		.913
	EP4		.905
	EP5		.919
	EP6		.909

Note. Output of SPSS 20

Table 5. Measurement Items Assessment/ Assessment of Survey Items

Variables	Items	Loadings	VIF	Construct
Capacity Building	CB1	0.786	2.757	3.89 / 1.623
	CB2	0.762	2.691	
	CB3	0.862	3.741	
	CB4	0.852	3.235	
	CB5	0.842	3.000	
	CB6	0.805	2.829	
Employee's Performance	EP1	0.901	4.055	5.40 / 1.360
	EP2	0.888	3.398	
	EP3	0.828	2.528	
	EP4	0.873	2.942	
	EP5	0.784	2.063	
	EP6	0.842	2.811	
Supervisor Support	SS1	0.725	1.923	5.34 / 1.258
	SS2	0.842	3.640	
	SS3	0.750	2.776	
	SS4	0.797	3.031	
	SS5	0.686	1.673	
	SS6	0.817	2.997	
	SS7	0.869	4.930	
	SS8	0.783	3.416	

Note. Output of SmartPLS.

Table 6. Construct Validity and Reliability Assessment

Variables	Cronbach's alpha	Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)
Capacity Building	0.904	0.933	0.924	0.671
Employee's performance	0.925	0.930	0.941	0.729
Supervisor support.	0.911	0.921	0.928	0.617

Note. Output of smart PLS

Table 7. Heterotrait-Monotrait Ratio of Correlations (HTMT)

variables	Capacity Building	Employee's performance	Supervisor support.
Capacity Building			
Employee's performance	0.401		
Supervisor support.	0.371	0.809	

Note. Output of smart PLS

Table 8. Hypothesis testing (direct effect)

Hypothesis	β	Mean	SD	t	P-value	Decision
Capacity Building $\rightarrow$ Employee's performance	0.139	0.141	0.044	3.132	0.002	accepted
Supervisor support. $\rightarrow$ Employee's performance	0.711	0.711	0.052	13.758	0.000	accepted

Note. Output of smart PLS

Table 9. Correlation

	Supervisor support	Capacity building	Employee's performance
Supervisor support	1	.336**	.738**
Capacity building		1	.363**
Employee's performance			1

Note. Output of SPSS 20

tant for human resource development (Asfaw et al., 2015; Guest, 1997). Likewise, training enhance employees performance, productivity, knowledge and skill in banking sector and it also helps in career development of any individual over the life span (Gull and Azam, 2012; Wanyama and Mutsotso, 2010; Otoo and Mishra, 2018). Regarding supervisory support, managers should maintain good working environment in workplace, consistent feedback to their subordinate, clear and effective communication with their employees regarding work related challenges and fair treatment to the employees in workplace help improve employees productivity and performance in banking sectors (Pulakos, 2009). The results of the research show that managers' support does not significantly affect the performance of textile industry employees (Wassem et al., 2019). capacity building and supervisor support improves the performance of employees is justified by this study. Based on the results, the study recommended that employees' capacity building is positively linked with the employees' performance. Therefore, firms should arrange training for employees with the collaboration of the HR department.

CONCLUSION

In conclusion, there is a positive effect of capacity building on employees' performance. Similarly, supervisor support also has a positive effect on employees' performance. It means providing training and supervisor help to the banking employees to increase their performance level of the employees. Furthermore, there is a positive and significant relationship between capacity building, supervisor support, and employee performance. This study emphasizes the value of capacity building and managerial support in improving employee performance in the banking industry. Policymakers could encourage this by organizing training programs for bank employees.

LIMITATIONS AND FUTURE RESEARCH

This study only focuses on the employees' performance of the commercial bank of Siddharth Nagar municipality, Bhairahawa. So, the finding may not be applicable in other areas. Future research should investigate how peers and family influence career choices and how these aspects improve employee performance. Organizations may apply this information to create strategies for training.

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CONFLICT OF INTEREST

The authors decide no conflict of interest related to this study.

AUTHORS' CONTRIBUTION

Asfaq Ahmad: Conceptualization, methodology, data curation, original draft preparation, software,

Govind Jnawali: Validation, investigation, formal analysis, writing-review and editing, supervision, software

DATA AVAILABILITY STATEMENT

Data were exclusively used for this paper.

ETHICAL STATEMENT

This study did not require ethical approval because it did not include any human or animal trials. This unique article has not been submitted to another publisher. Furthermore, this article meets all the relevant ethical considerations.

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